

Computer Vision with Deep Learning

Lecture 14: Light and shading- I

slides credit: Ioannis Gkioulekas, Lana Lazebnik, Raymond Yeh, David Lindell, (and whoever mentioned in specific slides)

Announcements

- HW2 due tomorrow!

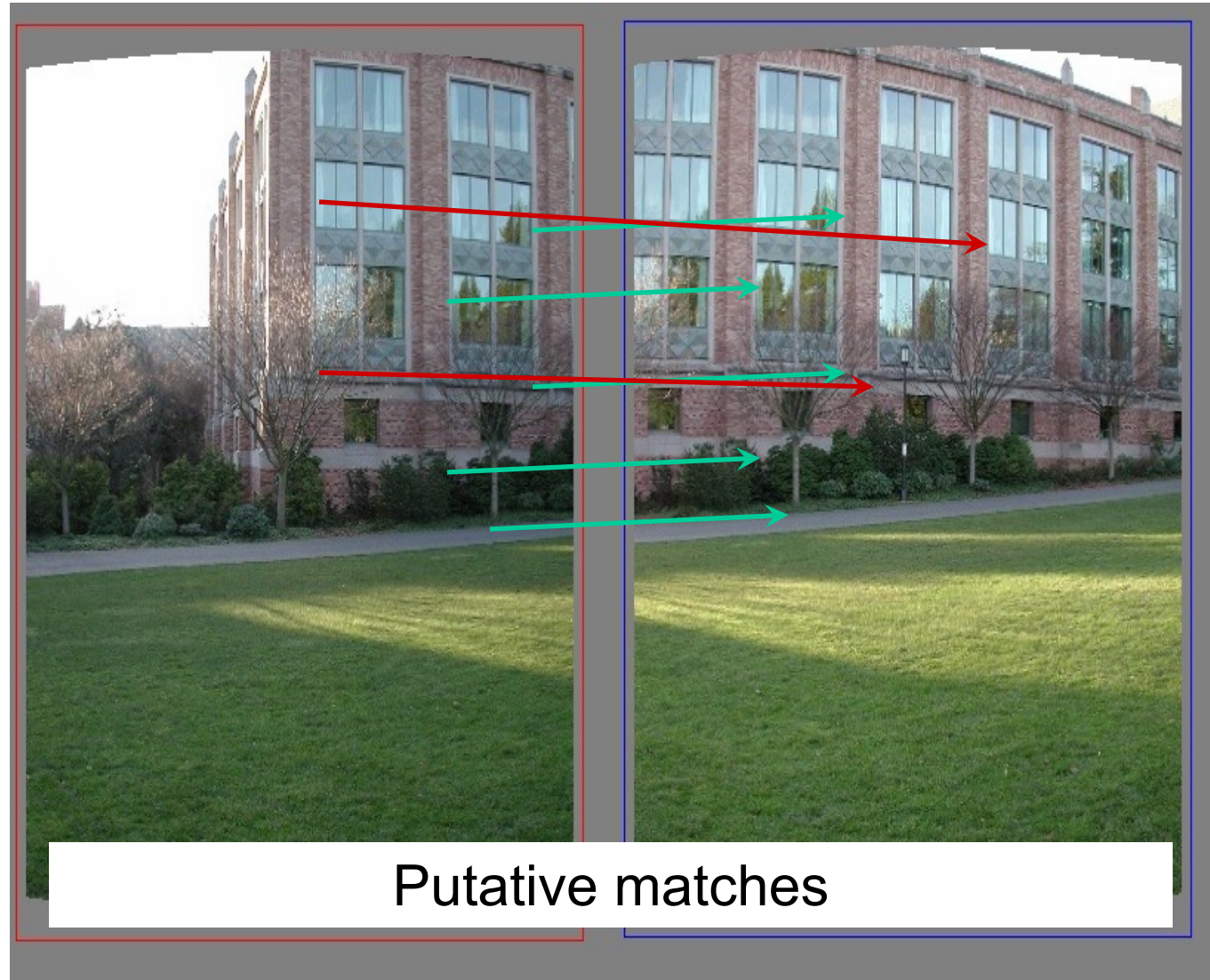
Today

- Pinhole cameras
- Cameras with lenses
- Digital sensors

PREVIOUSLY ON

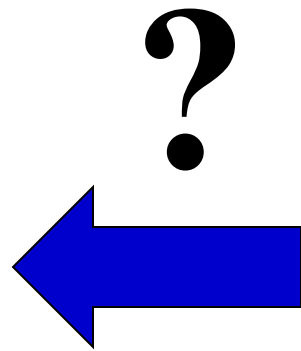
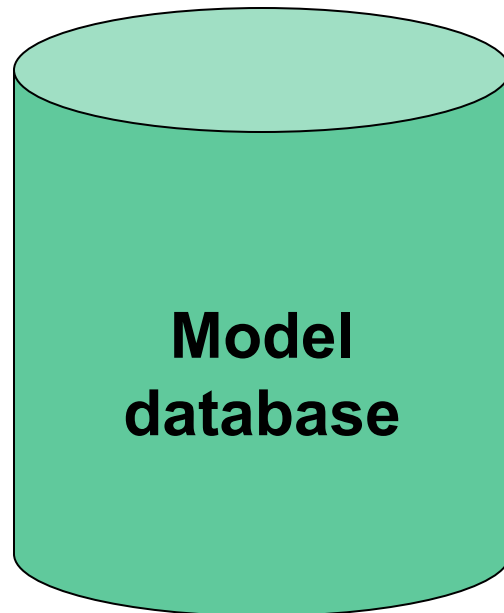
COMPUTER VISION WITH DEEP LEARNING

RANSAC example: Translation



Scalability: Alignment to large databases

- What if we need to align a test image with thousands or millions of images in a model database?
 - Efficient putative match generation: approximate descriptor similarity search, inverted indices



Test image



Large-scale visual search

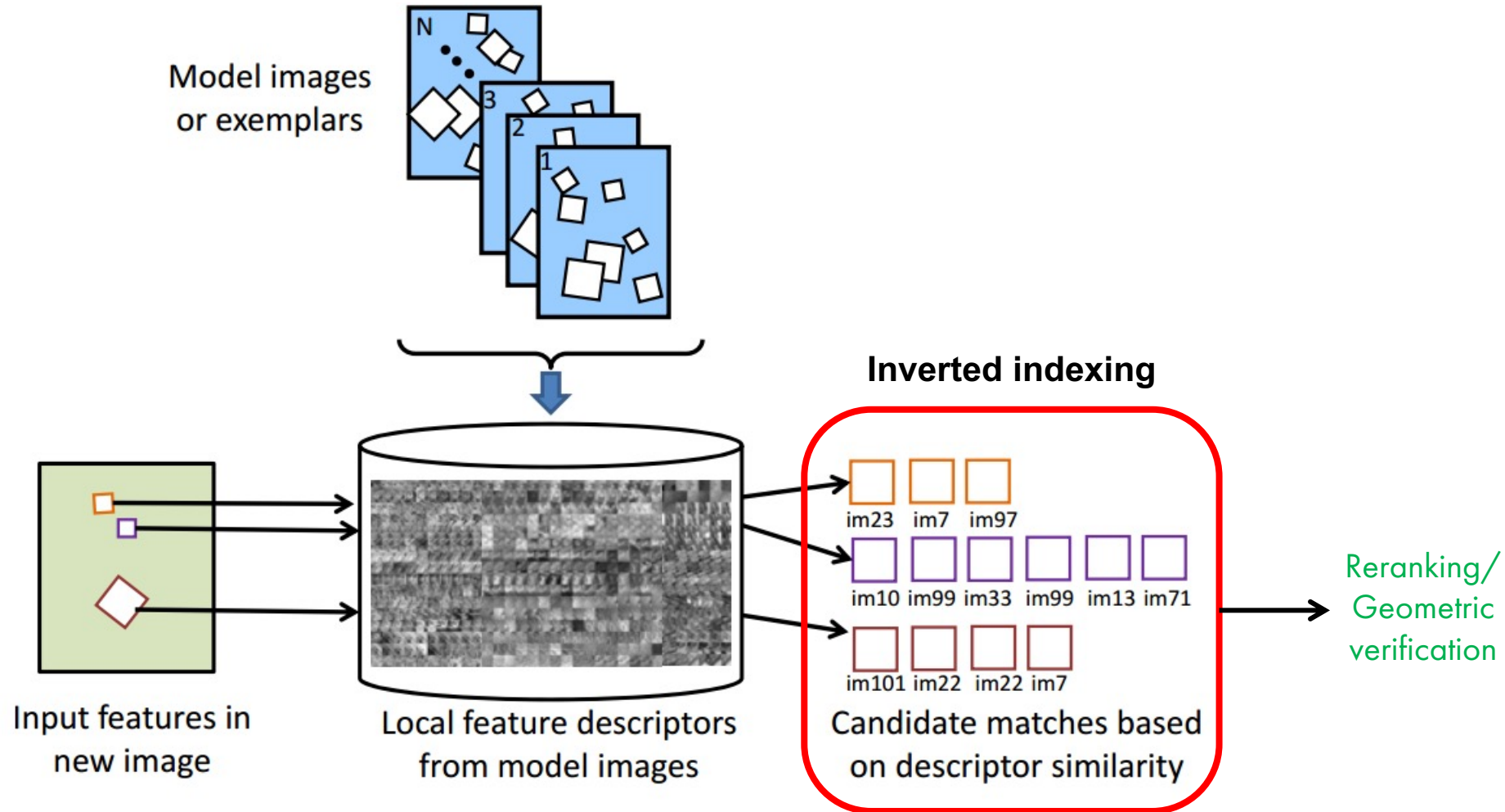
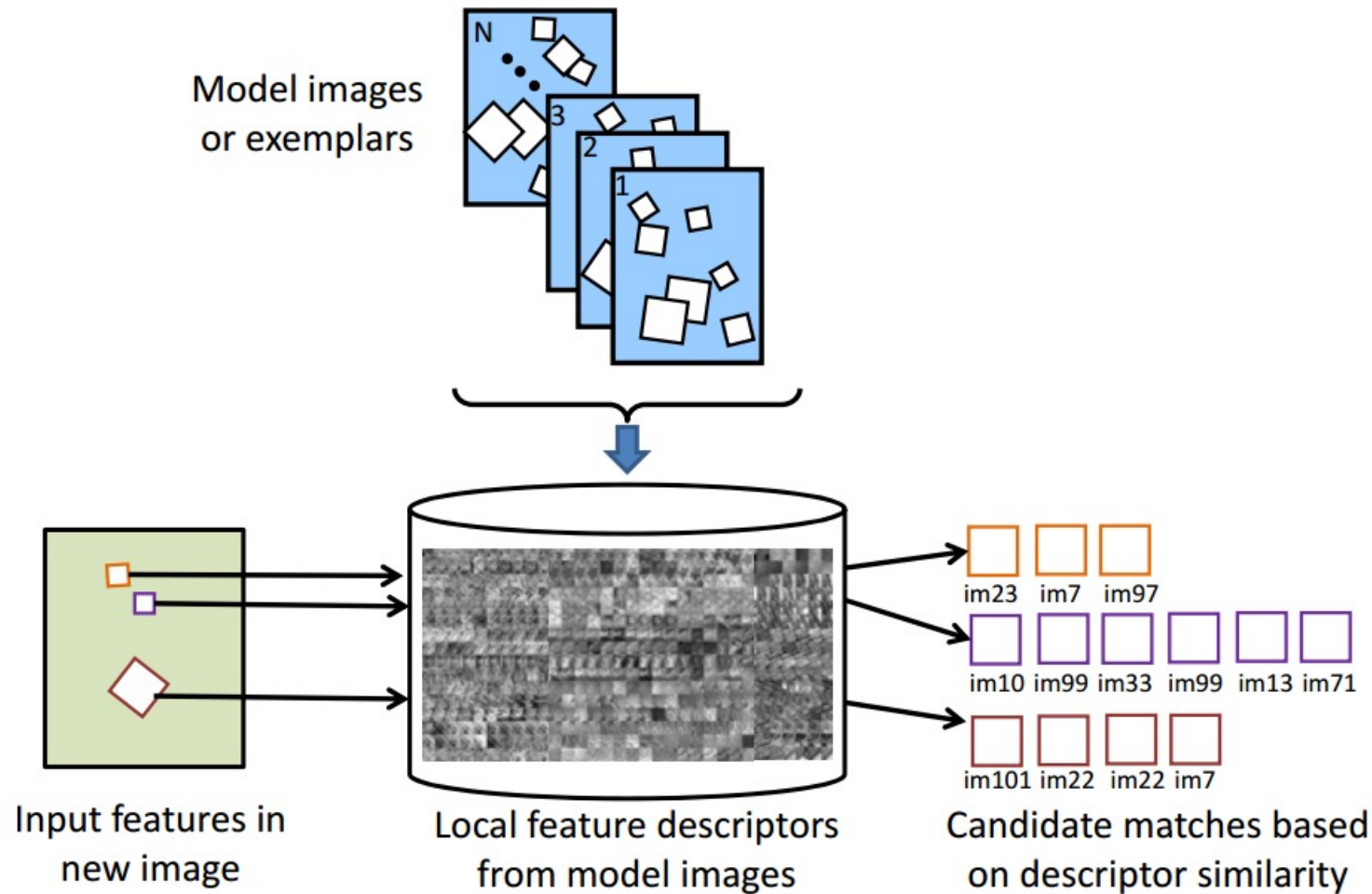


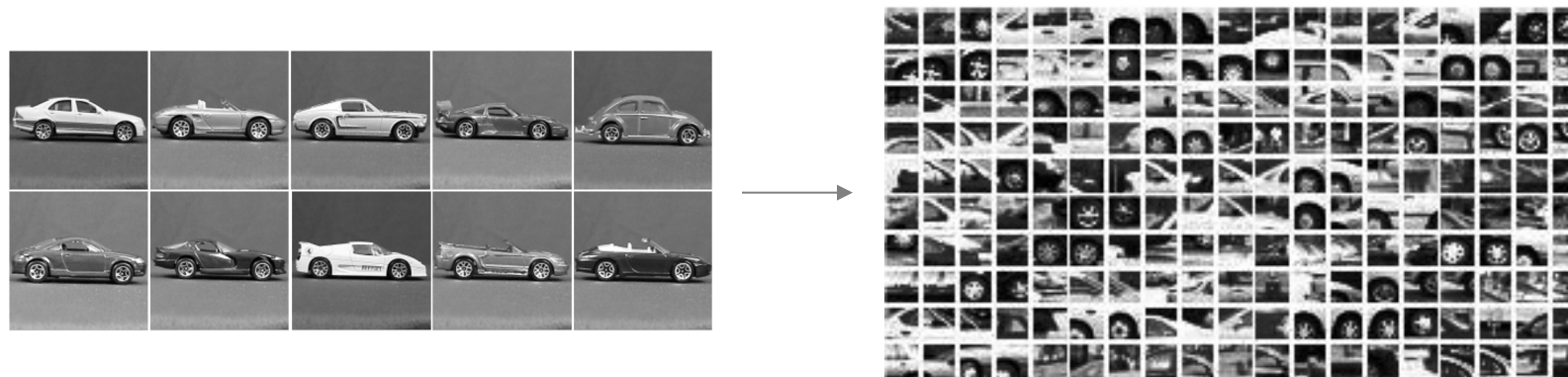
Figure from: Kristen Grauman and Bastian Leibe, [Visual Object Recognition](#), Synthesis Lectures on Artificial Intelligence and Machine Learning, April 2011, Vol. 5, No. 2, pp. 1-181

How to do the indexing?

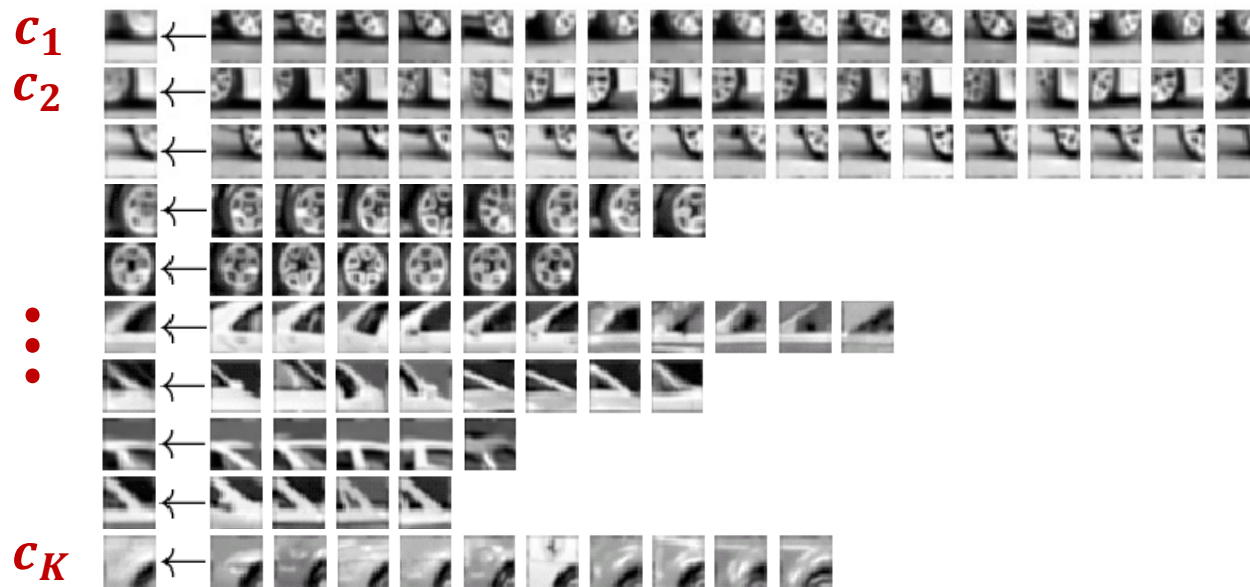


- Idea: find a set of *visual codewords* to which descriptors can be *quantized*

Recall: Visual codebook for implicit shape models



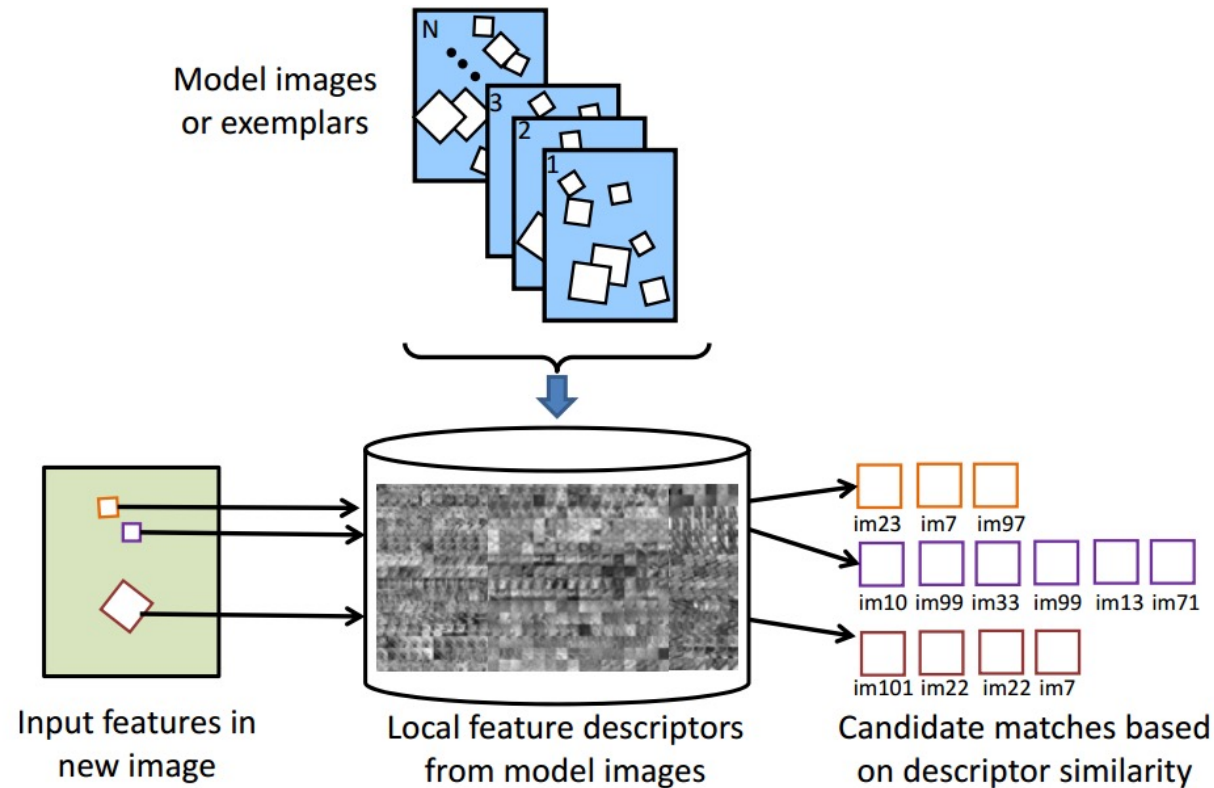
Appearance codebook



K-means example

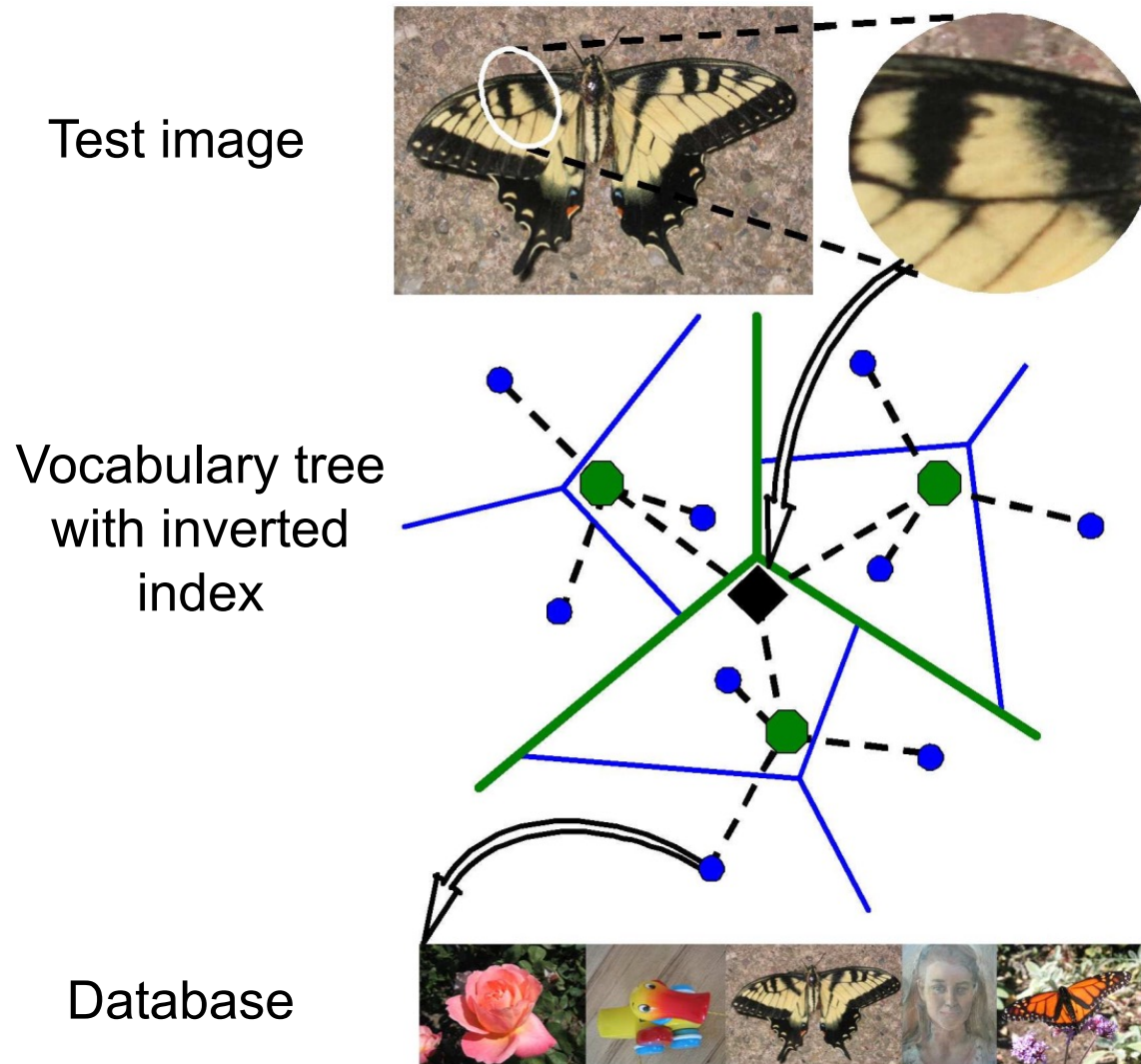


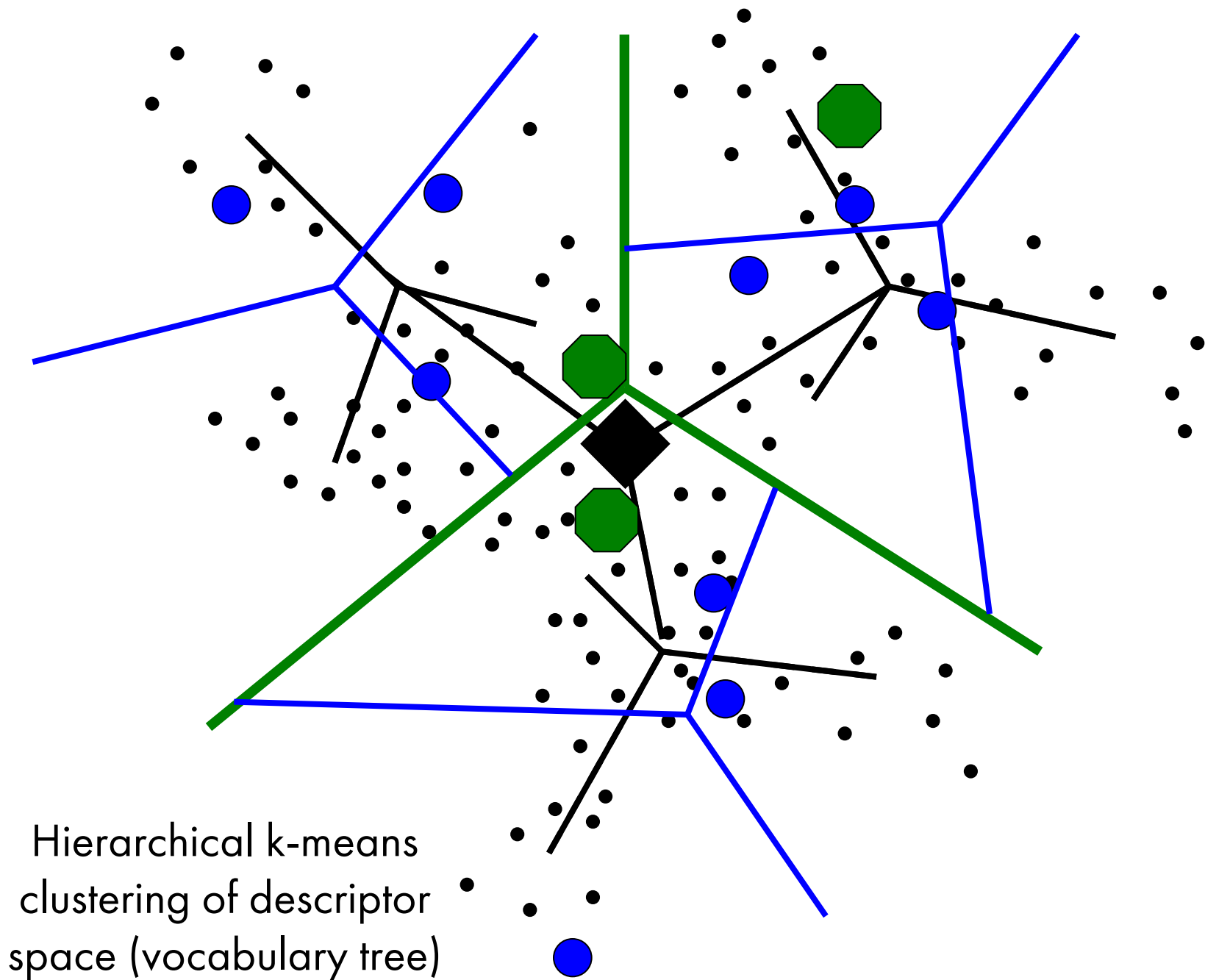
How to do the indexing?



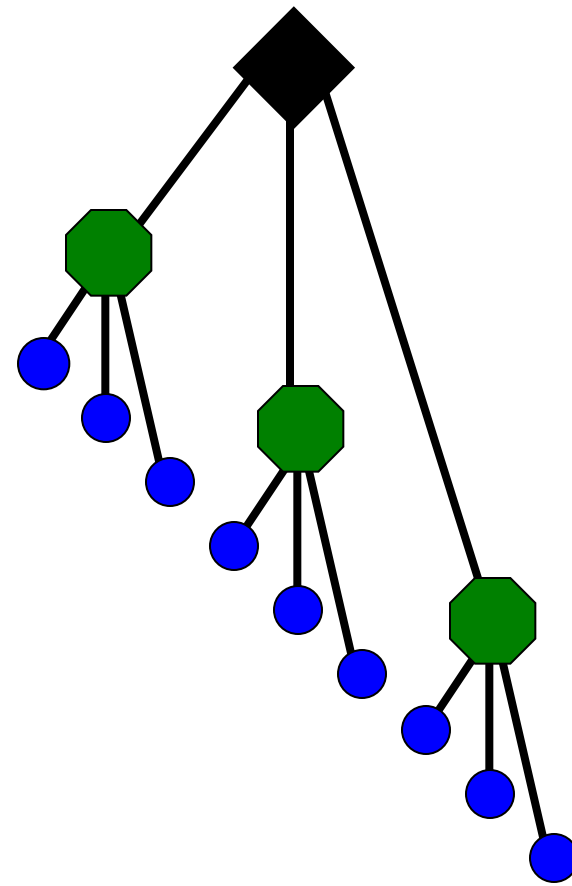
- Cluster descriptors in the database to form codebook
- At query time, quantize descriptors in query image to nearest codevectors
- Problem solved?

Efficient indexing technique: Vocabulary trees





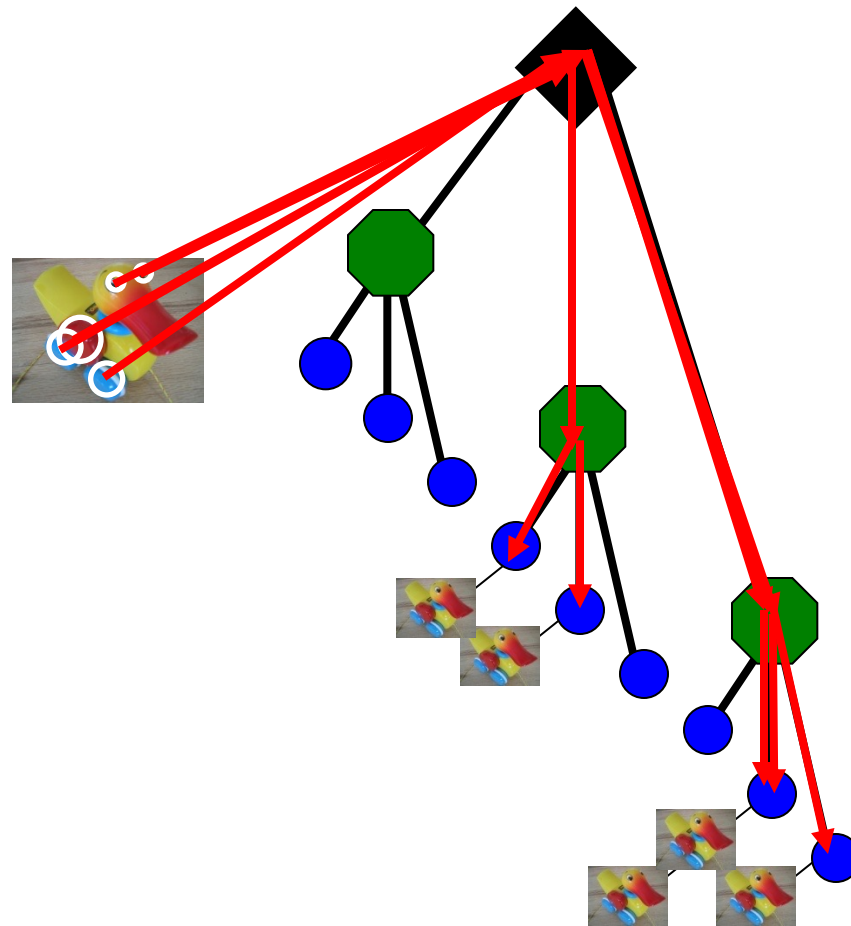
Hierarchical k-means
clustering of descriptor
space (vocabulary tree)



Vocabulary tree/inverted index

Slide credit: D. Nister

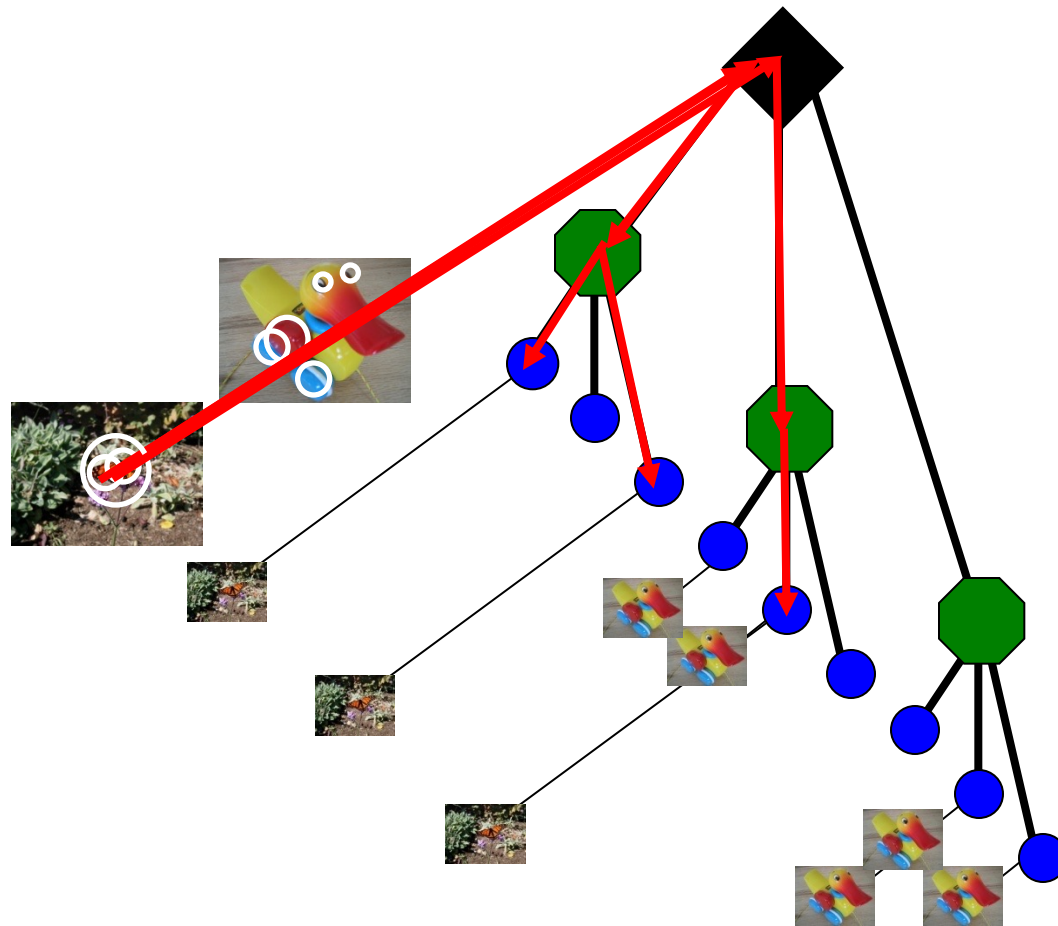
Model images



Populating the vocabulary tree/inverted index

Slide credit: D. Nister

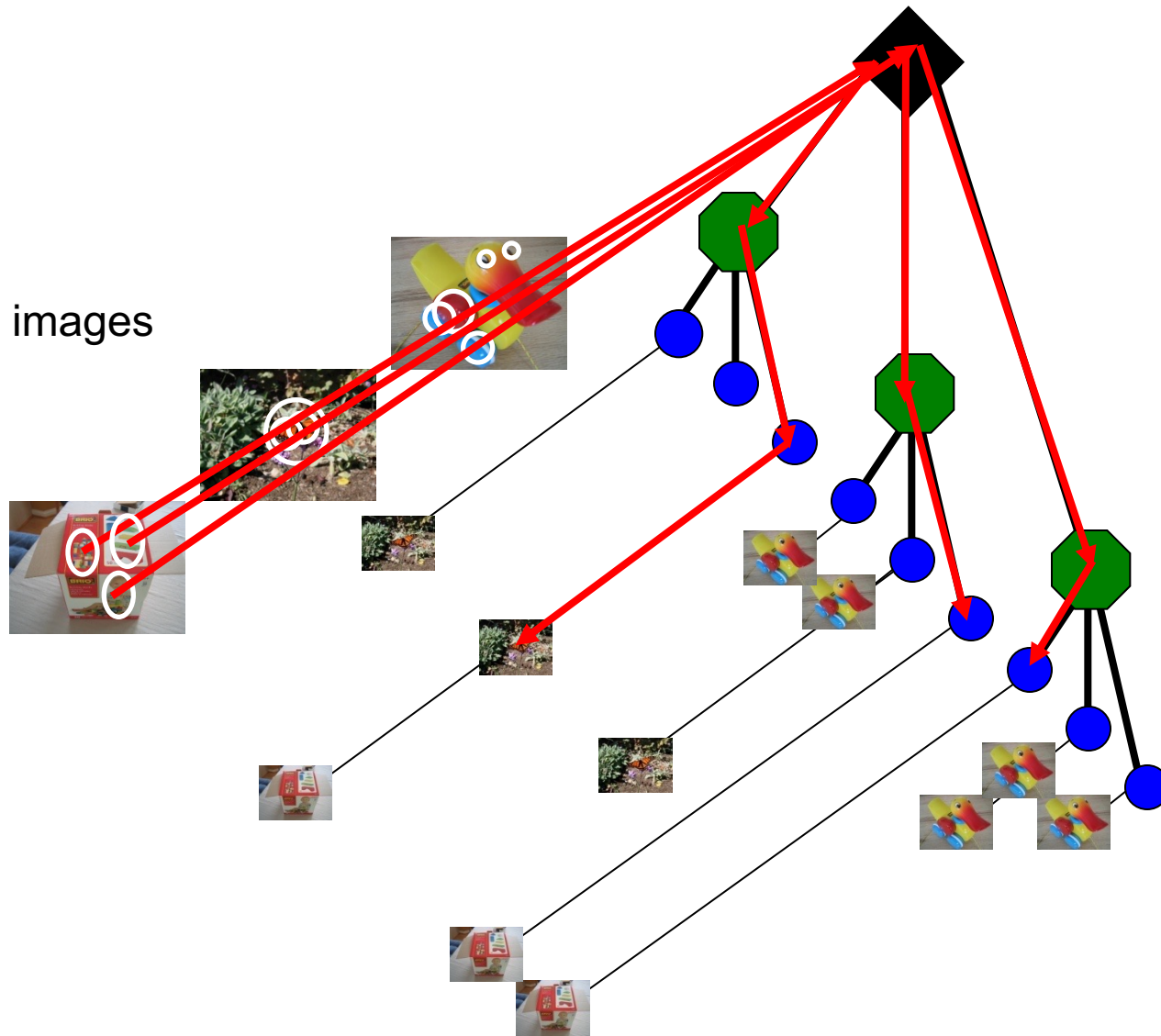
Model images



Populating the vocabulary tree/inverted index

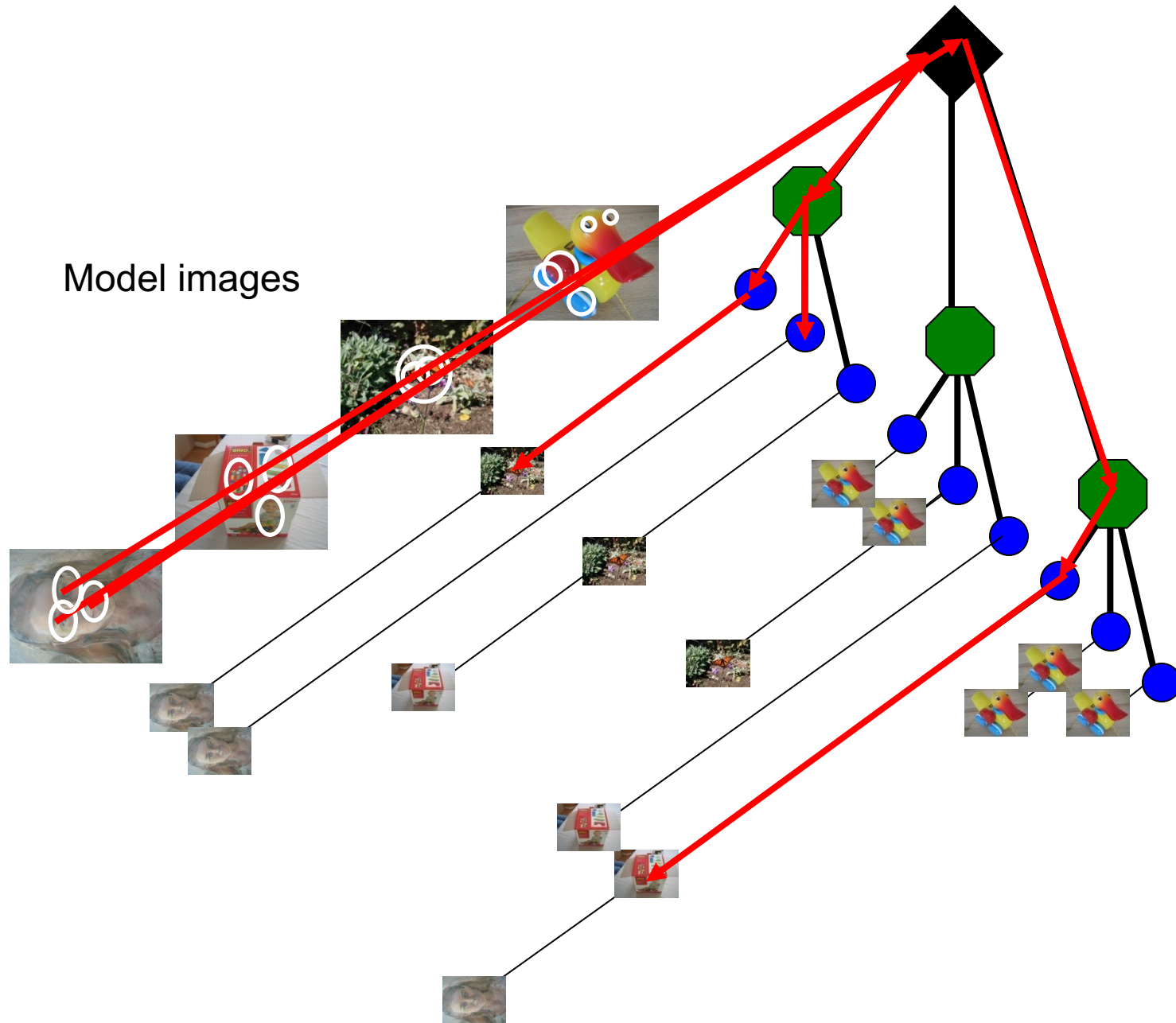
Slide credit: D. Nister

Model images



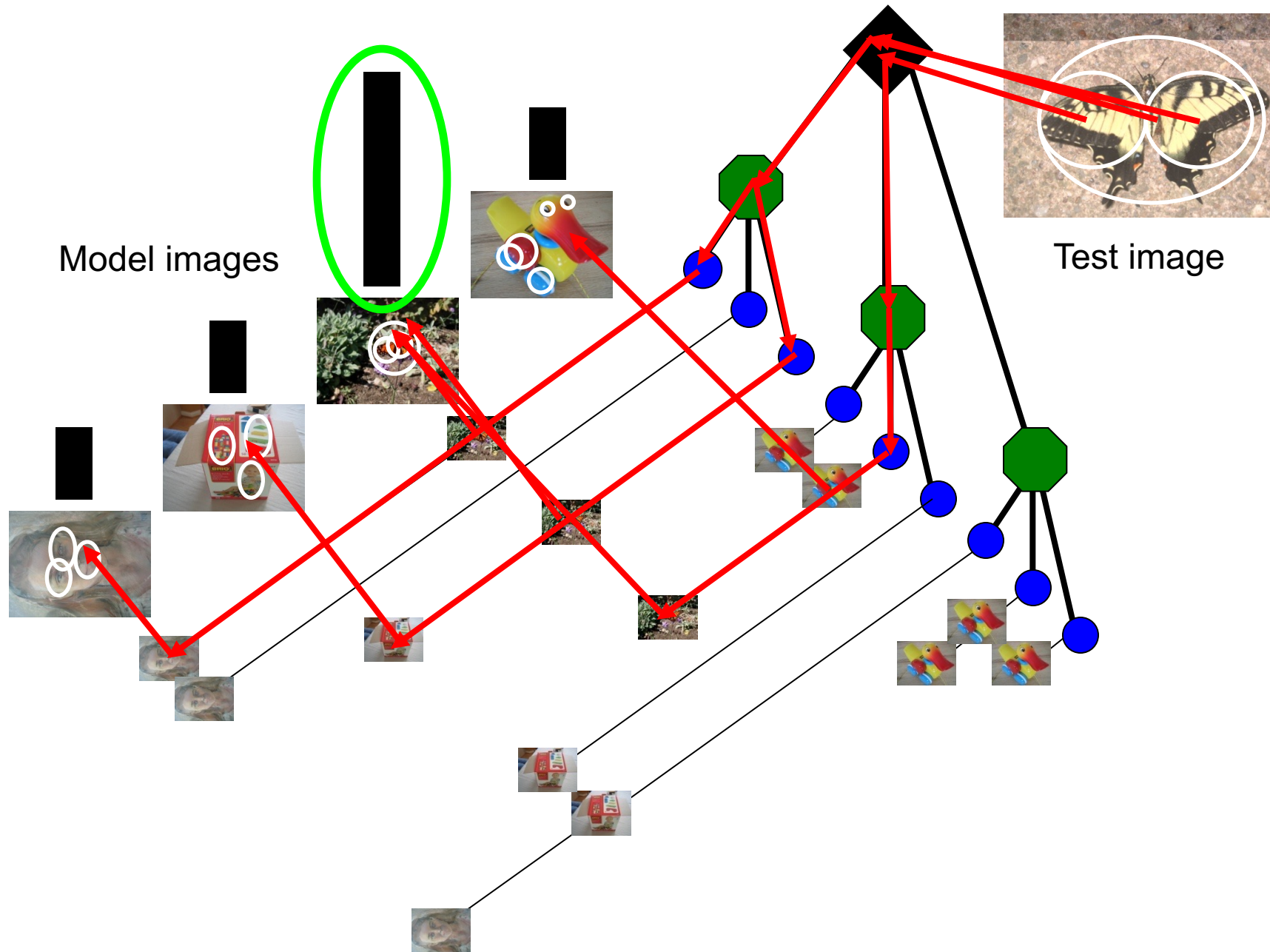
Populating the vocabulary tree/inverted index

Slide credit: D. Nister



Populating the vocabulary tree/inverted index

Slide credit: D. Nister



Model images

Test image

Looking up a test image

Slide credit: D. Nister

how do capture an image?

Let's say we have a sensor...



digital sensor
(CCD or
CMOS)

... and an object we like to photograph

real-world
object



digital sensor
(CCD or
CMOS)



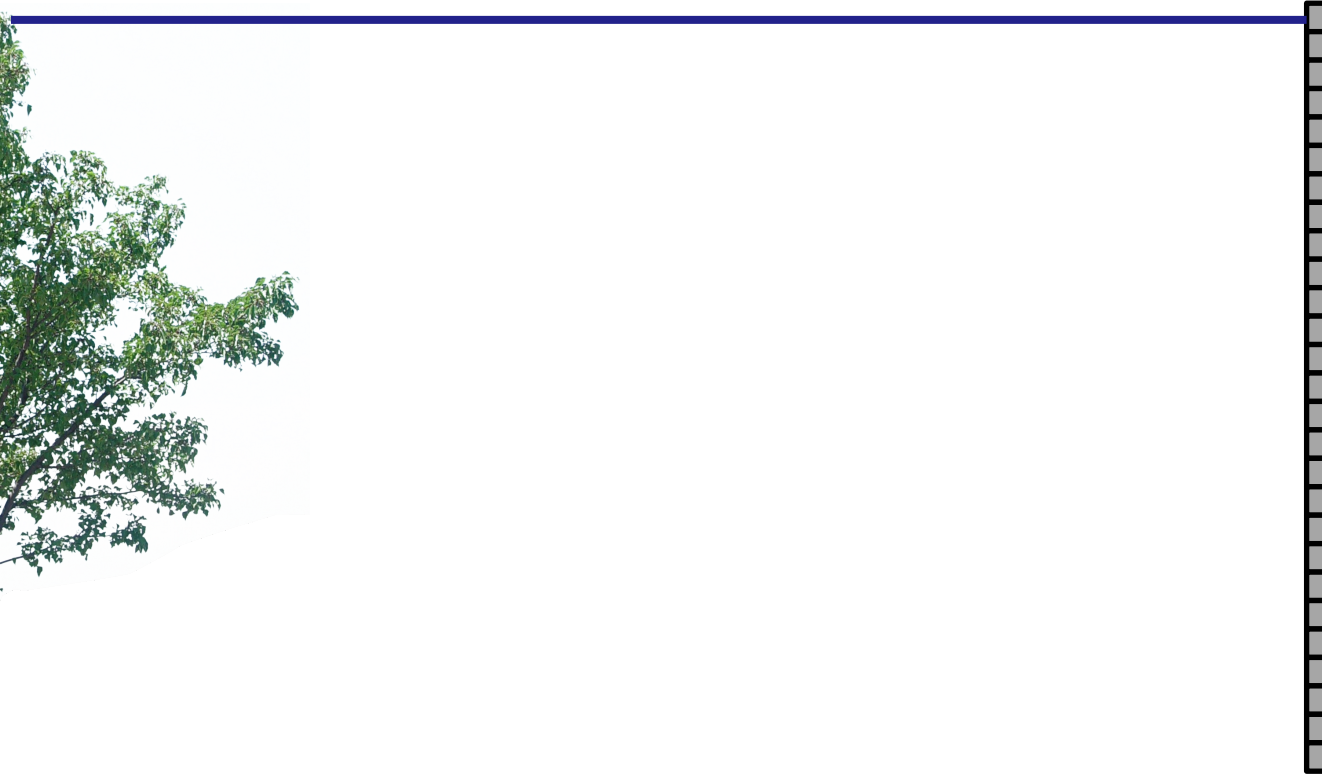
What would an image taken like this look like?

Bare-sensor imaging

real-world
object

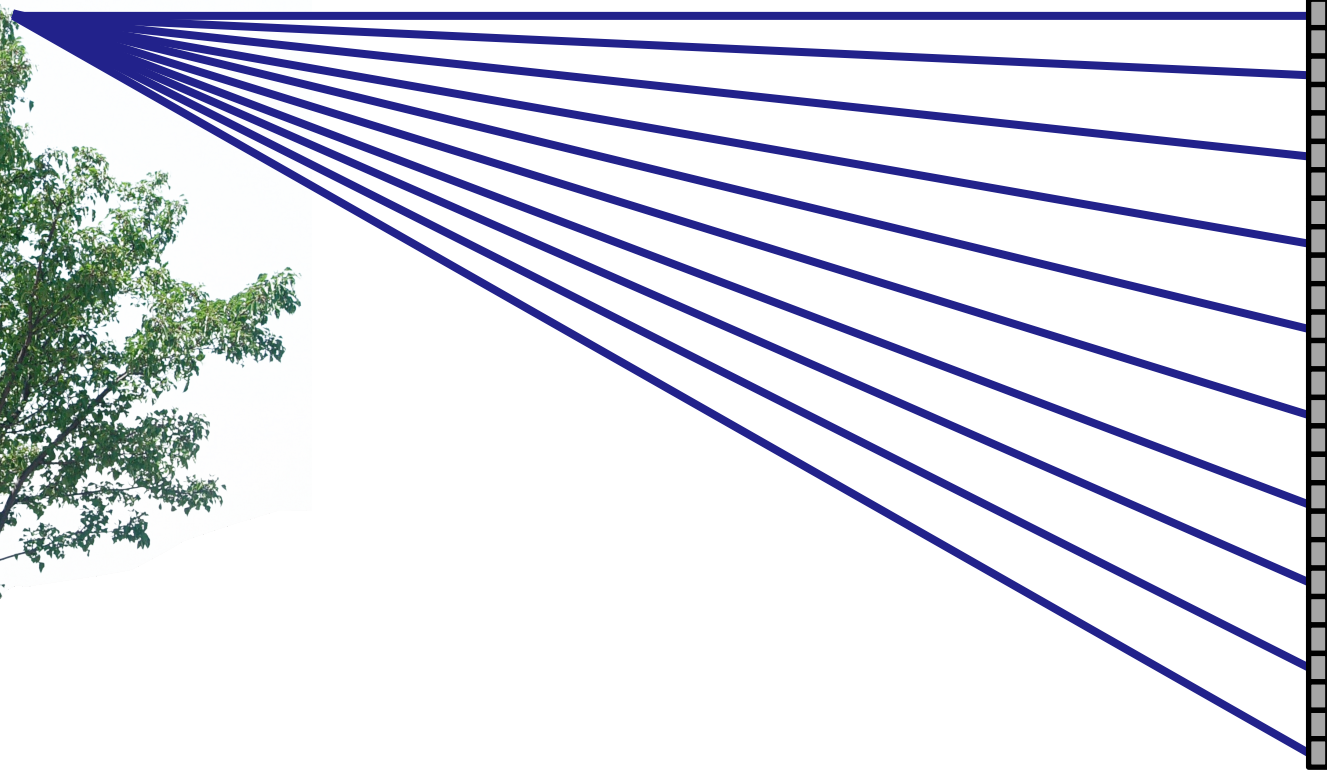


digital sensor
(CCD or
CMOS)



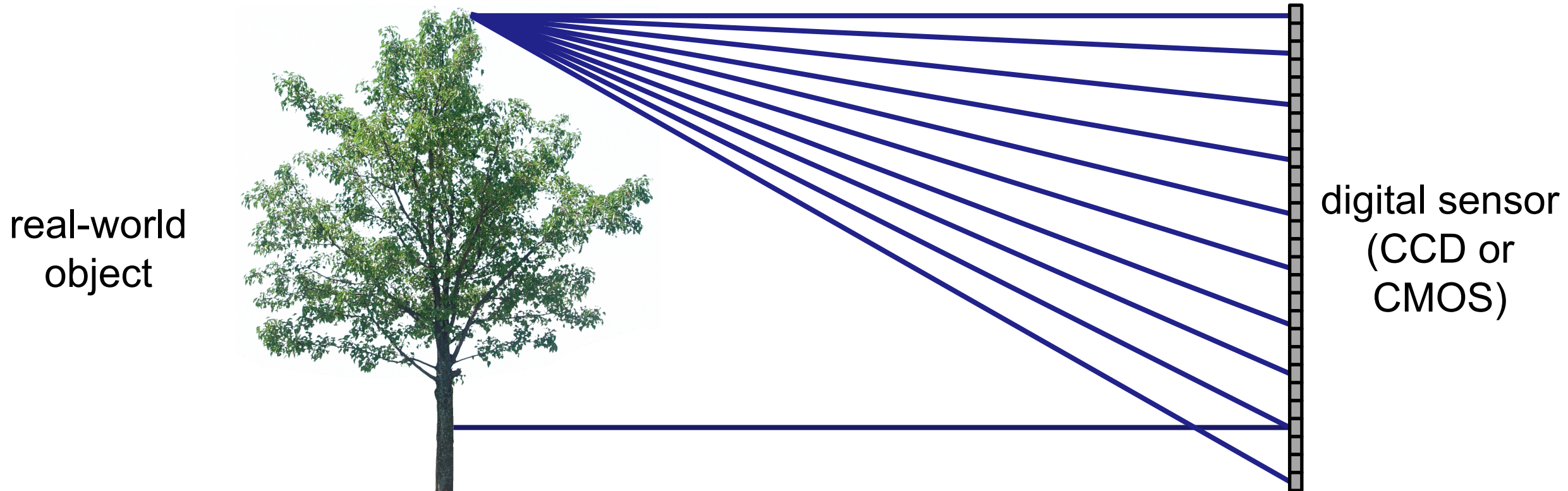
Bare-sensor imaging

real-world
object

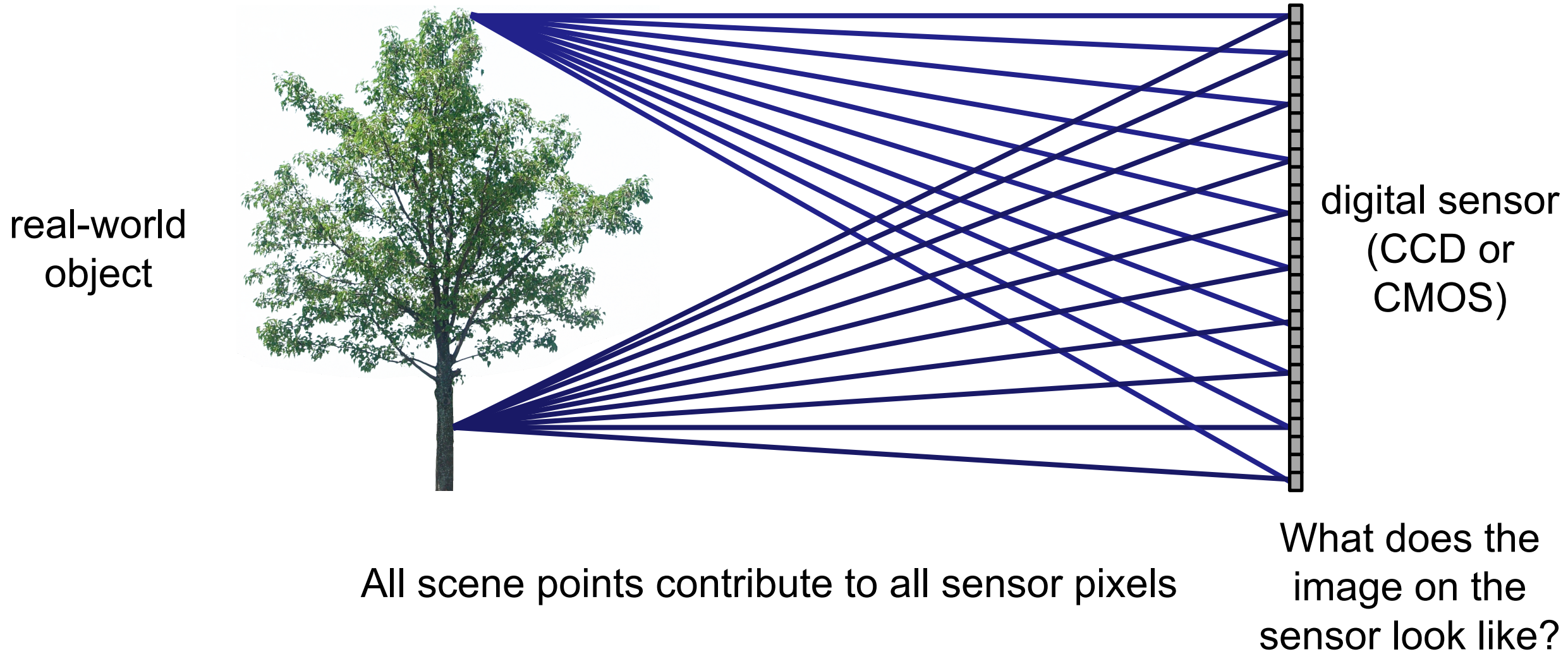


digital sensor
(CCD or
CMOS)

Bare-sensor imaging



Bare-sensor imaging



Bare-sensor imaging



All scene points contribute to all sensor pixels

What can we do to make our image look better?

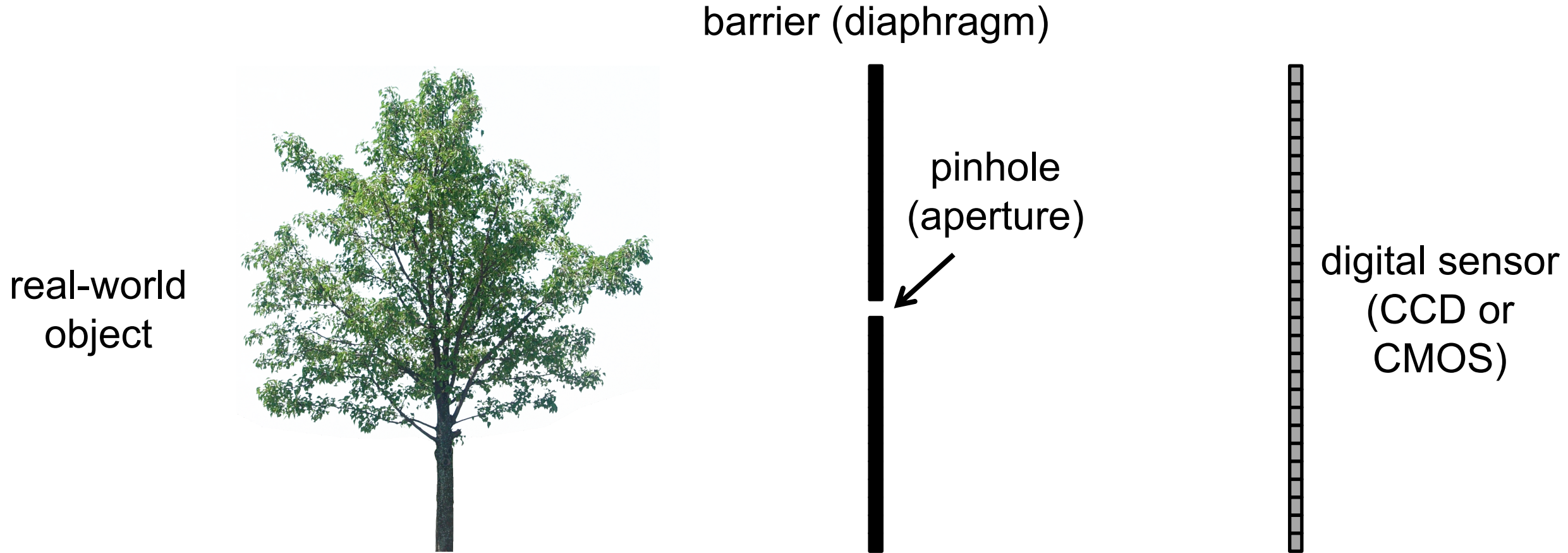
real-world
object



digital sensor
(CCD or
CMOS)



Let's add something to this scene



What would an image taken like this look like?

Pinhole imaging

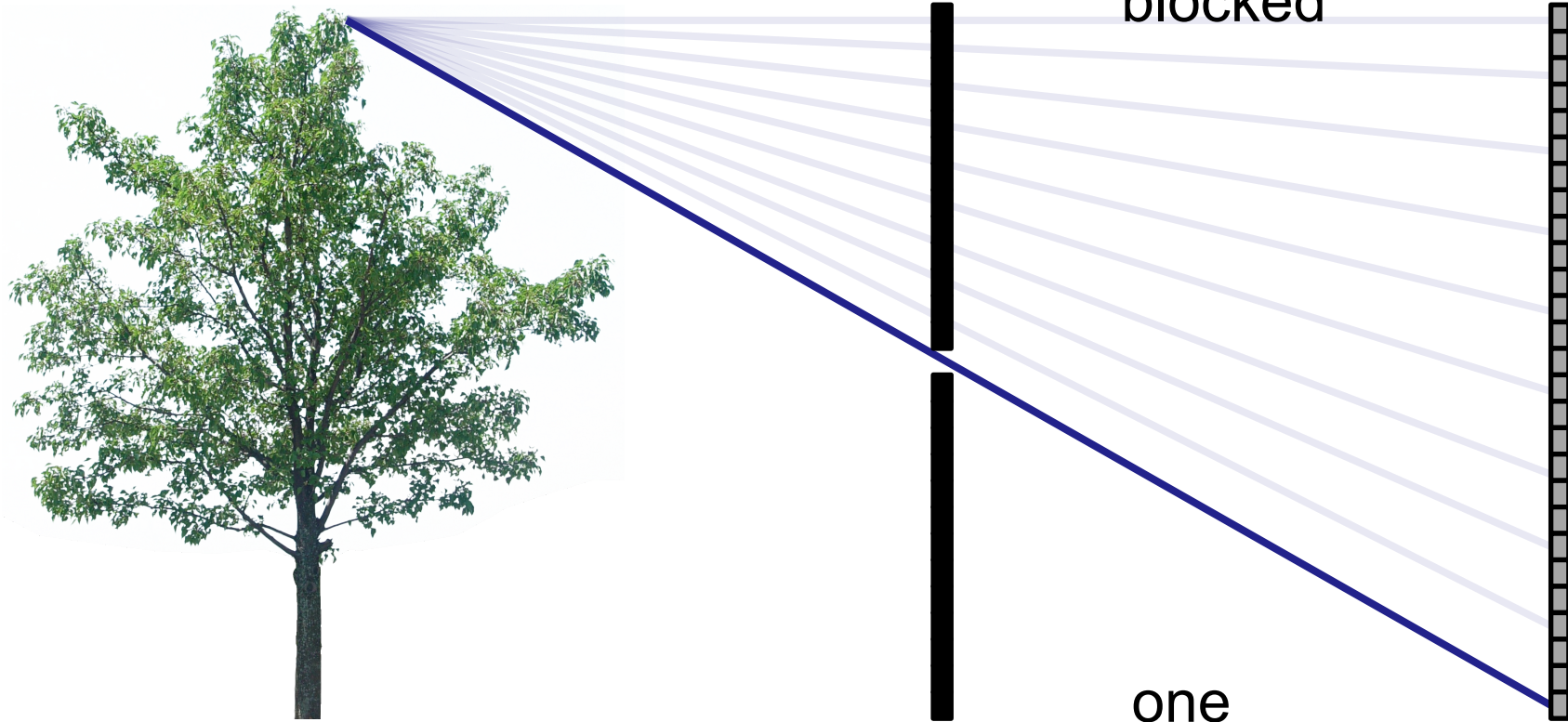
real-world
object



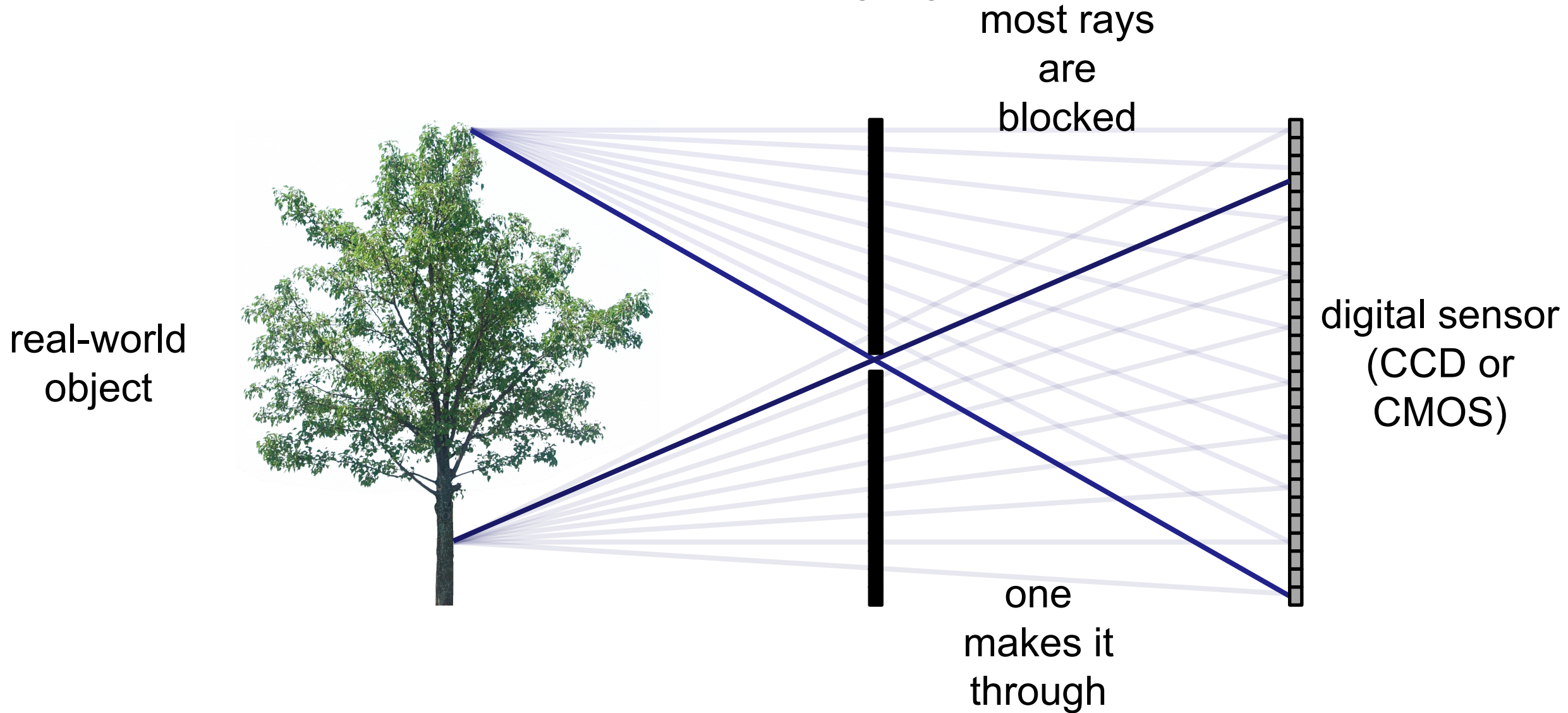
most rays
are
blocked

one
makes it
through

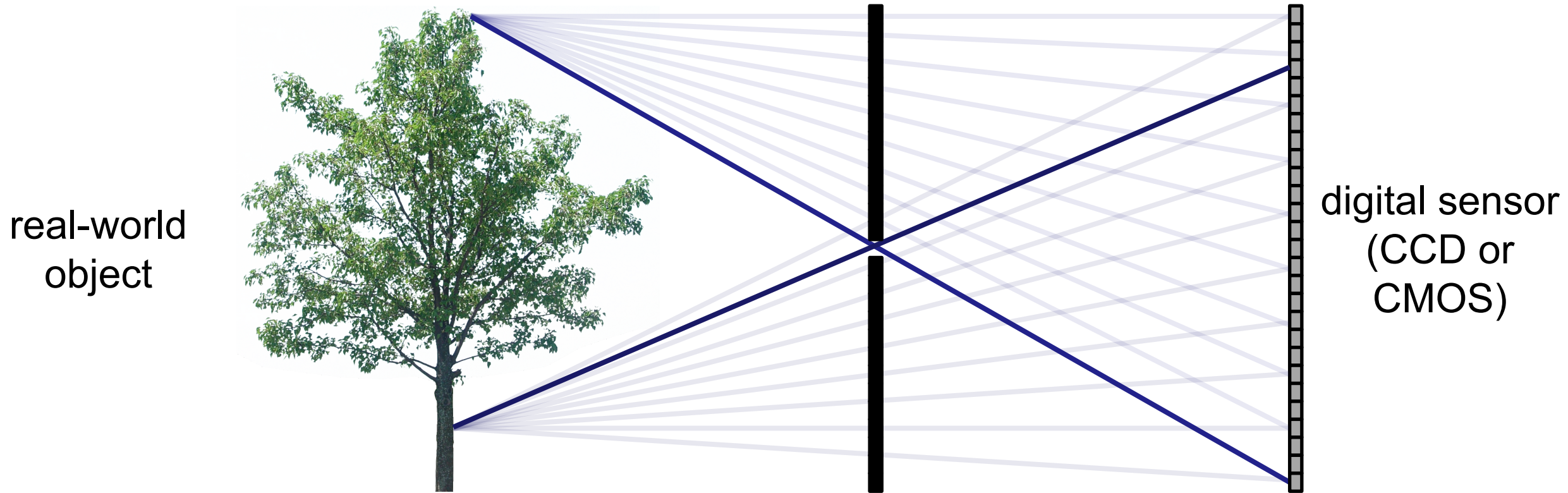
digital sensor
(CCD or
CMOS)



Pinhole imaging



Pinhole imaging

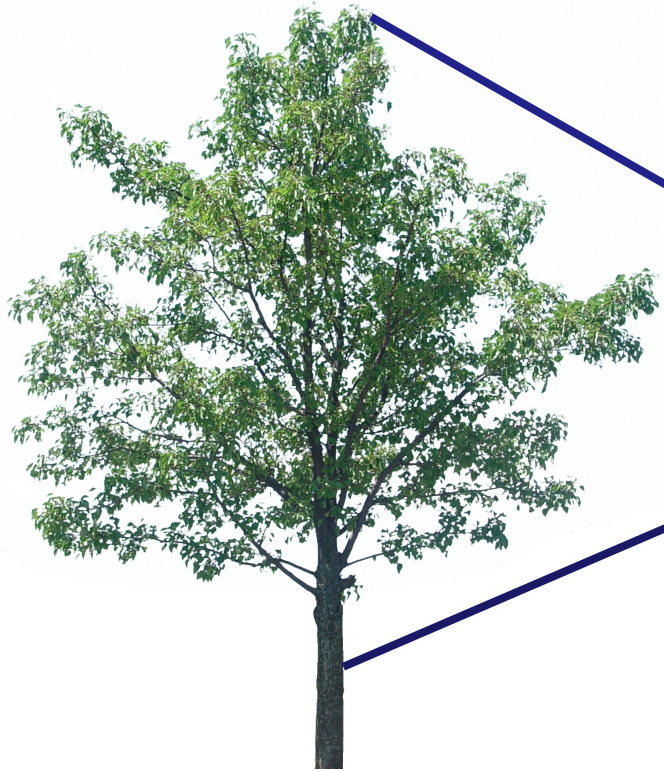


Each scene point contributes to only one sensor pixel

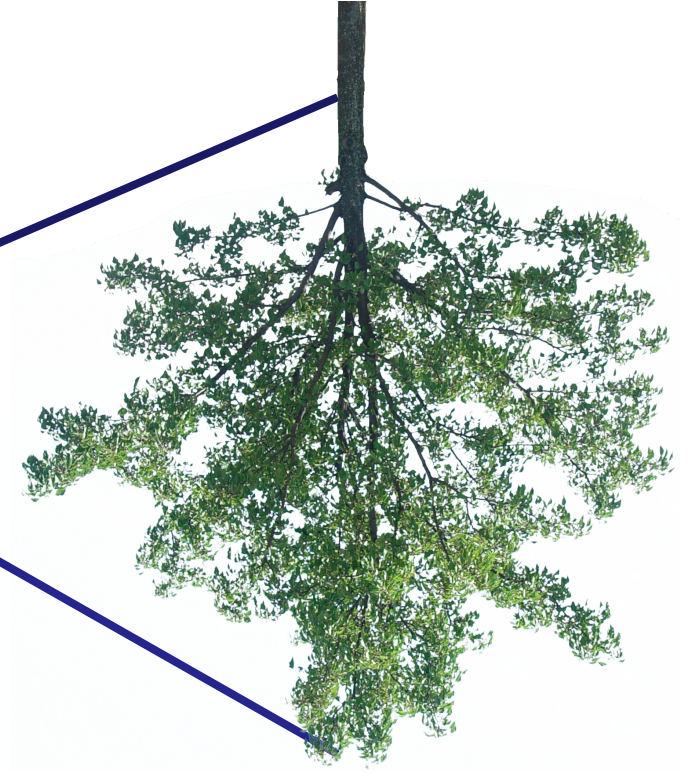
What does the image on the sensor look like?

Pinhole imaging

real-world
object

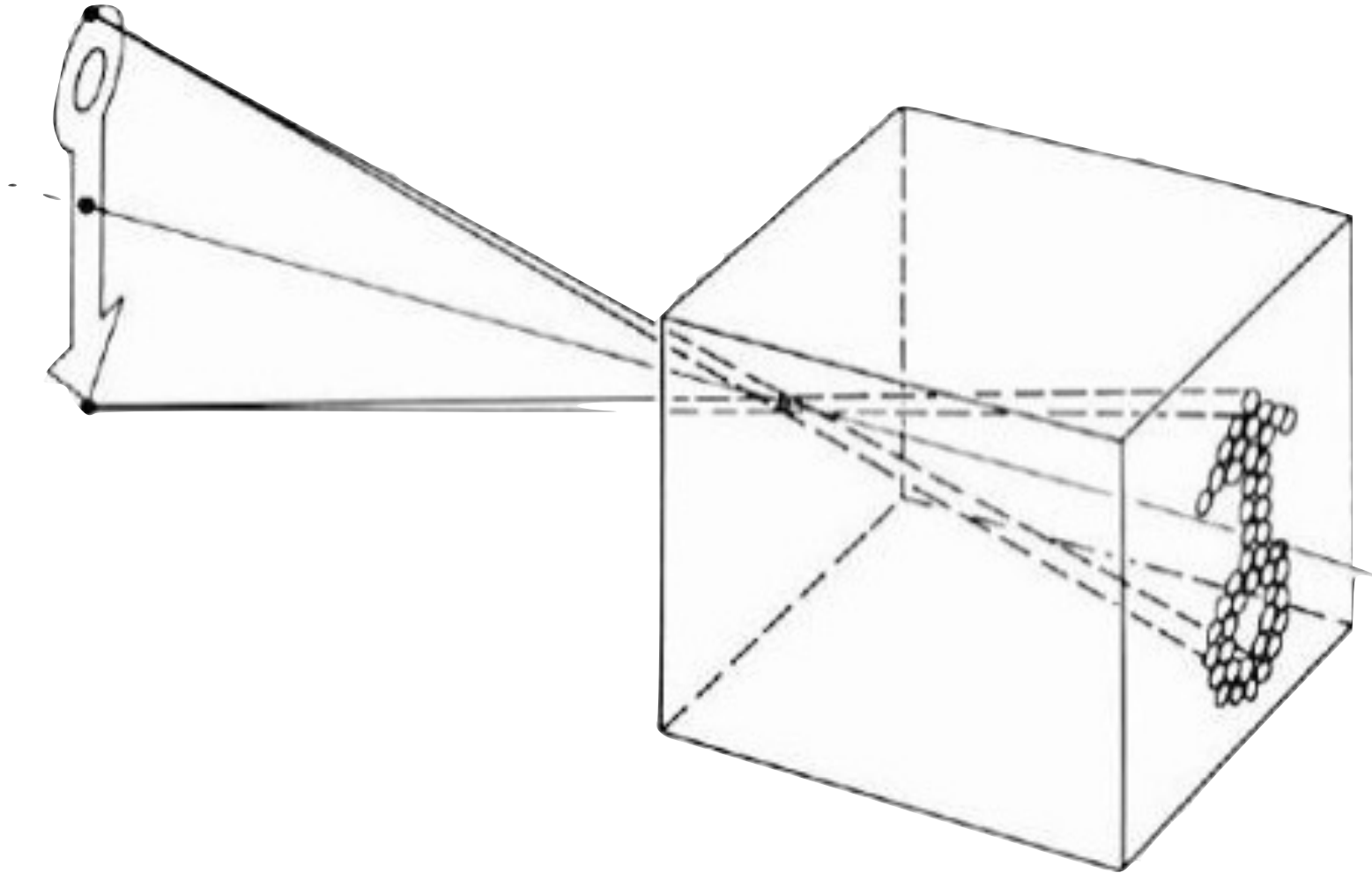


copy of real-world object
(inverted and scaled)



Pinhole camera

Pinhole camera a.k.a. camera obscura



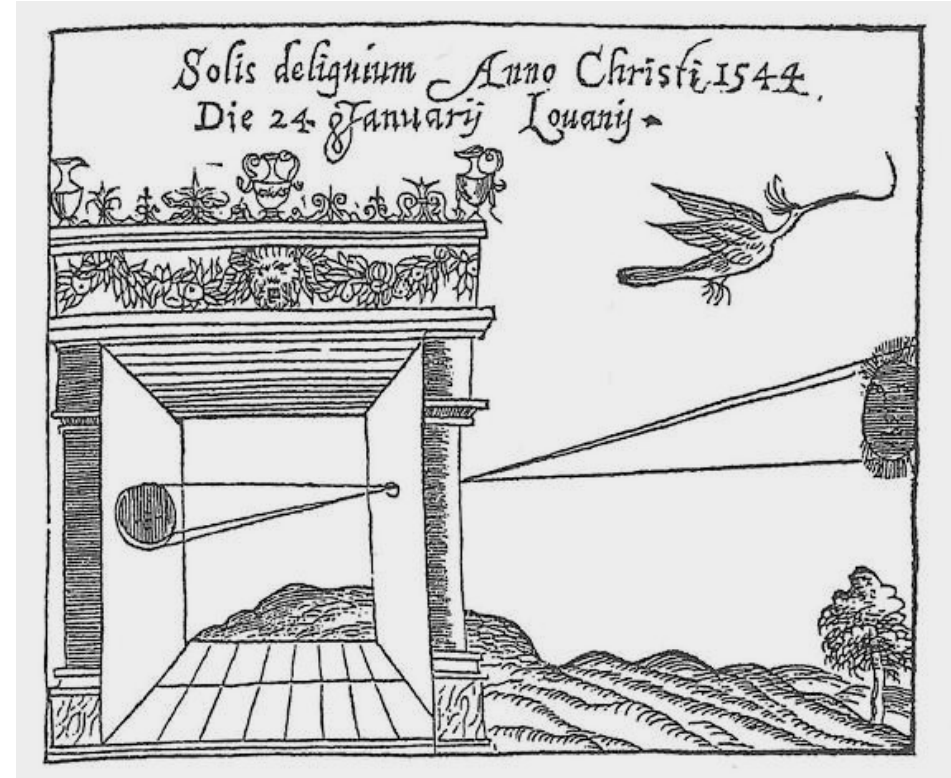
Pinhole camera a.k.a. camera obscura

First mention ...



Chinese philosopher Mozi
(470 to 390 BC)

First camera ...



Greek philosopher Aristotle
(384 to 322 BC)

Pinhole camera terms

real-world
object



barrier (diaphragm)

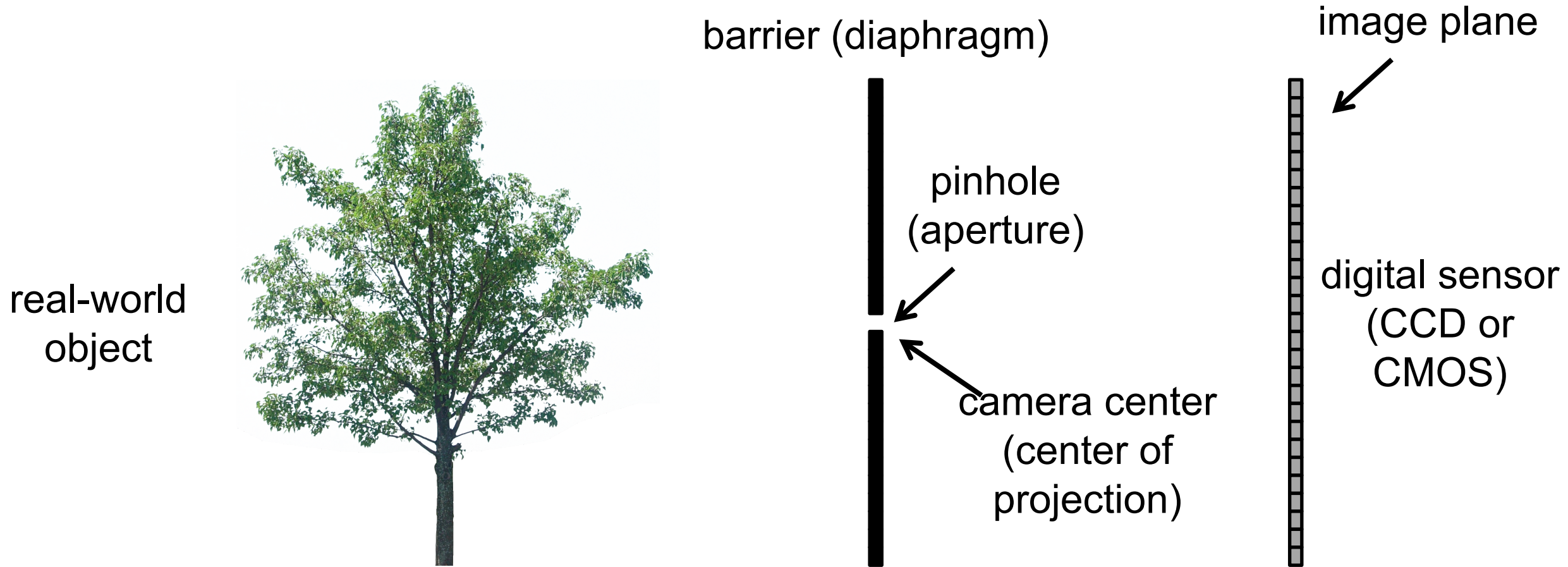


pinhole
(aperture)



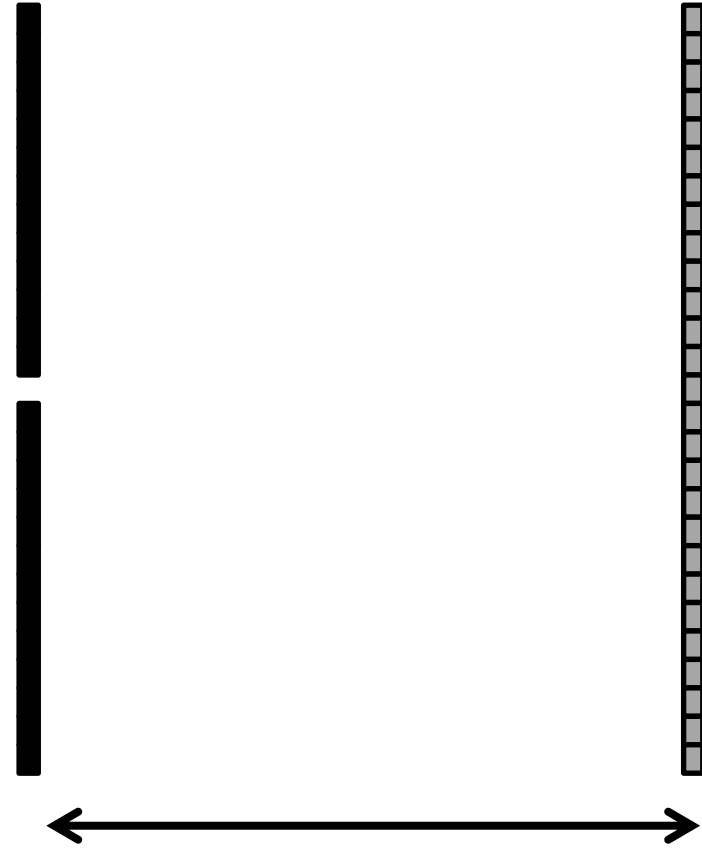
digital sensor
(CCD or
CMOS)

Pinhole camera terms



Focal length

real-world
object

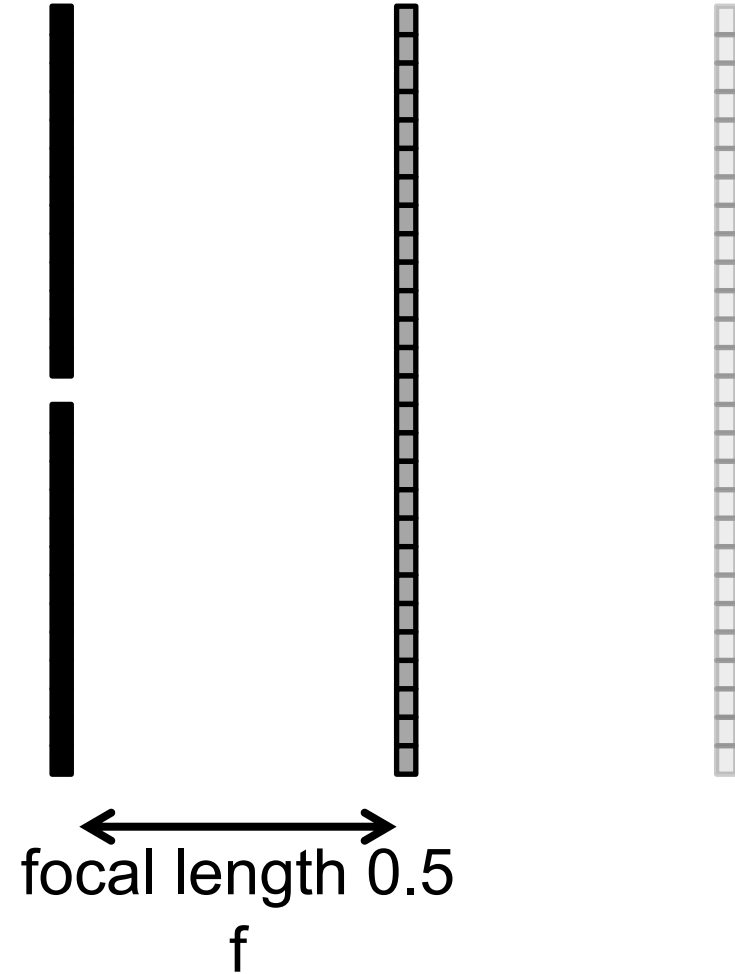


focal length f

Focal length

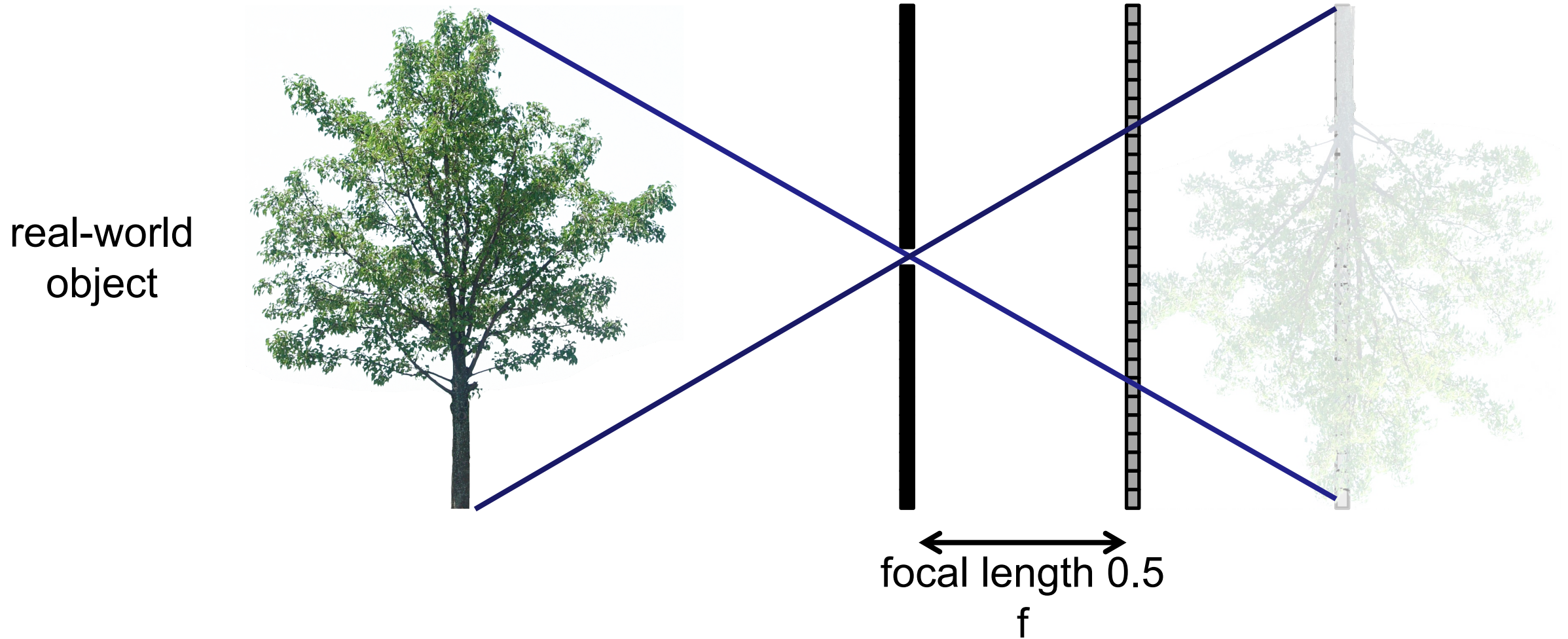
What happens as we change the focal length?

real-world
object



Focal length

What happens as we change the focal length?

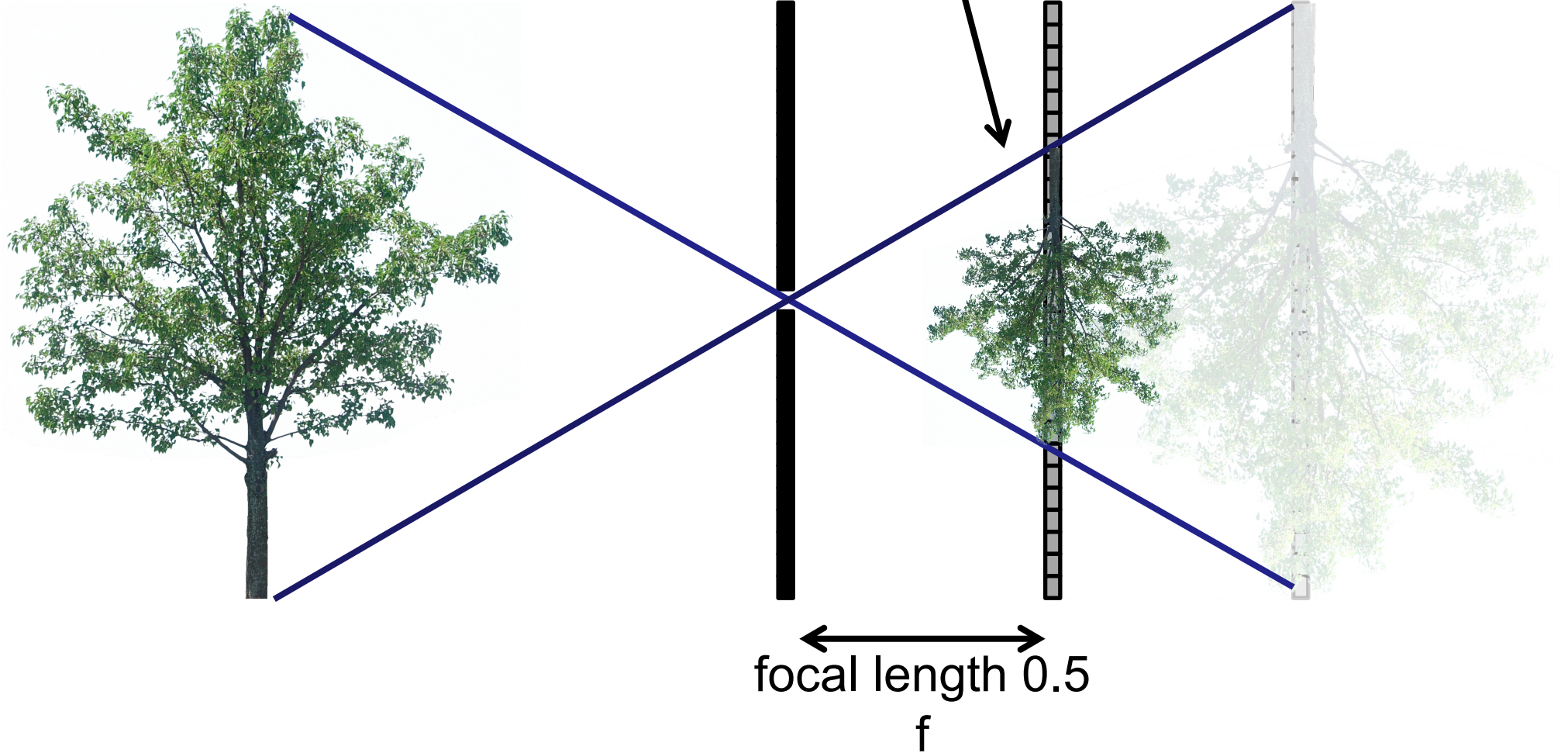


Focal length

What happens as we change the focal length?

object projection is half the size

real-world
object



Pinhole size

real-world
object



pinhole
diameter



- Ideal pinhole has infinitesimally small size
- In practice that is impossible.

Pinhole size

What happens as we change the pinhole diameter?

real-world
object



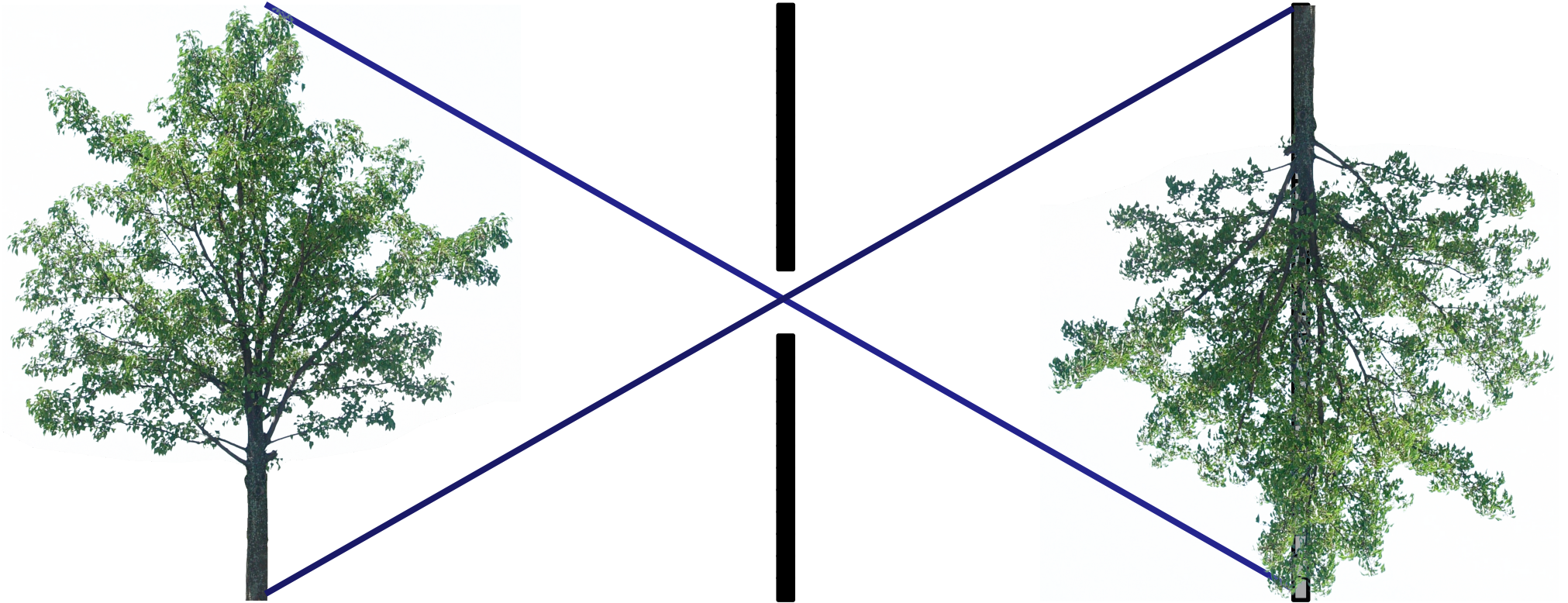
pinhole
diameter



Pinhole size

What happens as we change the pinhole diameter?

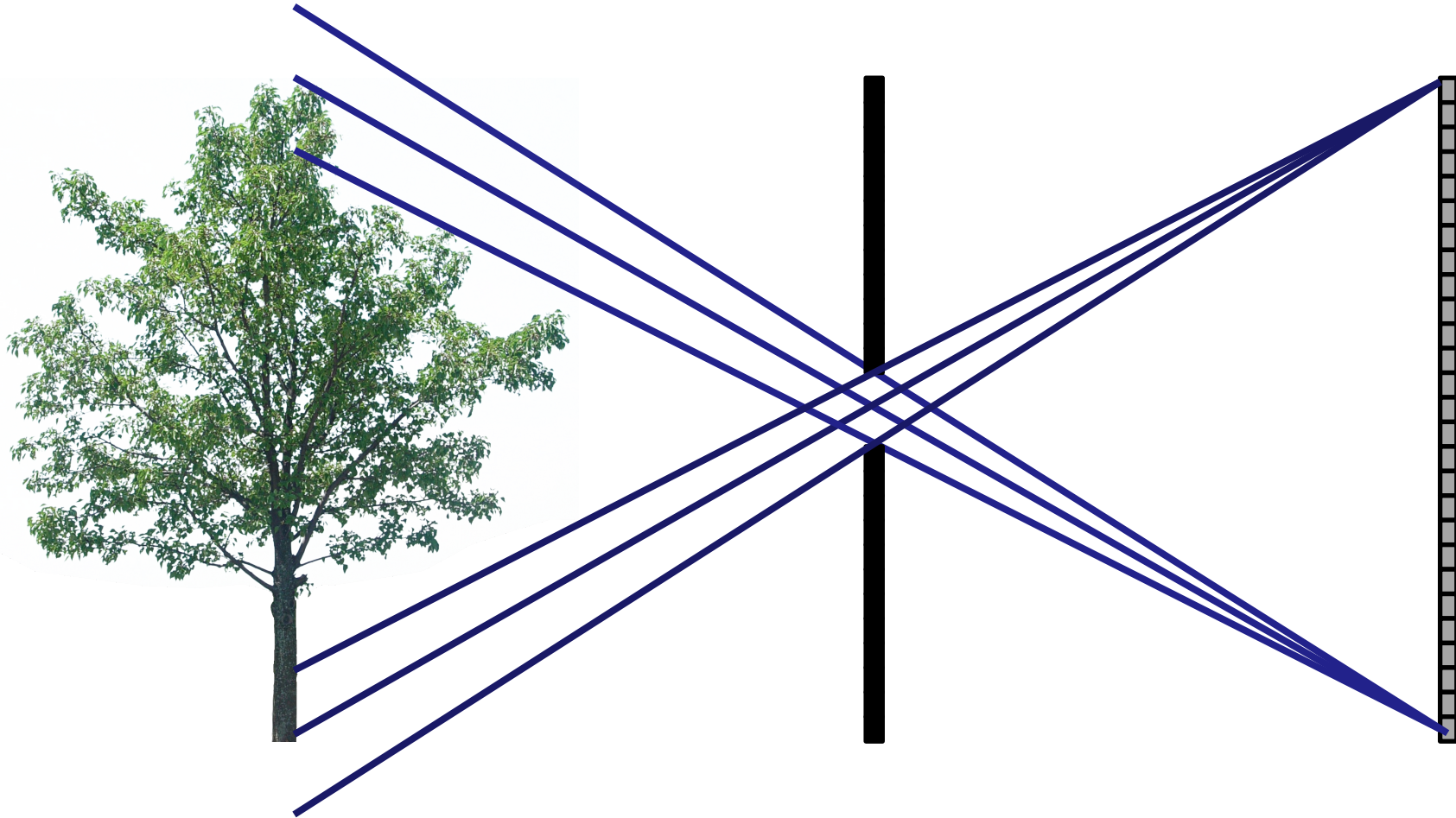
real-world
object



Pinhole size

What happens as we change the pinhole diameter?

real-world
object

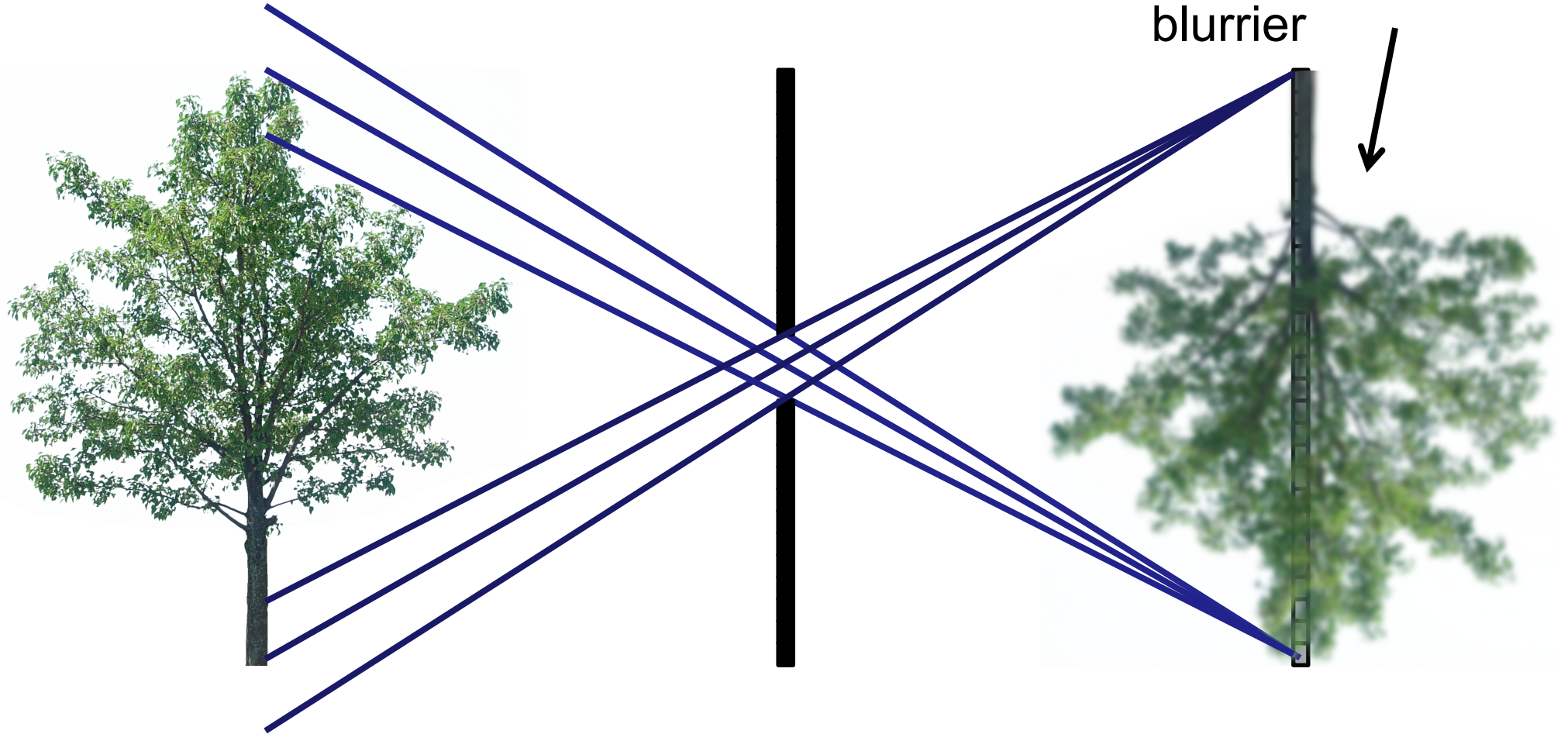


Pinhole size

What happens as we change the pinhole diameter?

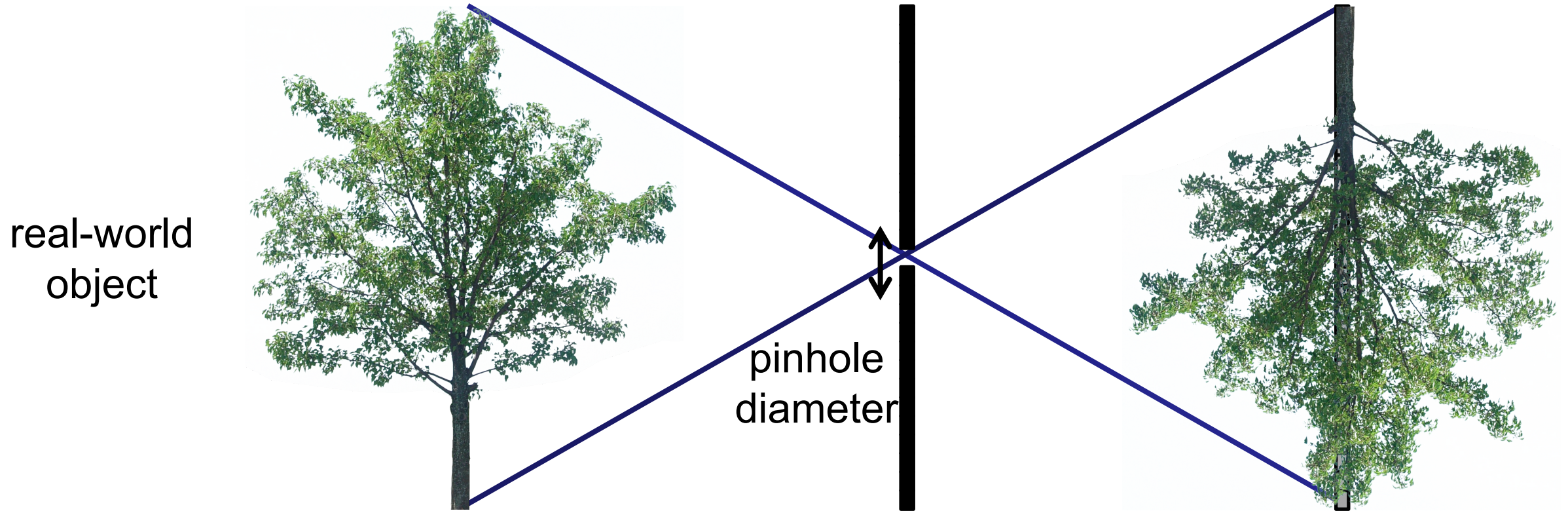
object projection becomes blurrier

real-world
object



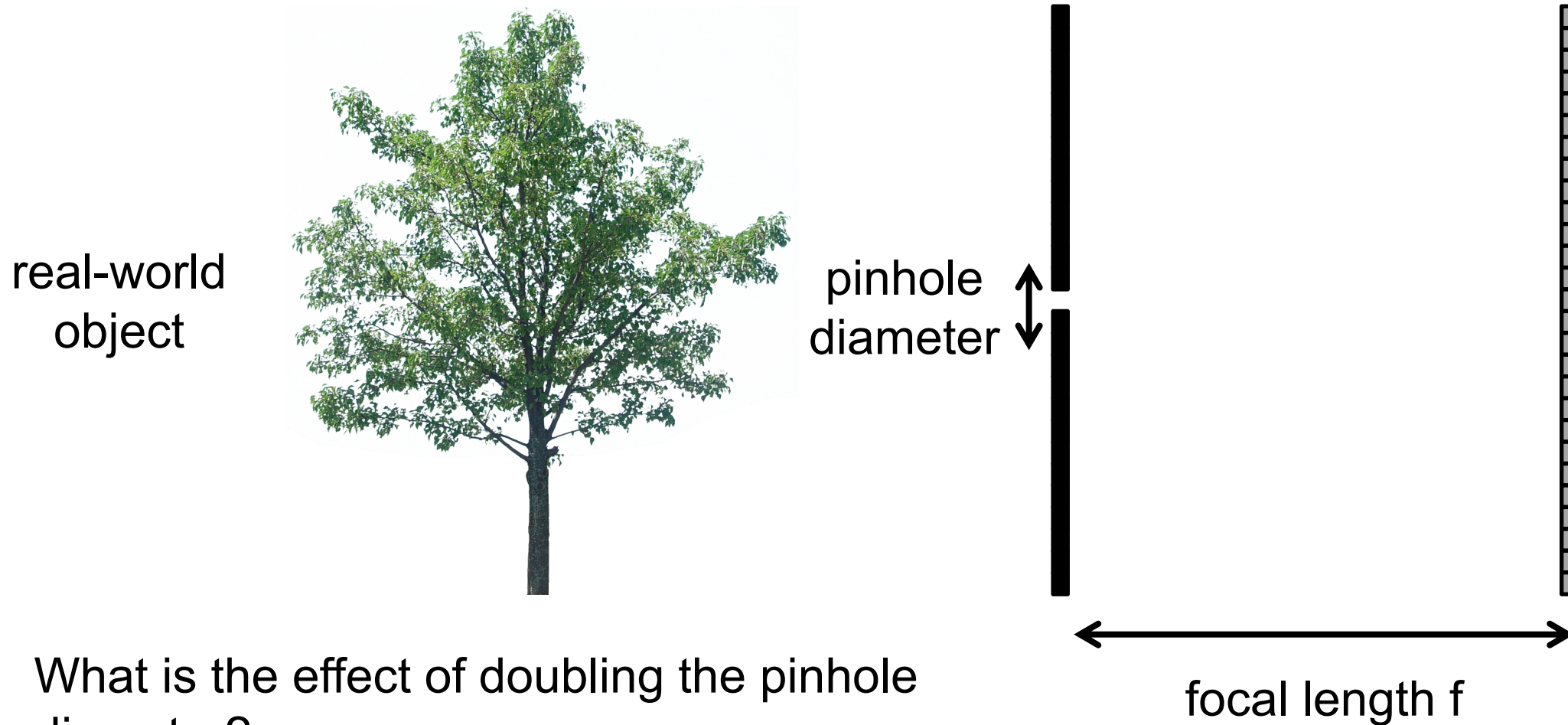
Pinhole size

What happens as we change the pinhole diameter?



Will the image keep getting sharper the smaller we make the pinhole?

What about light efficiency?



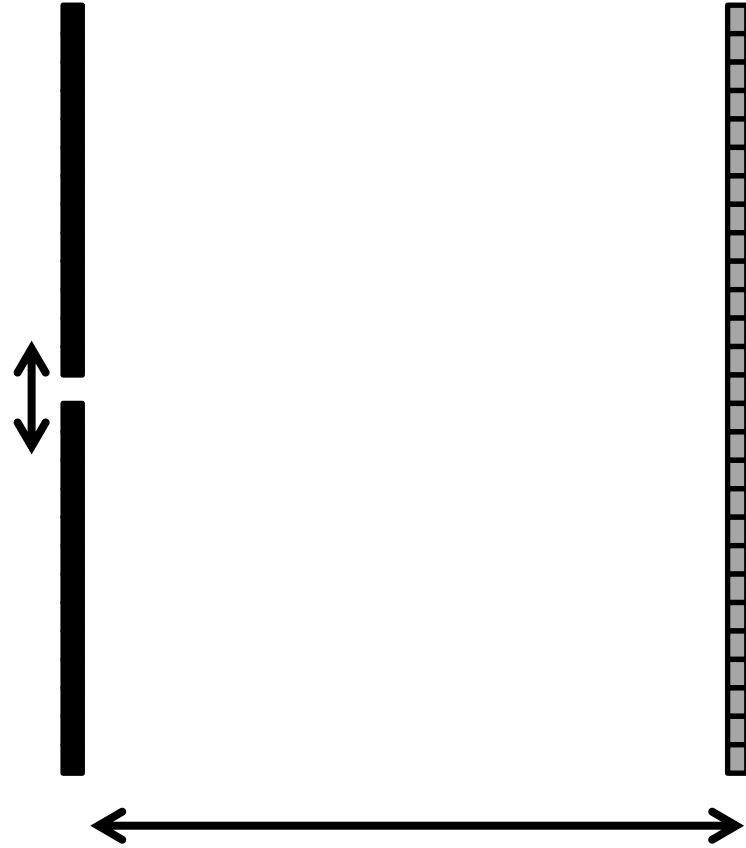
- What is the effect of doubling the pinhole diameter?
- What is the effect of doubling the focal length?

What about light efficiency?

real-world
object



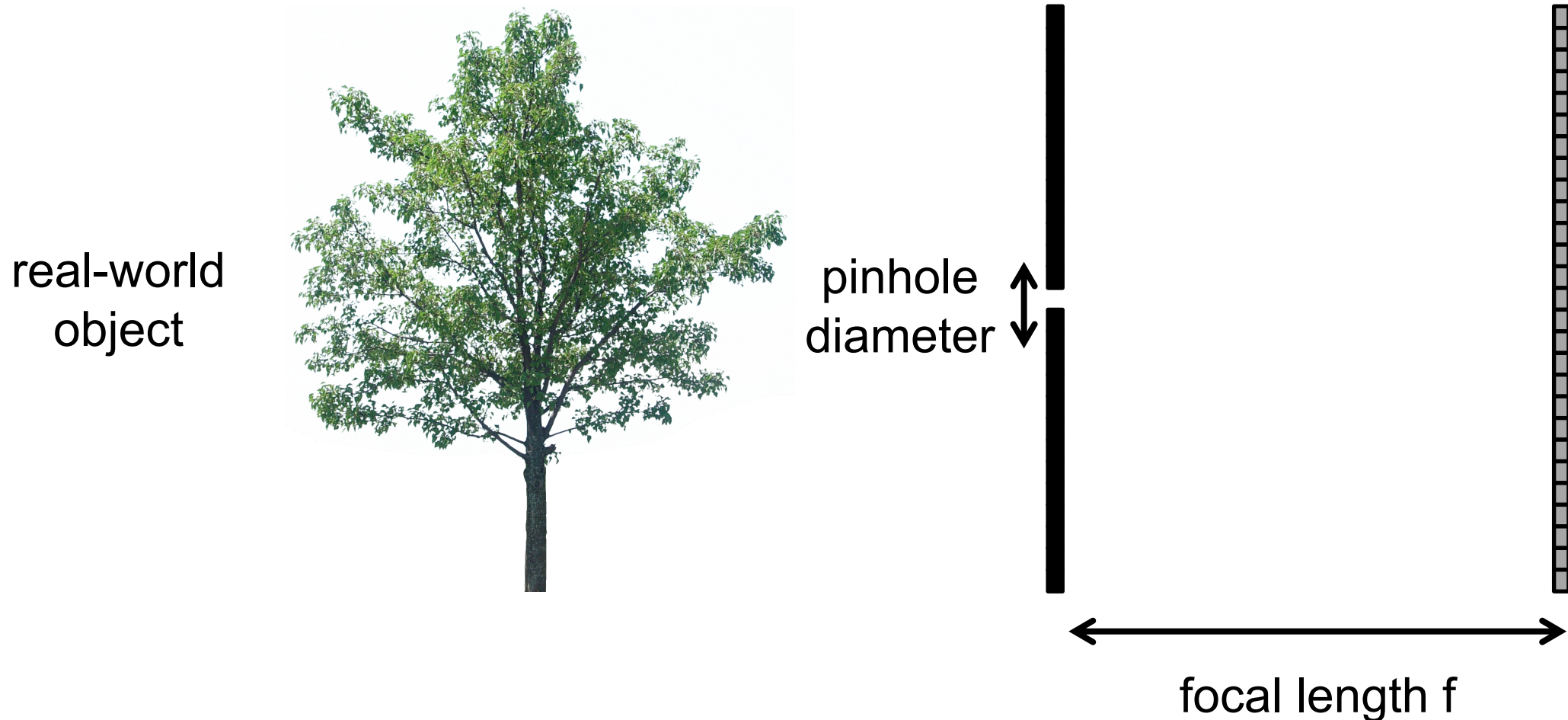
pinhole
diameter



- 2x pinhole diameter $\rightarrow$ 4x light
- 2x focal length $\rightarrow$ $\frac{1}{4}$ x light

Some terminology notes

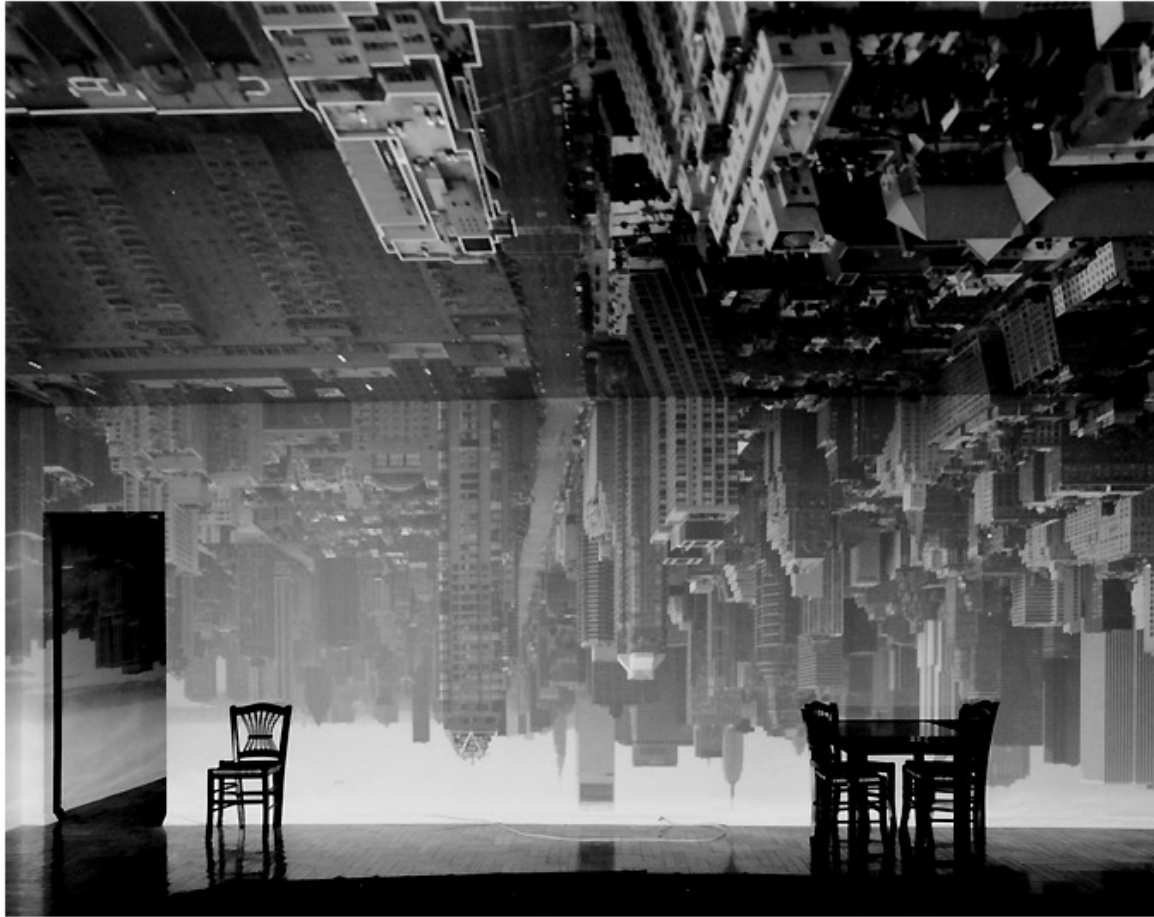
A “stop” is a change in camera settings that changes amount of light by a factor of 2



The “f-number” is the ratio: focal length / pinhole diameter

Accidental pinholes

Turning a room into a camera obscura



Abelardo Morell, Camera Obscura Image of
Manhattan View Looking South in Large Room,
1996

<https://www.abelardomorell.net/camera-obscura>

After scouting rooms and reserving one for at least a day, Morell masks the windows except for the aperture. He controls three elements: the size of the hole, with a smaller one yielding a sharper but dimmer image; the length of the exposure, usually eight hours; and the distance from the hole to the surface on which the outside image falls and which he will photograph. He used 4 x 5 and 8 x 10 view cameras and lenses ranging from 75 to 150 mm.

After he's done inside, it gets harder. "I leave the room and I am constantly checking the weather, I'm hoping the maid reads my note not to come in, I'm worrying that the sun will hit the plastic masking and it will fall down, or that I didn't trigger the lens."

From *Grand Images Through a Tiny Opening*, **Photo District News**, February 2005





What does this image say about the world outside?



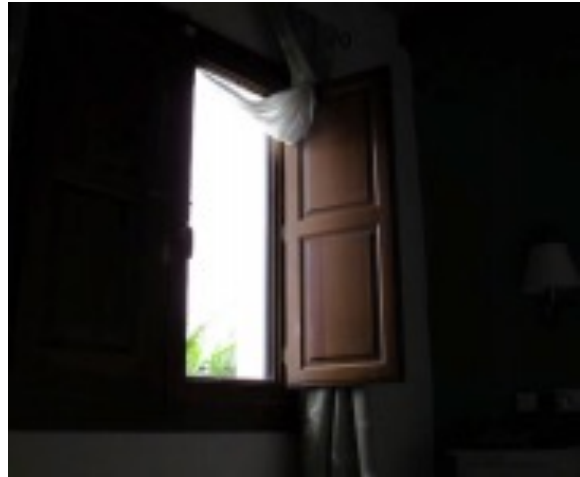
Accidental pinhole camera



Antonio Torralba, William T. Freeman
Computer Science and Artificial Intelligence Laboratory (CSAIL)
MIT
torralba@mit.edu, billf@mit.edu

Accidental pinhole camera

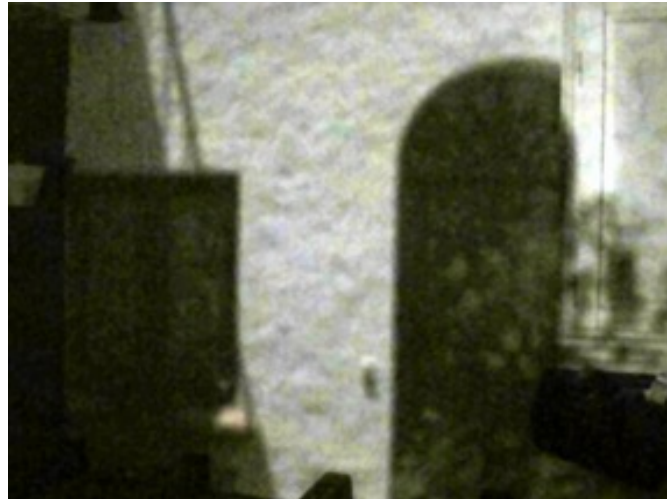
projected pattern on the wall



window is an
aperture



upside down



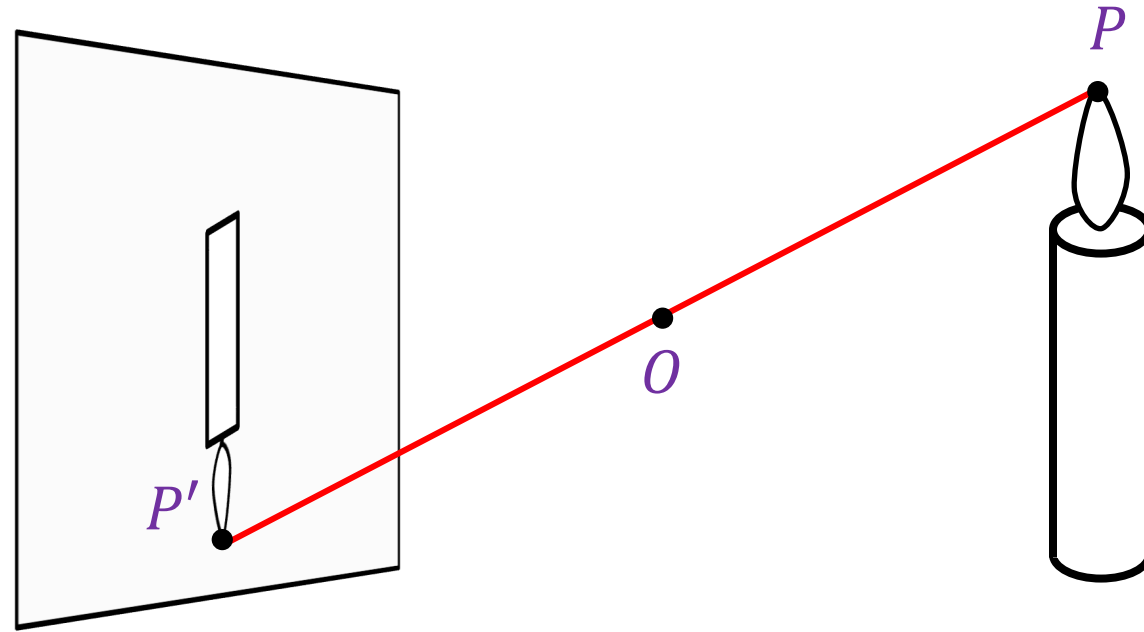
window with smaller
gap



view outside window

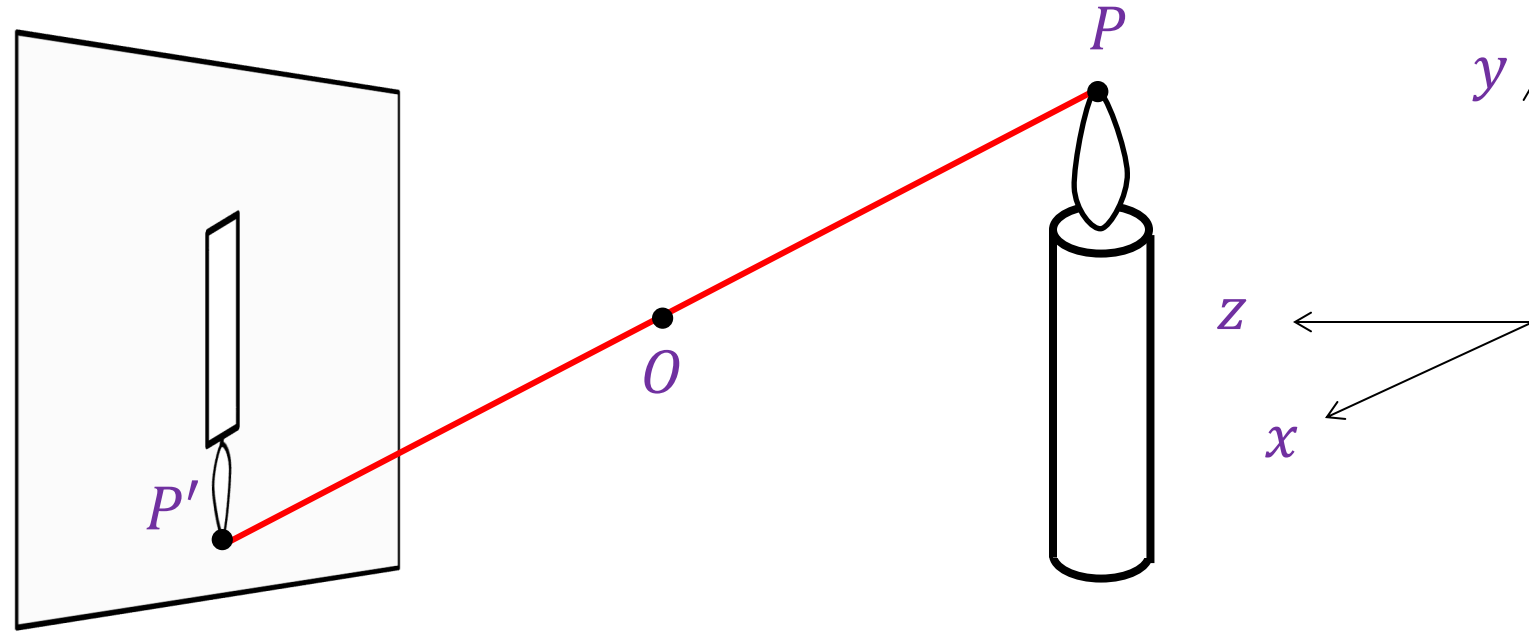


Modeling projection



- How do we find the projection P' of a scene point P ?
 - Form the **visual ray** connecting P to the camera center O and find where it intersects the image plane
- All scene points that lie on this visual ray have the same projection in the image
- Are there scene points for which this projection is undefined?

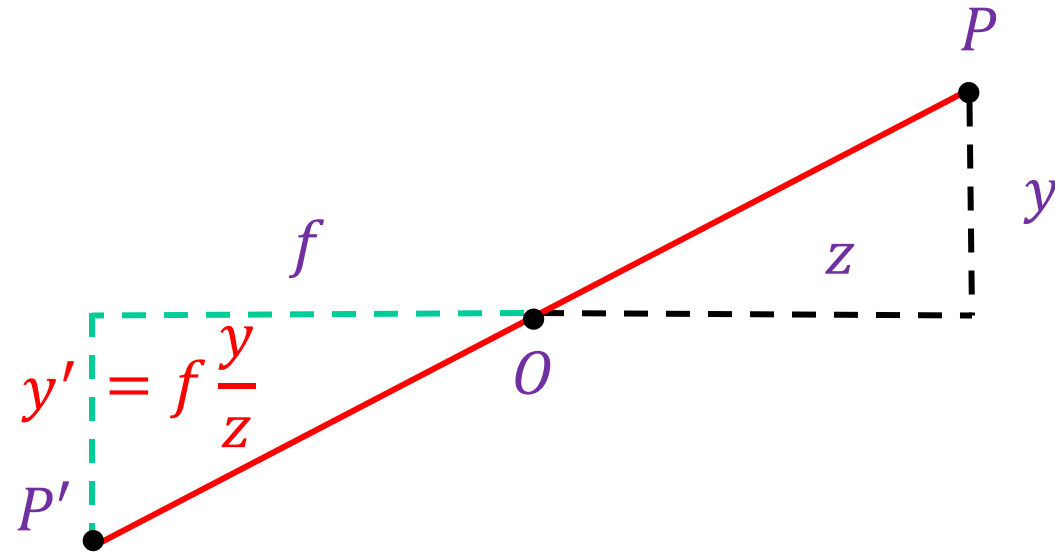
Modeling projection



Canonical coordinate system

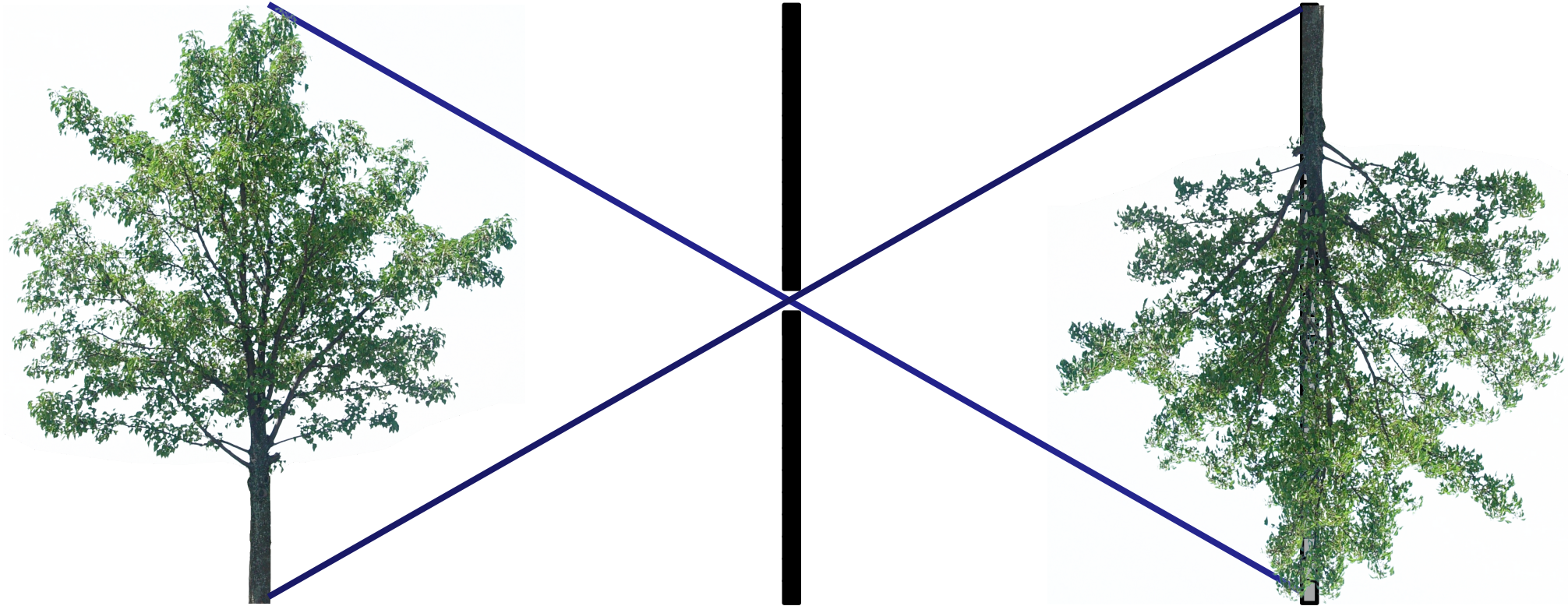
- The optical center (O) is at the origin
- The z axis is the *optical axis* perpendicular to the image plane
- The xy plane is parallel to the image plane, x and y axes are horizontal and vertical directions of the image plane

Deriving projected coordinates



$$(x, y, z) \rightarrow \left(f \frac{x}{z}, f \frac{y}{z} \right)$$

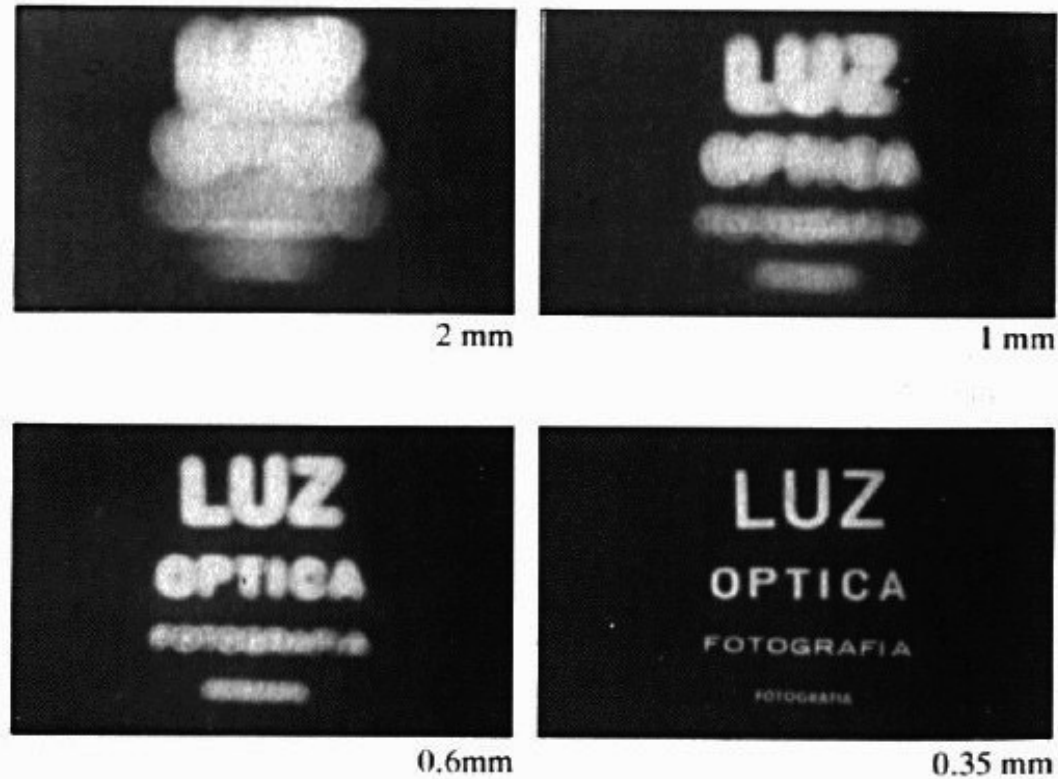
Pinhole camera trade-off



Small (ideal) pinhole:

1. Image is sharp.
2. Signal-to-noise ratio is low.

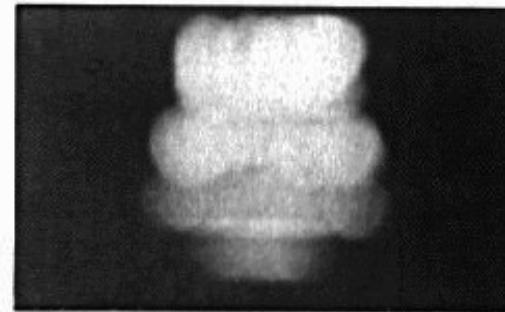
Shrinking the aperture



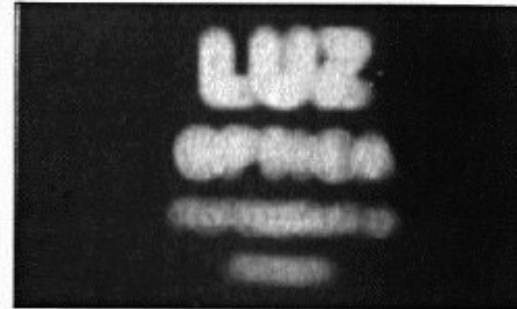
Why not make the aperture as small as possible?

- Less light gets through
- Diffraction!

Shrinking the aperture



2 mm



1 mm



0.6mm



0.35 mm



0.15 mm

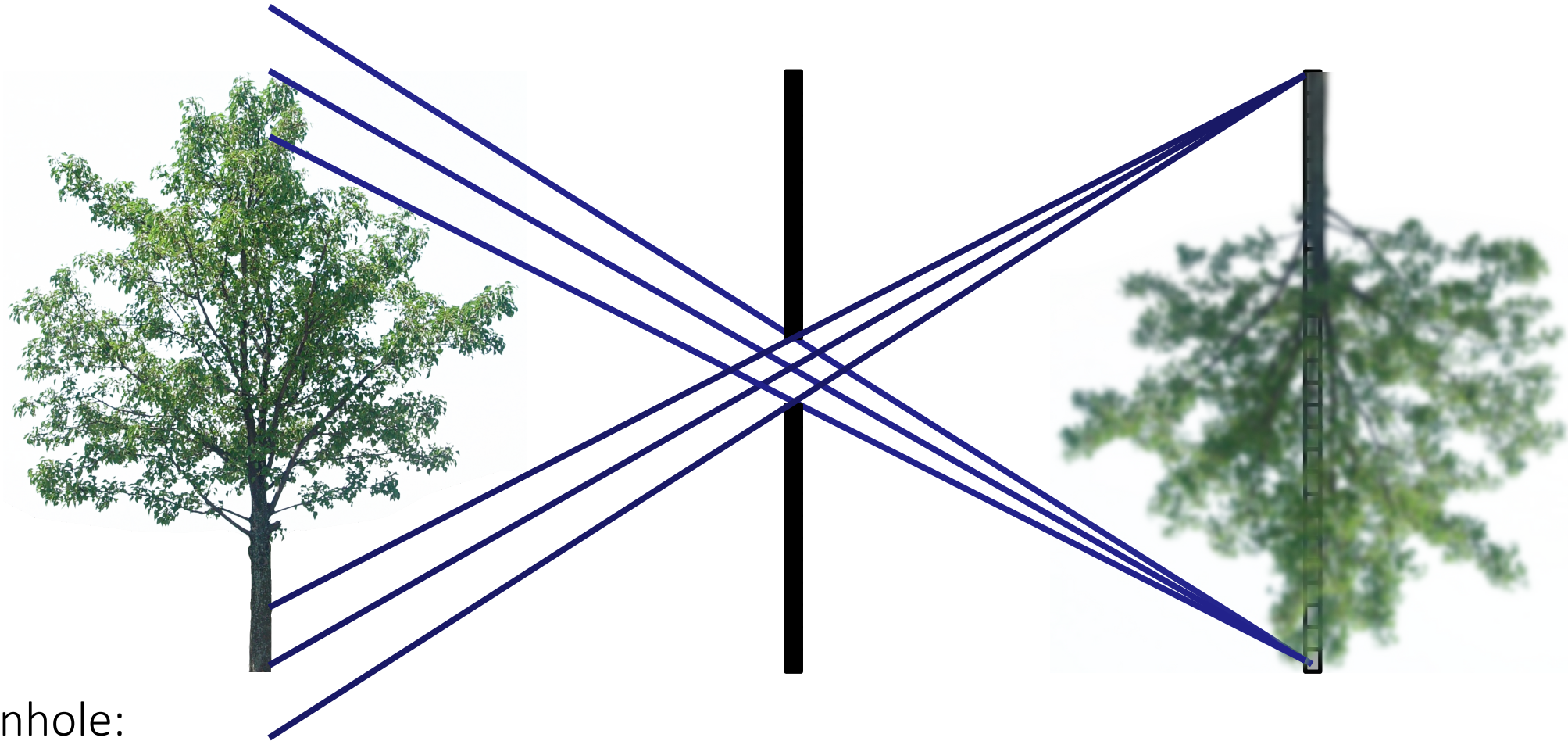


0.07 mm

diffraction



Pinhole camera trade-off

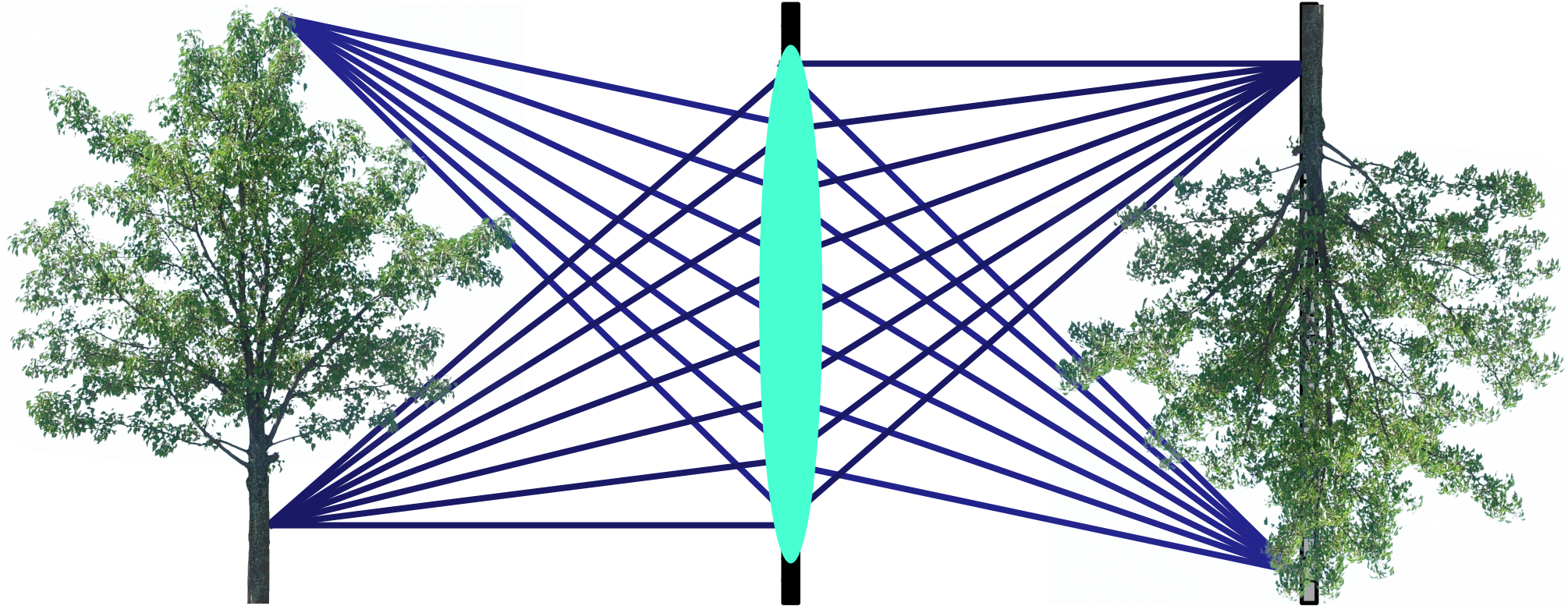


Large pinhole:

1. Image is blurry.
2. Signal-to-noise ratio is high.

Can we get best of both worlds?

Almost, by using lenses



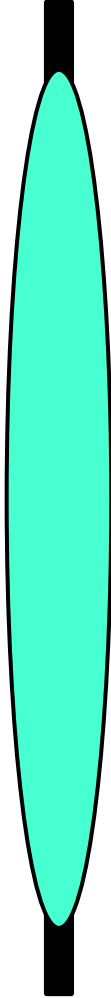
Lenses map “bundles” of rays from points on the scene to the sensor.

How does this mapping work exactly?

Lens (very) basics

What is a lens?

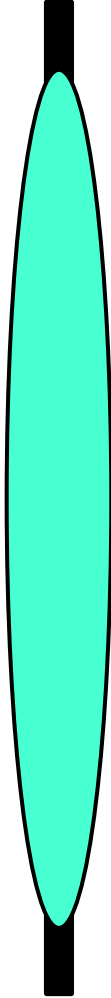
A piece of glass manufactured to have a specific shape



The lens on your camera

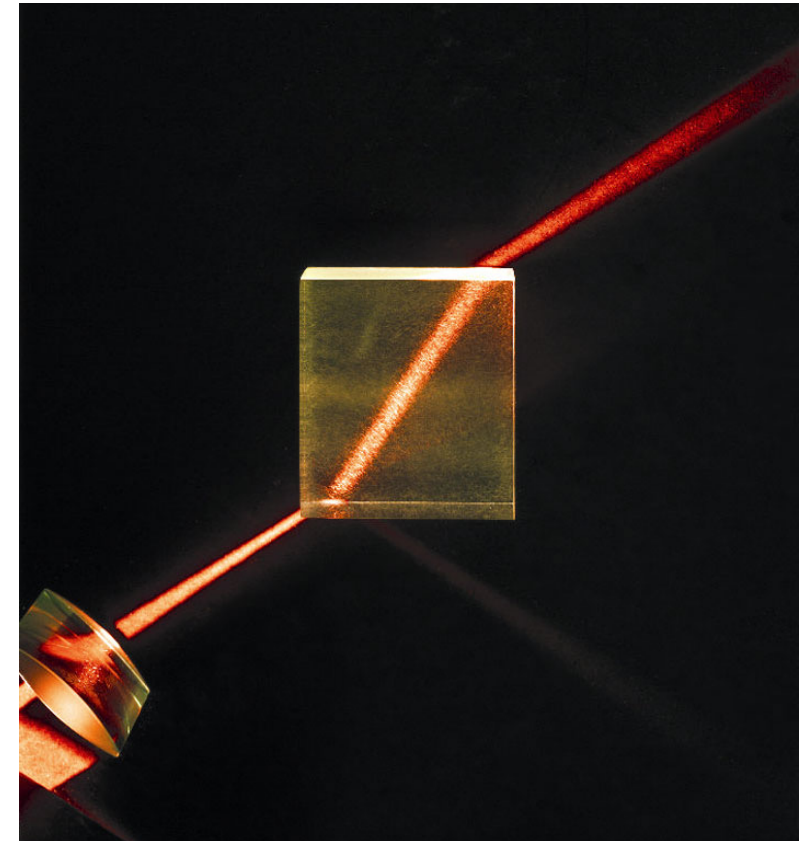
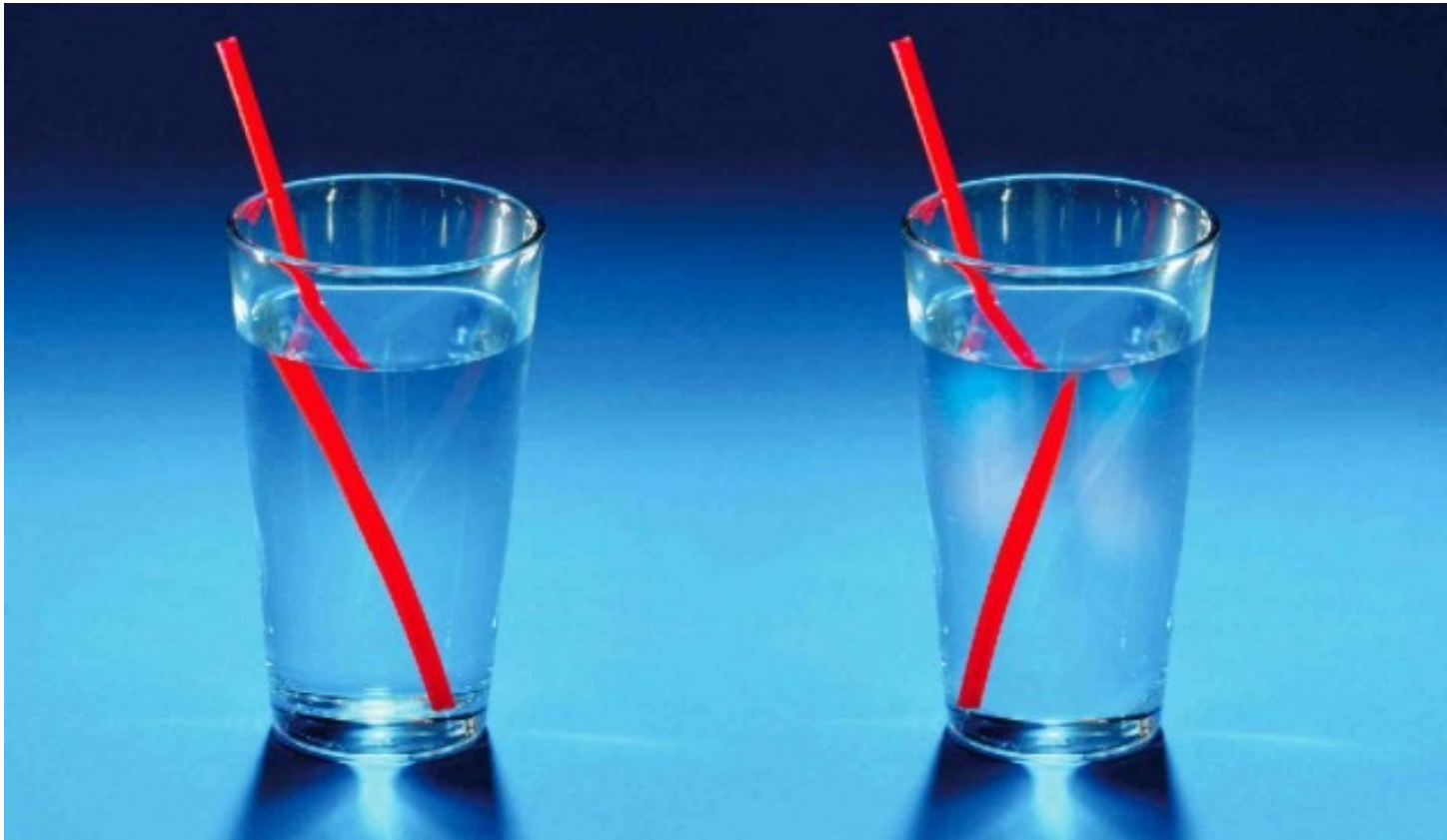


How does a lens work?



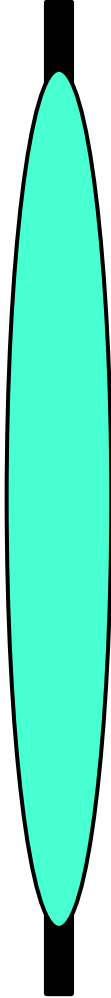
Refraction

Refraction is the bending of rays of light when they move from one material to another



How does a lens work?

Lenses are designed so that their refraction makes light rays bend in a very specific way.



The thin lens model

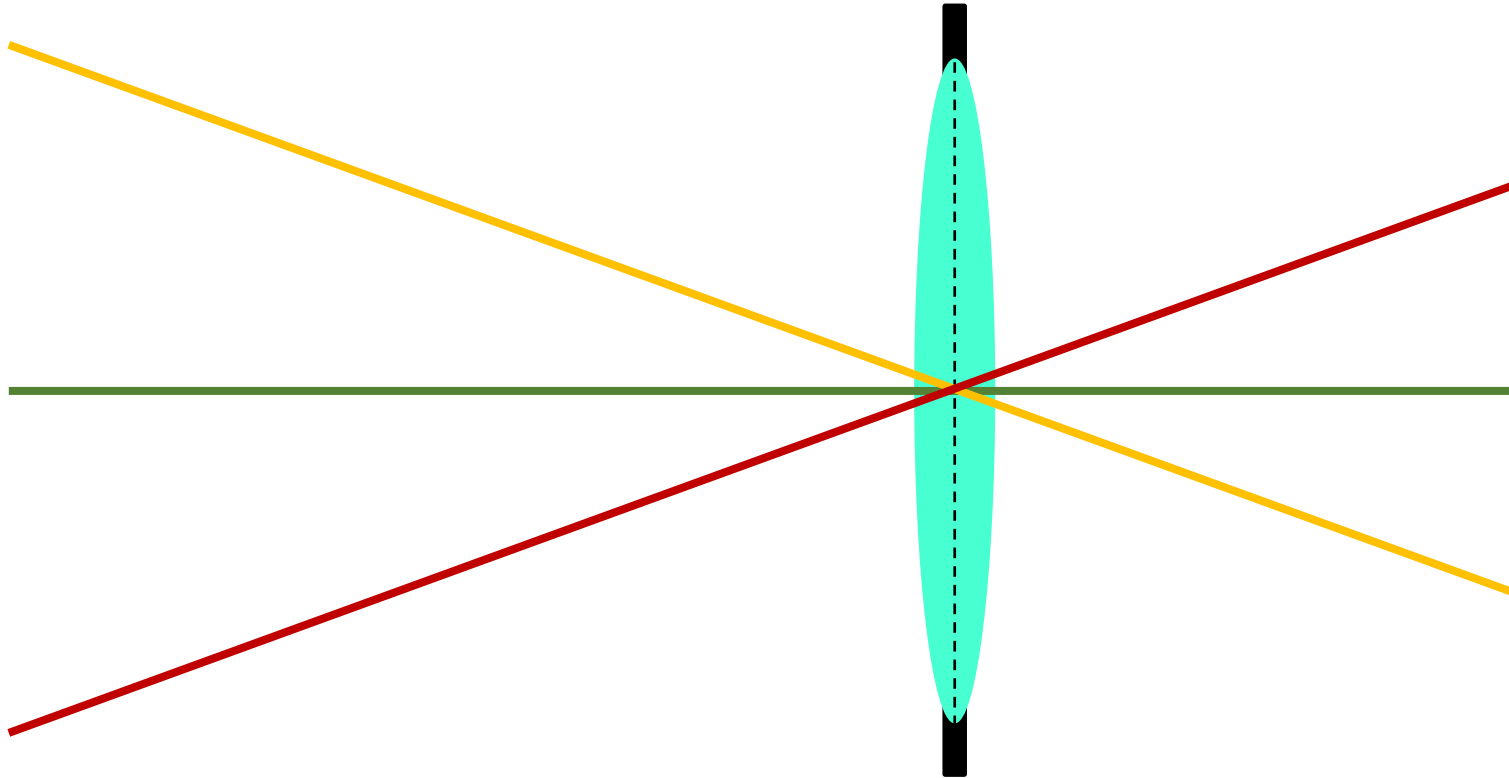
Thin lens model

Idealized geometric optics model for well-designed lenses.



Thin lens model

Simplification of geometric optics for well-designed lenses.

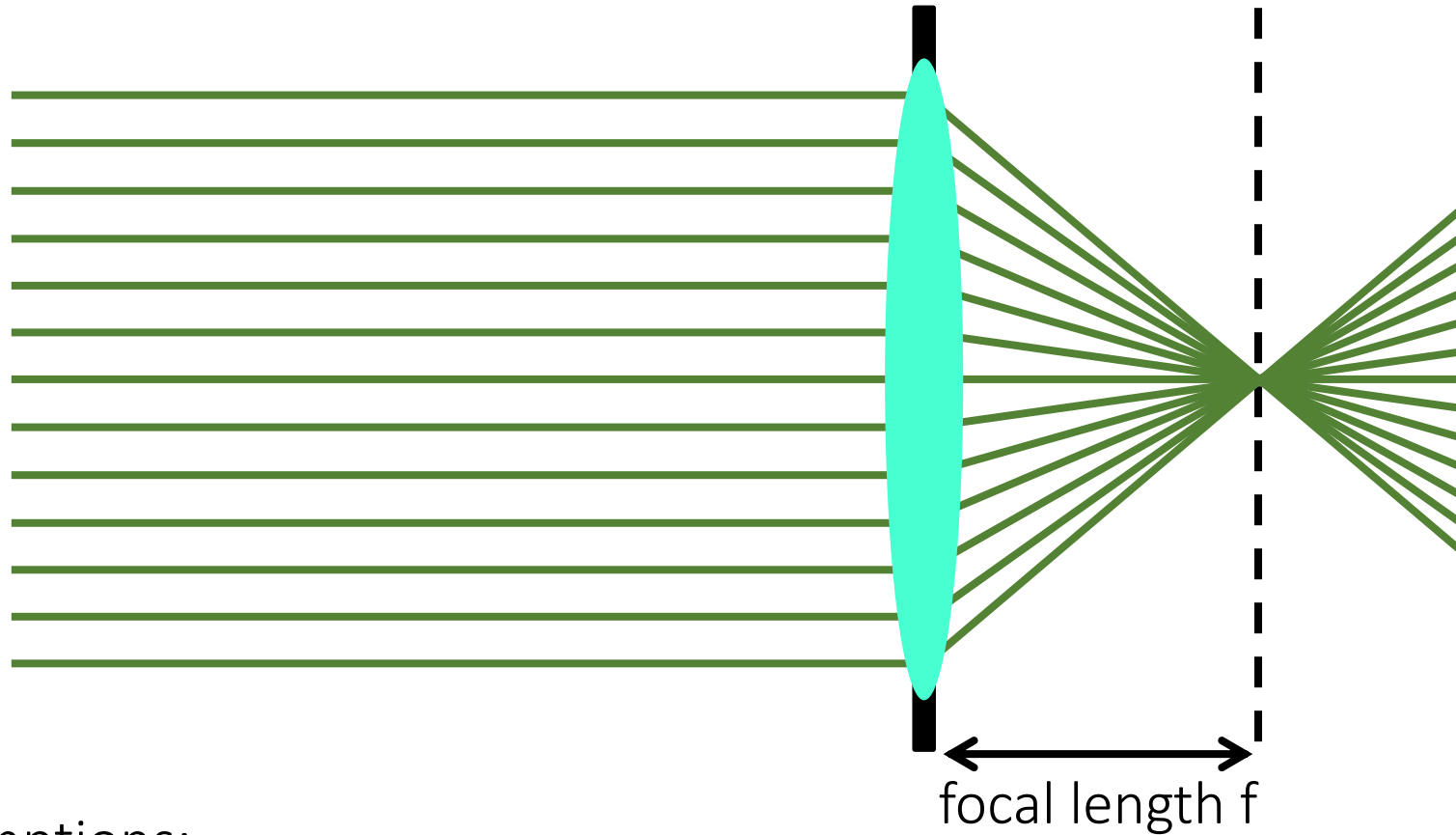


Two assumptions:

1. Rays passing through lens center are unaffected.

Thin lens model

Simplification of geometric optics for well-designed lenses.

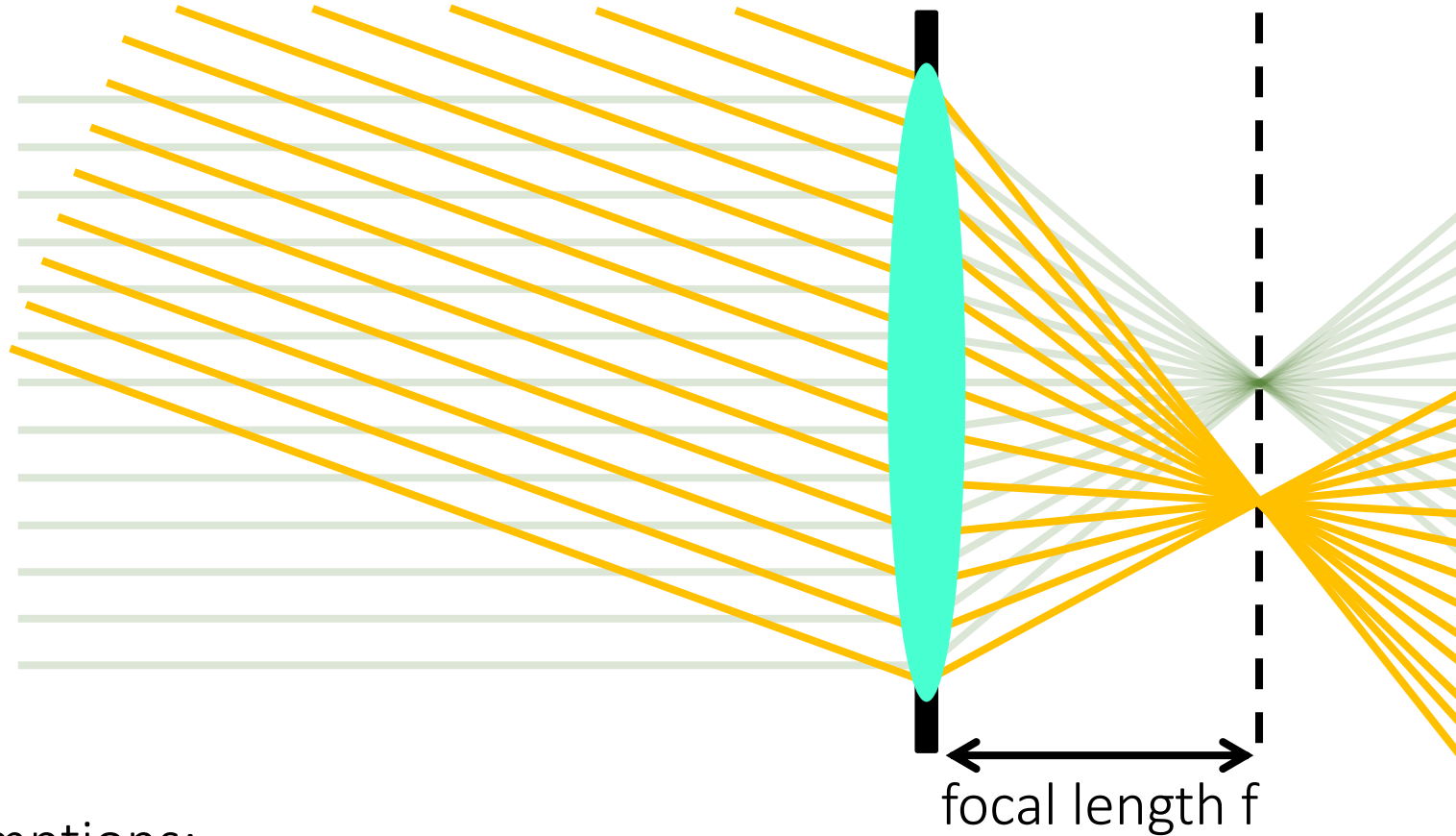


Two assumptions:

1. Rays passing through lens center are unaffected.
2. Parallel rays converge to a single point located on focal plane.

Thin lens model

Simplification of geometric optics for well-designed lenses.

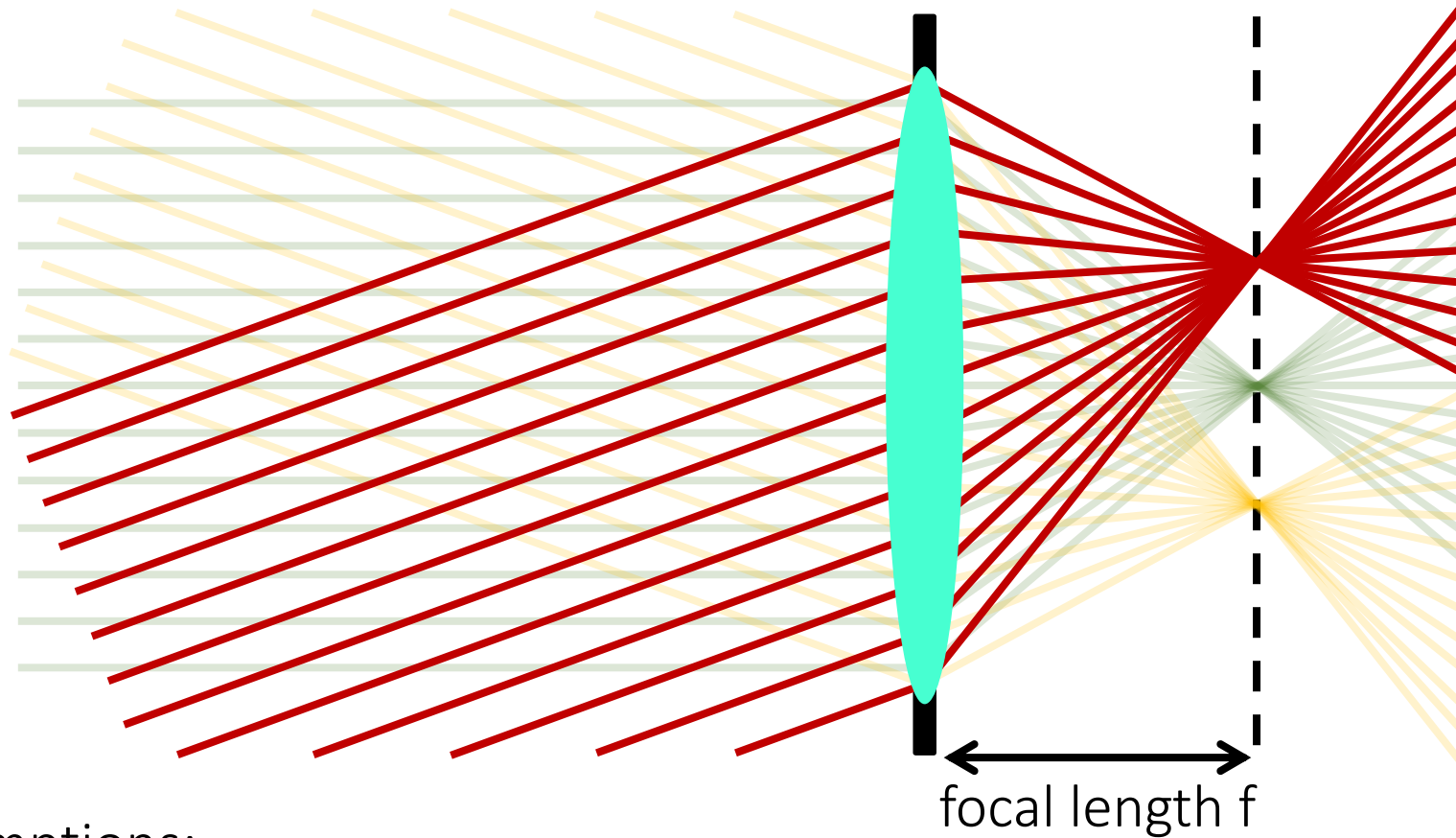


Two assumptions:

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Thin lens model

Simplification of geometric optics for well-designed lenses.

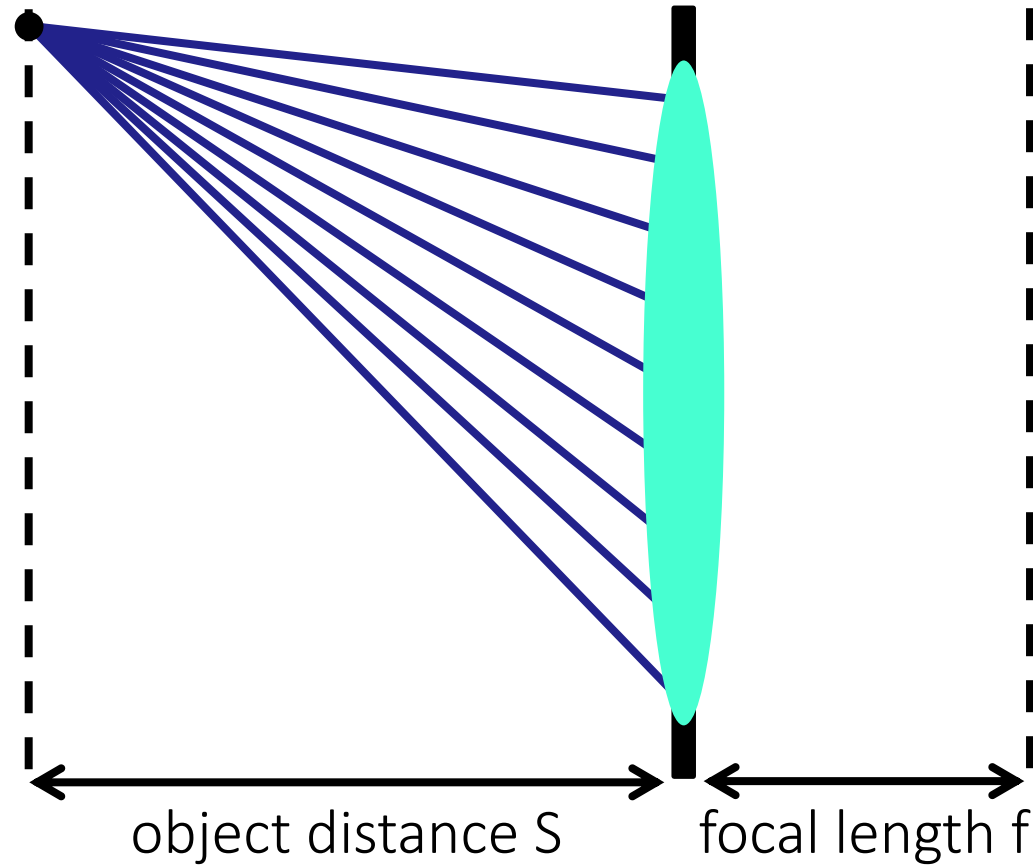


Two assumptions:

1. Rays passing through lens center are unaffected.
2. Parallel rays converge to a single point located on focal plane.

Tracing rays through a thin lens

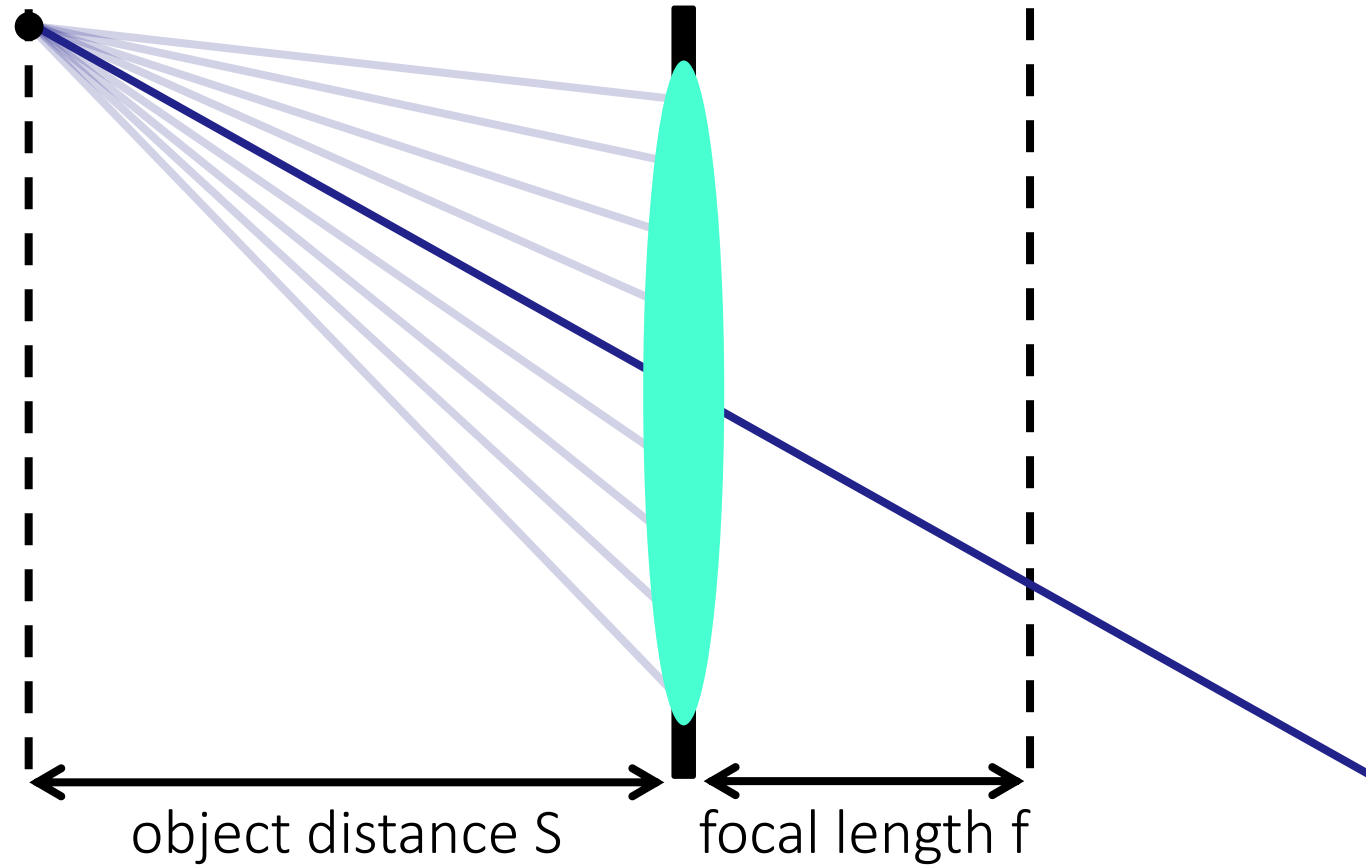
Consider an object emitting a bundle of rays. How do they propagate through the lens?



Tracing rays through a thin lens

Consider an object emitting a bundle of rays. How do they propagate through the lens?

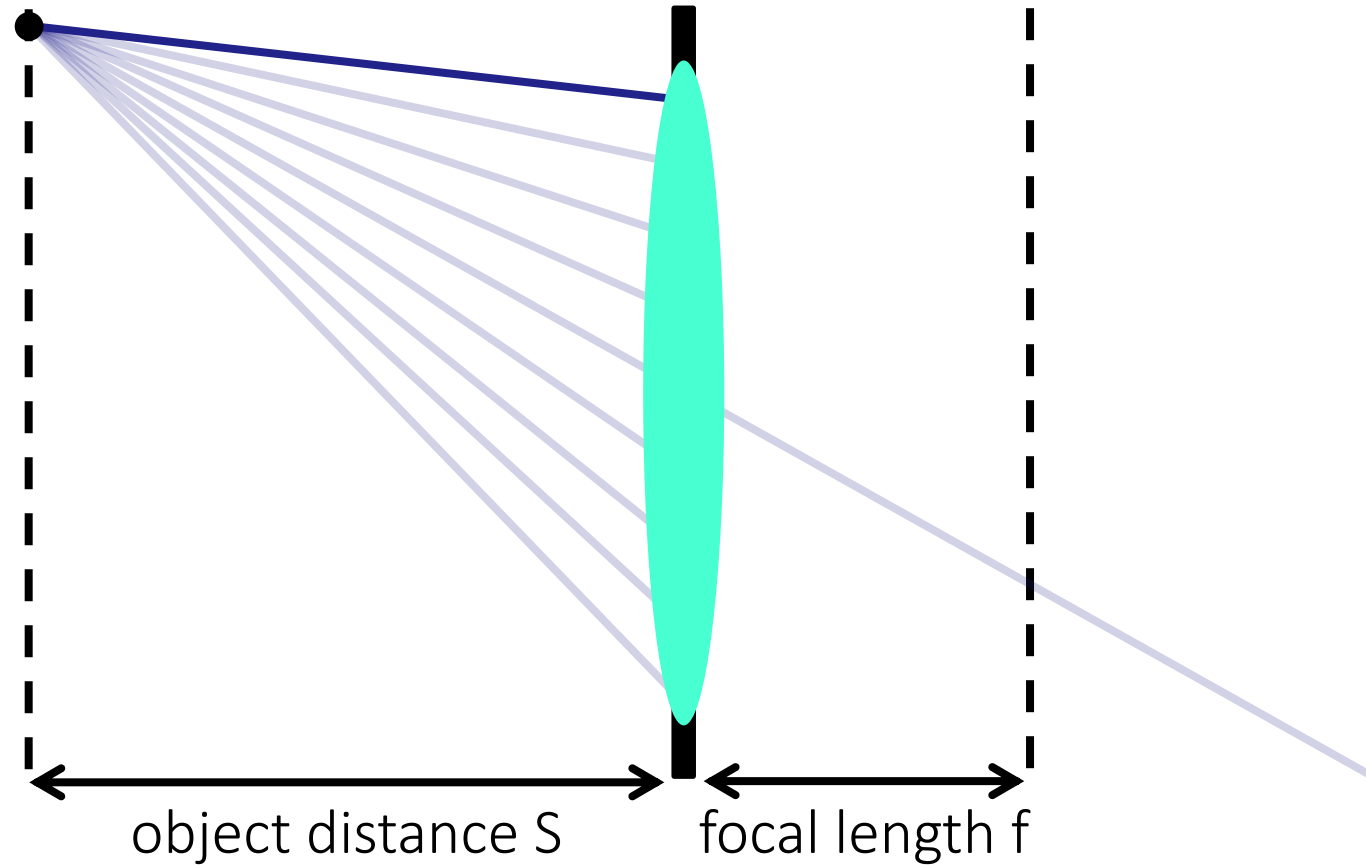
1. Trace rays through lens center.



Tracing rays through a thin lens

Consider an object emitting a bundle of rays. How do they propagate through the lens?

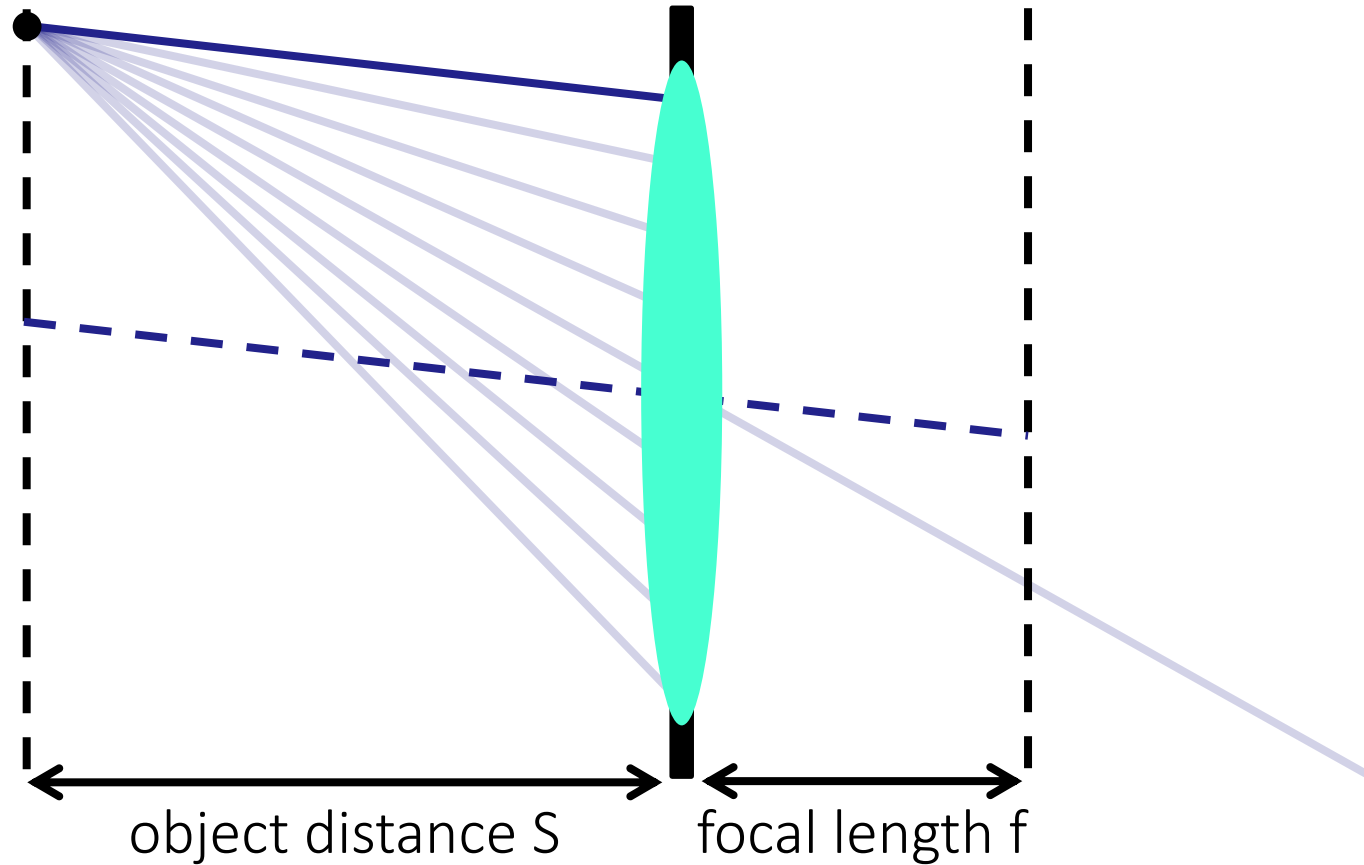
1. Trace rays through lens center.
2. For all other rays:



Tracing rays through a thin lens

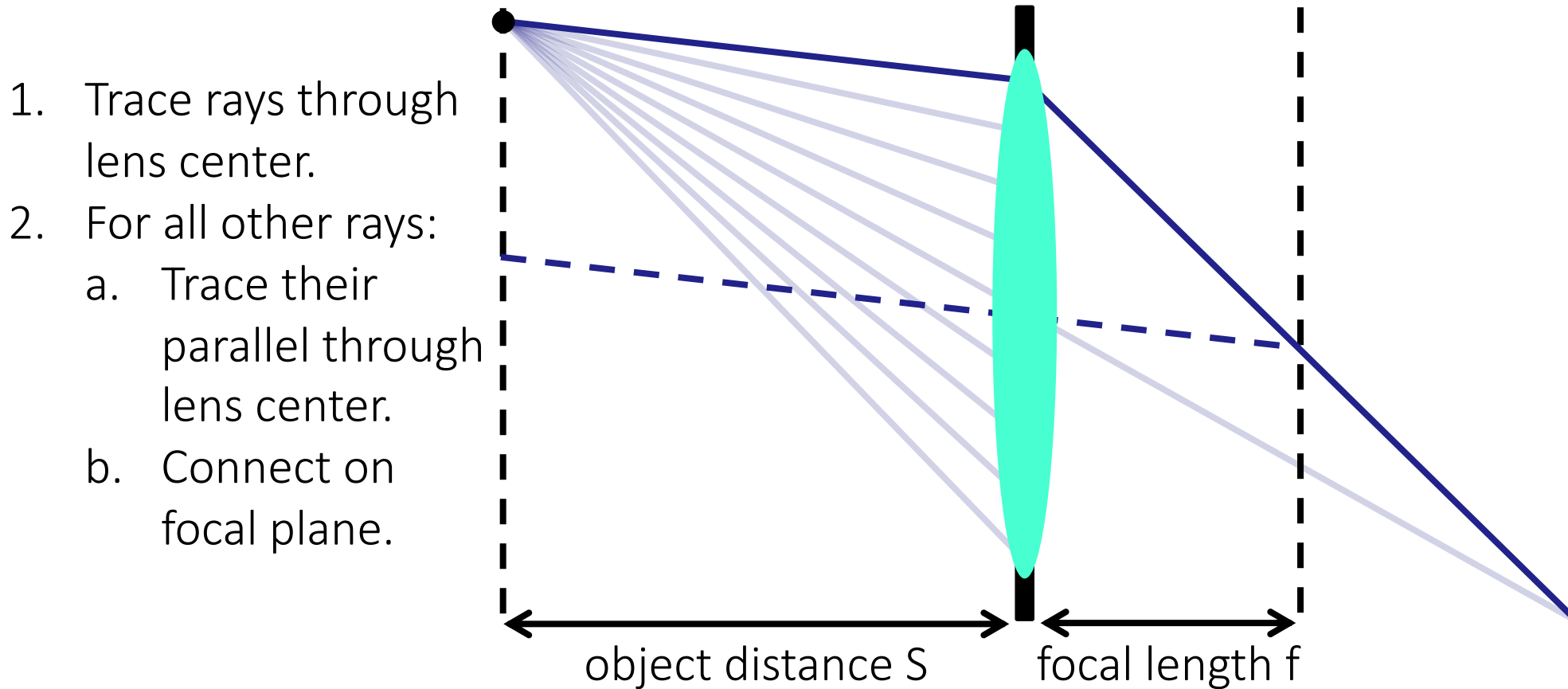
Consider an object emitting a bundle of rays. How do they propagate through the lens?

1. Trace rays through lens center.
2. For all other rays:
 - a. Trace their parallel through lens center.



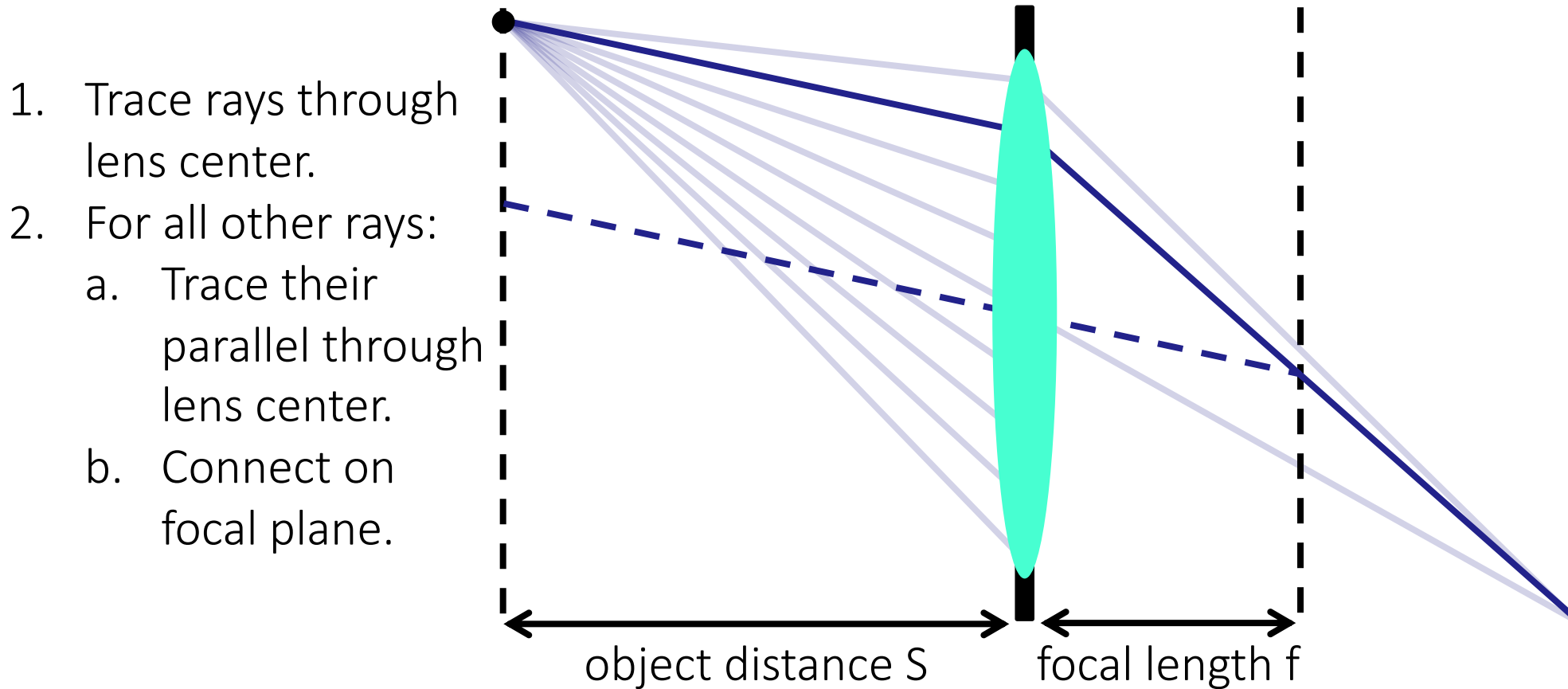
Tracing rays through a thin lens

Consider an object emitting a bundle of rays. How do they propagate through the lens?



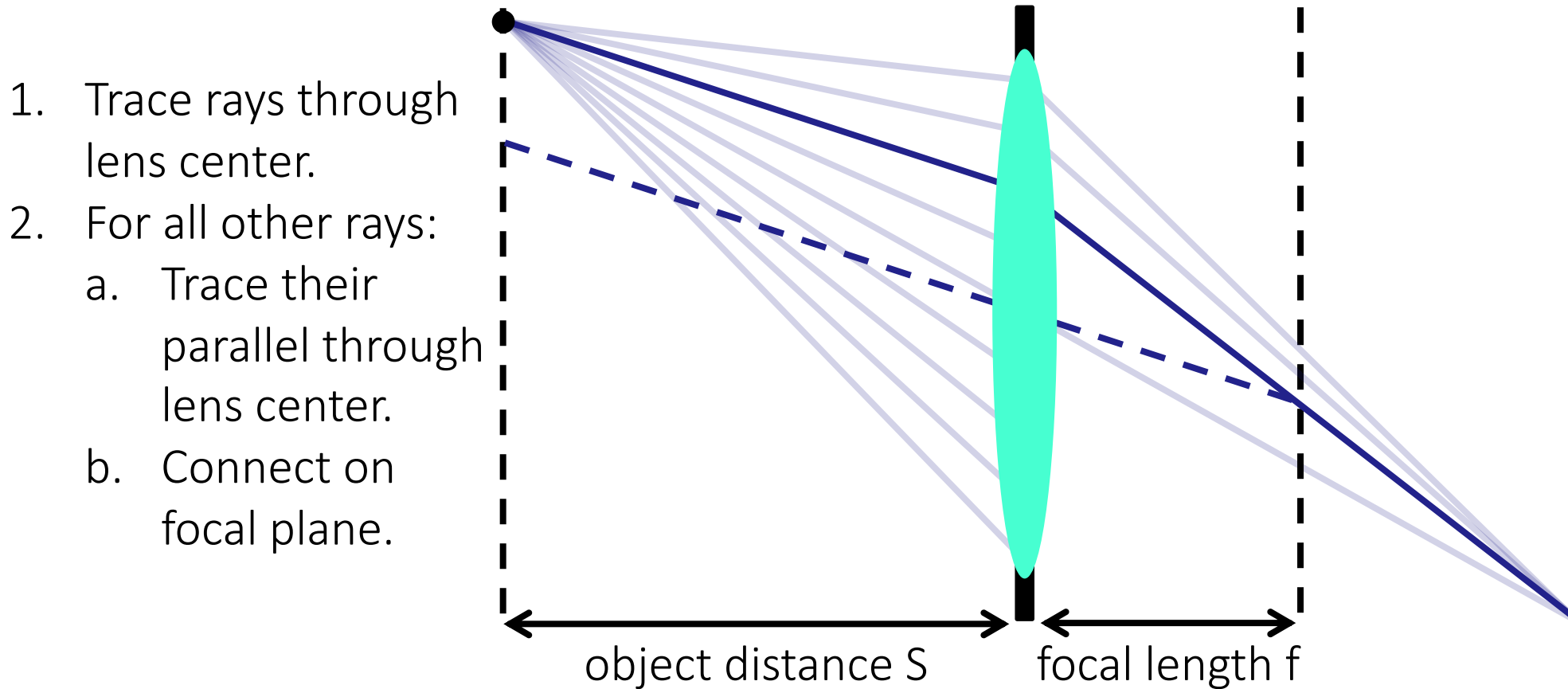
Tracing rays through a thin lens

Consider an object emitting a bundle of rays. How do they propagate through the lens?



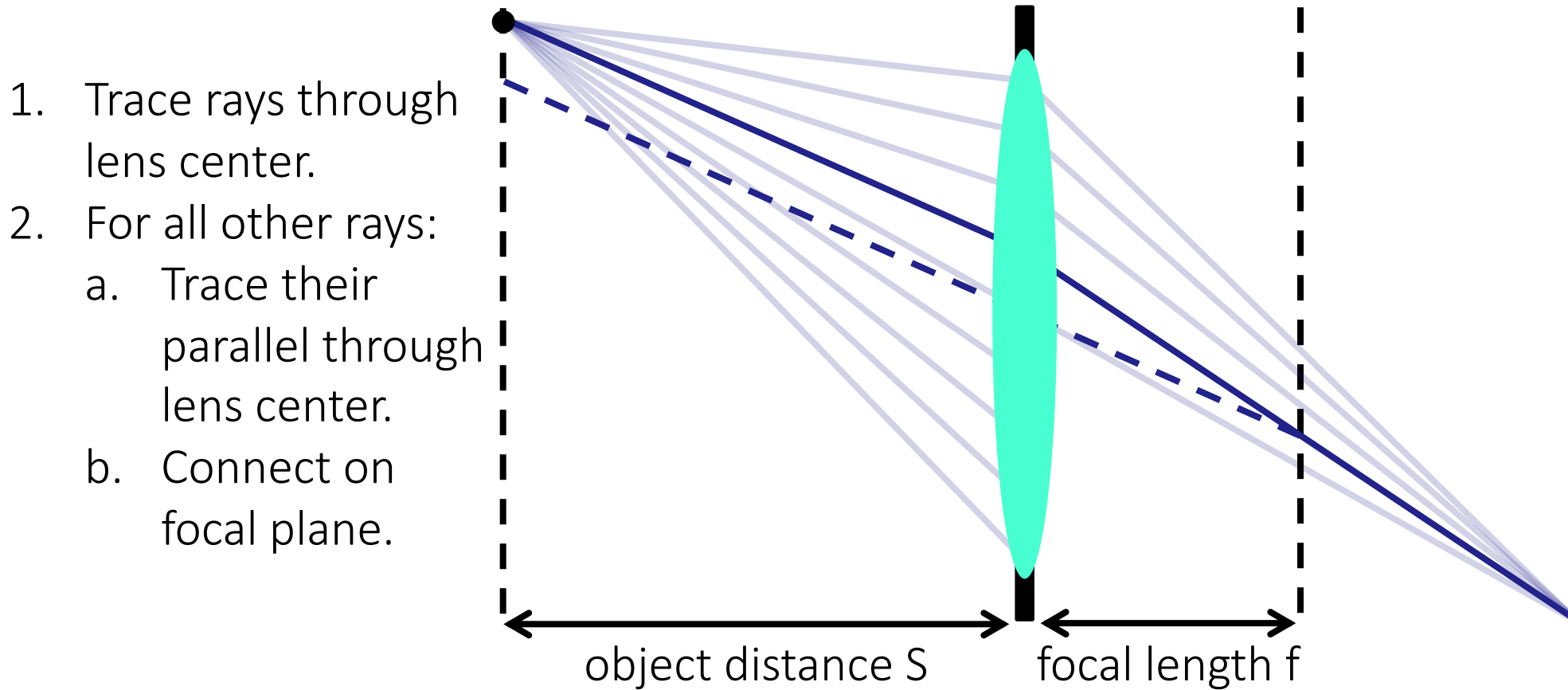
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Tracing rays through a thin lens

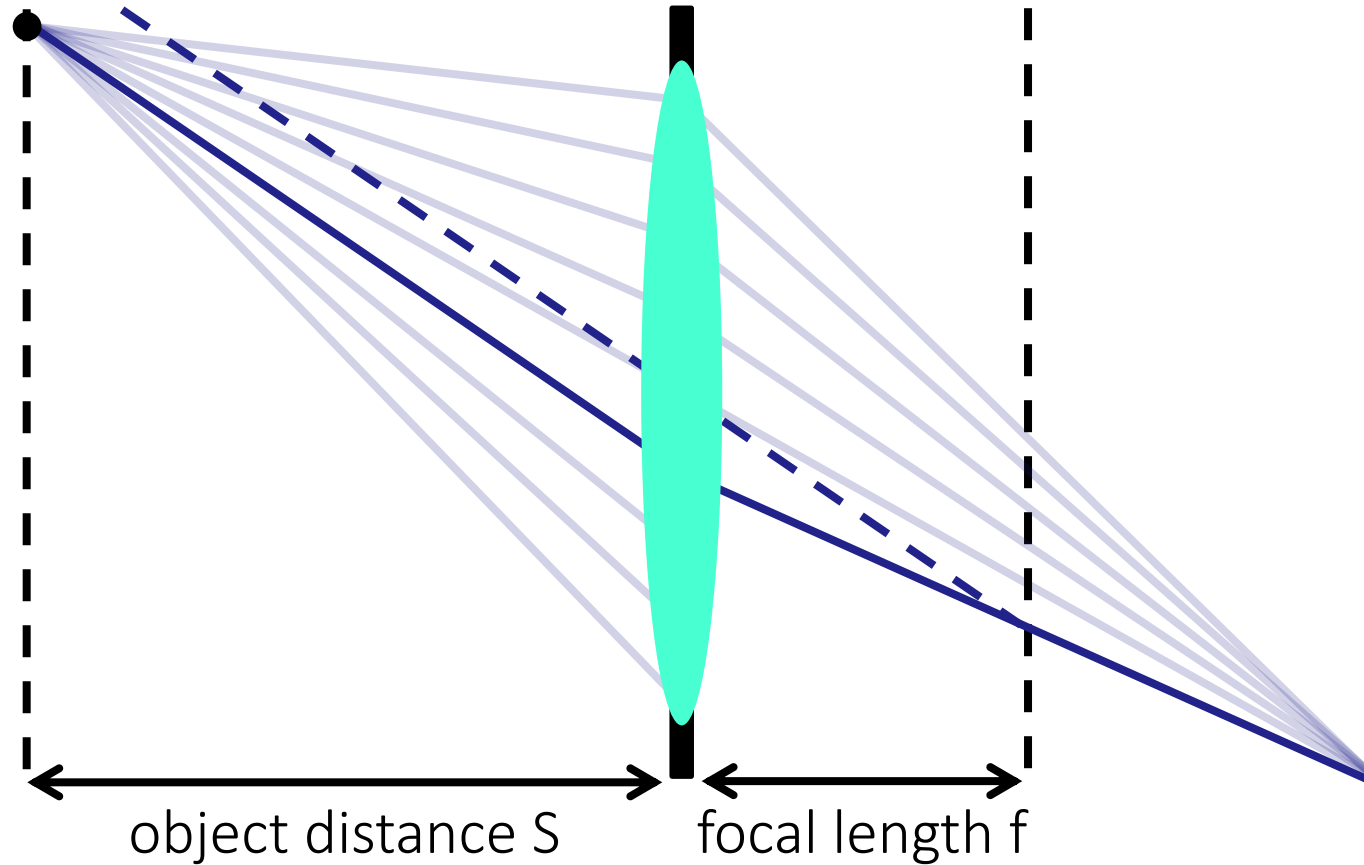
Consider an object emitting a bundle of rays. How do they propagate through the lens?



Tracing rays through a thin lens

Consider an object emitting a bundle of rays. How do they propagate through the lens?

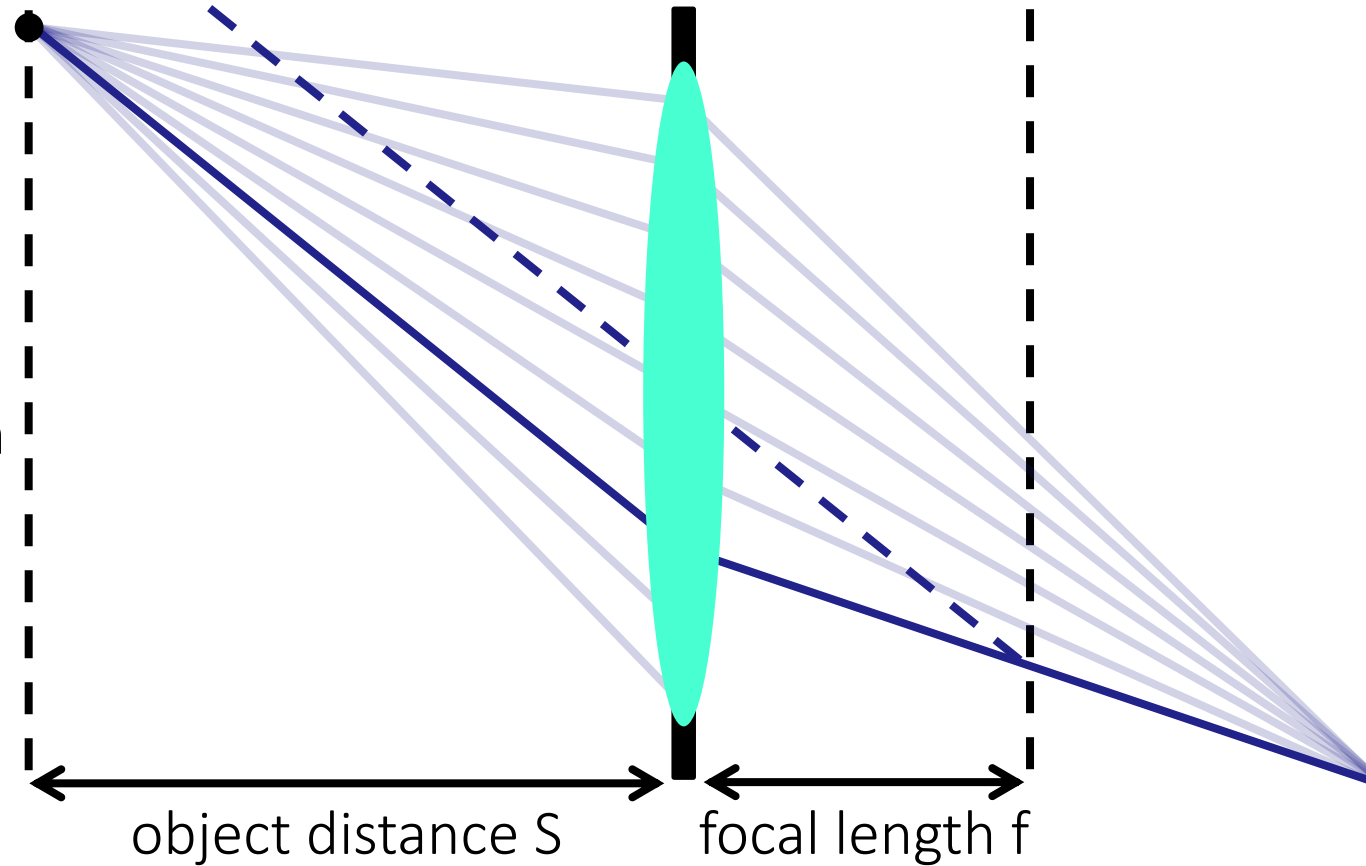
1. Trace rays through lens center.
2. For all other rays:
 - a. Trace their parallel through lens center.
 - b. Connect on focal plane.



Tracing rays through a thin lens

Consider an object emitting a bundle of rays. How do they propagate through the lens?

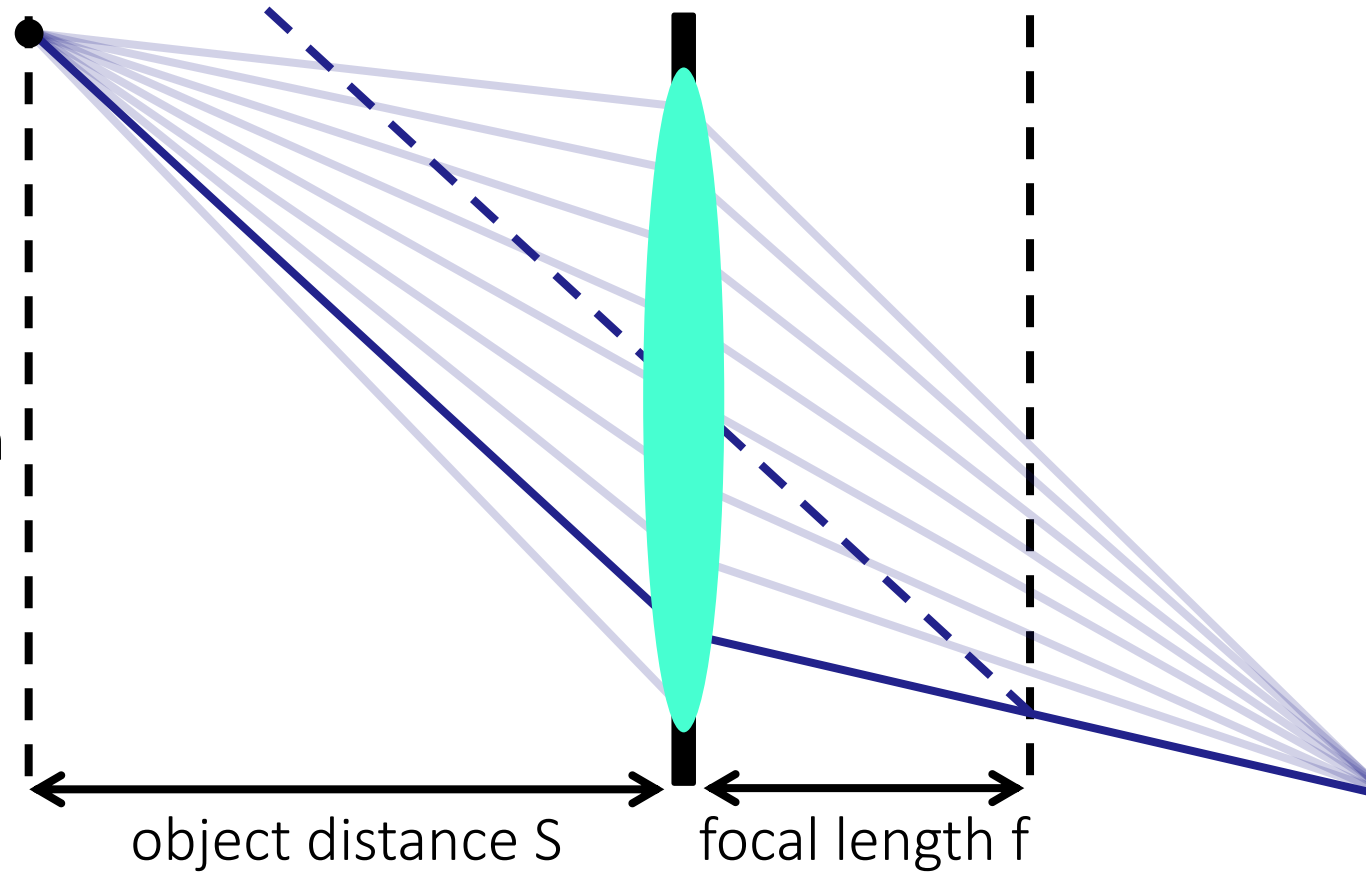
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Tracing rays through a thin lens

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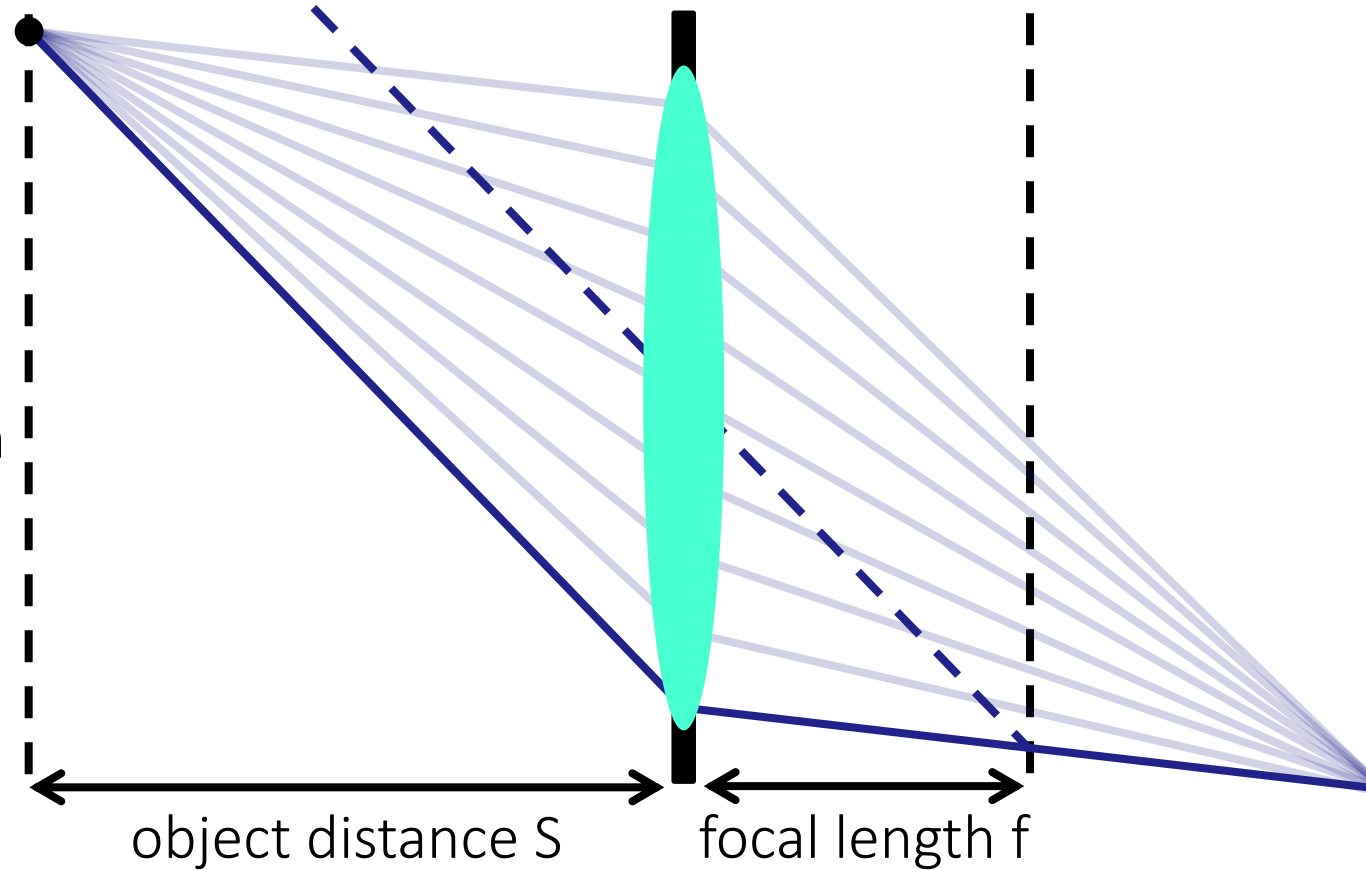
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 - a. Trace their parallel through lens center.
 - b. Connect on focal plane.



Tracing rays through a thin lens

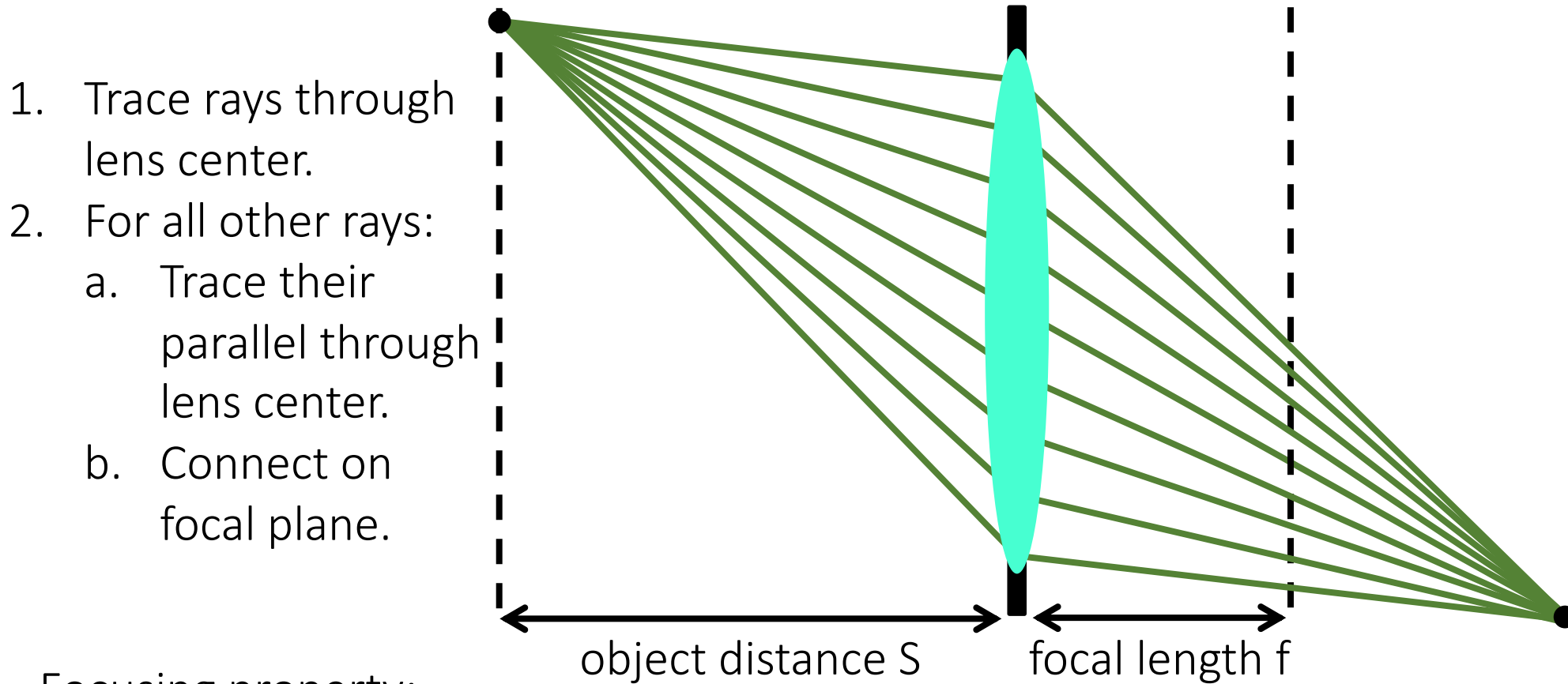
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 - a. Trace their parallel through lens center.
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Tracing rays through a thin lens

Consider an object emitting a bundle of rays. How do they propagate through the lens?

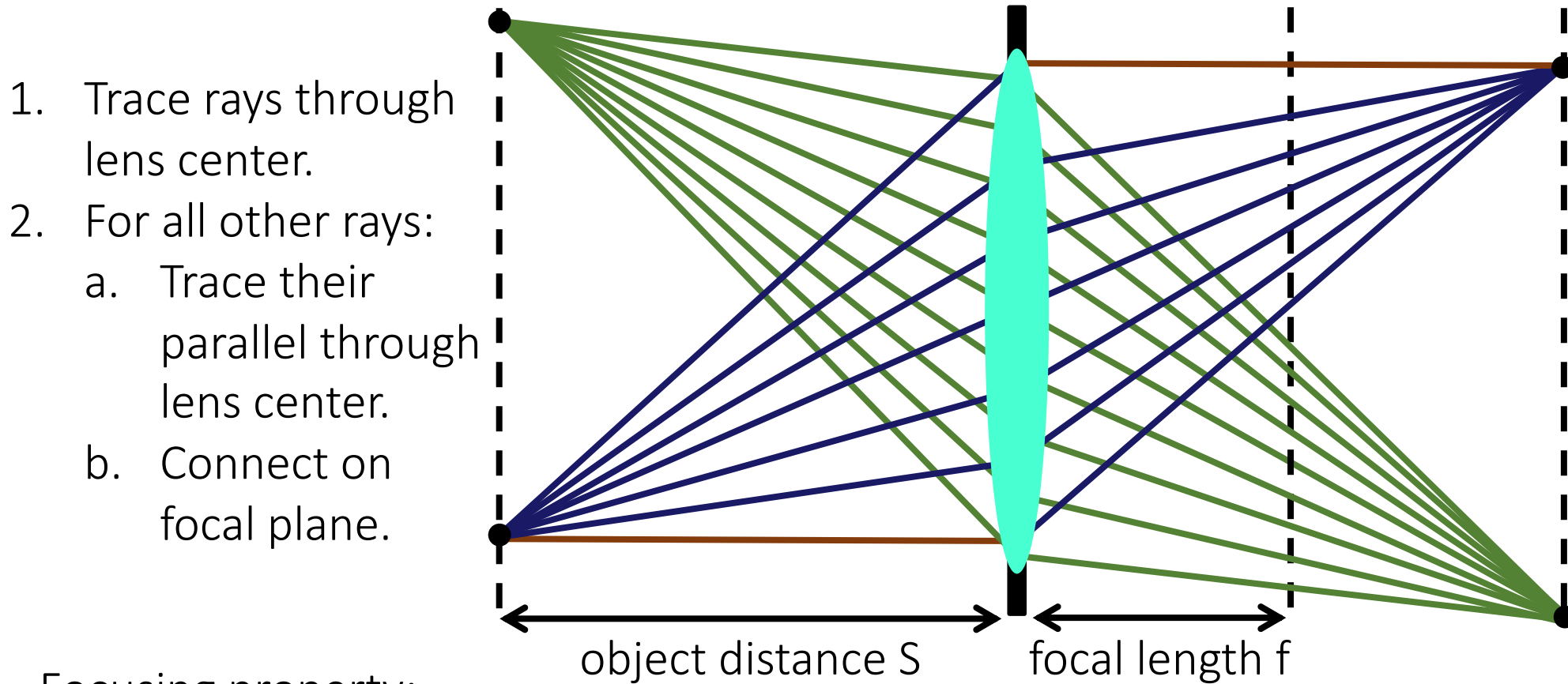


Focusing property:

1. Rays emitted from a point on one side converge to a point on the other side.

Tracing rays through a thin lens

Consider an object emitting a bundle of rays. How do they propagate through the lens?

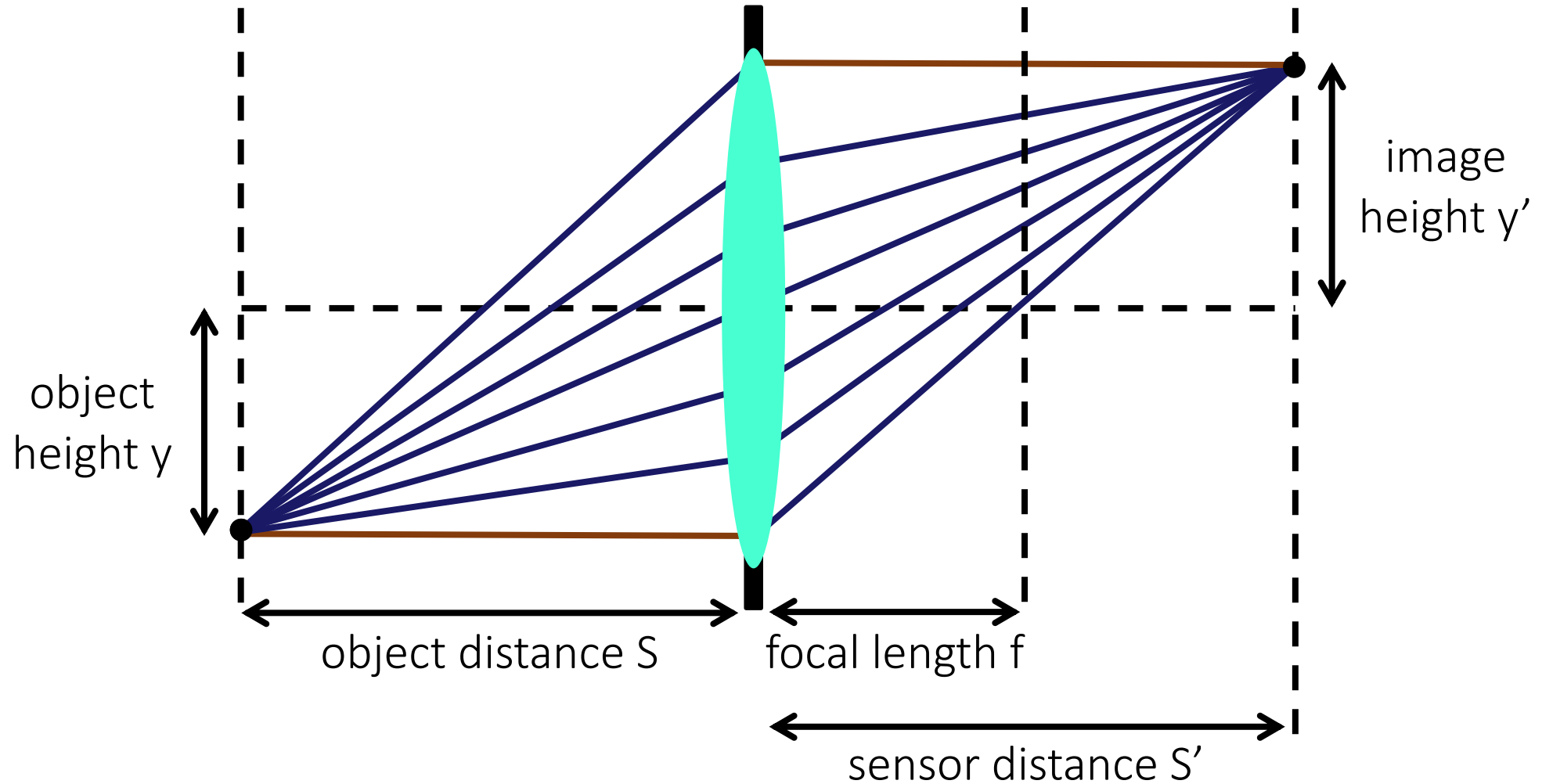


Focusing property:

1. Rays emitted from a point on one side converge to a point on the other side.
2. Bundles emitted from a plane parallel to the lens converge on a common plane.

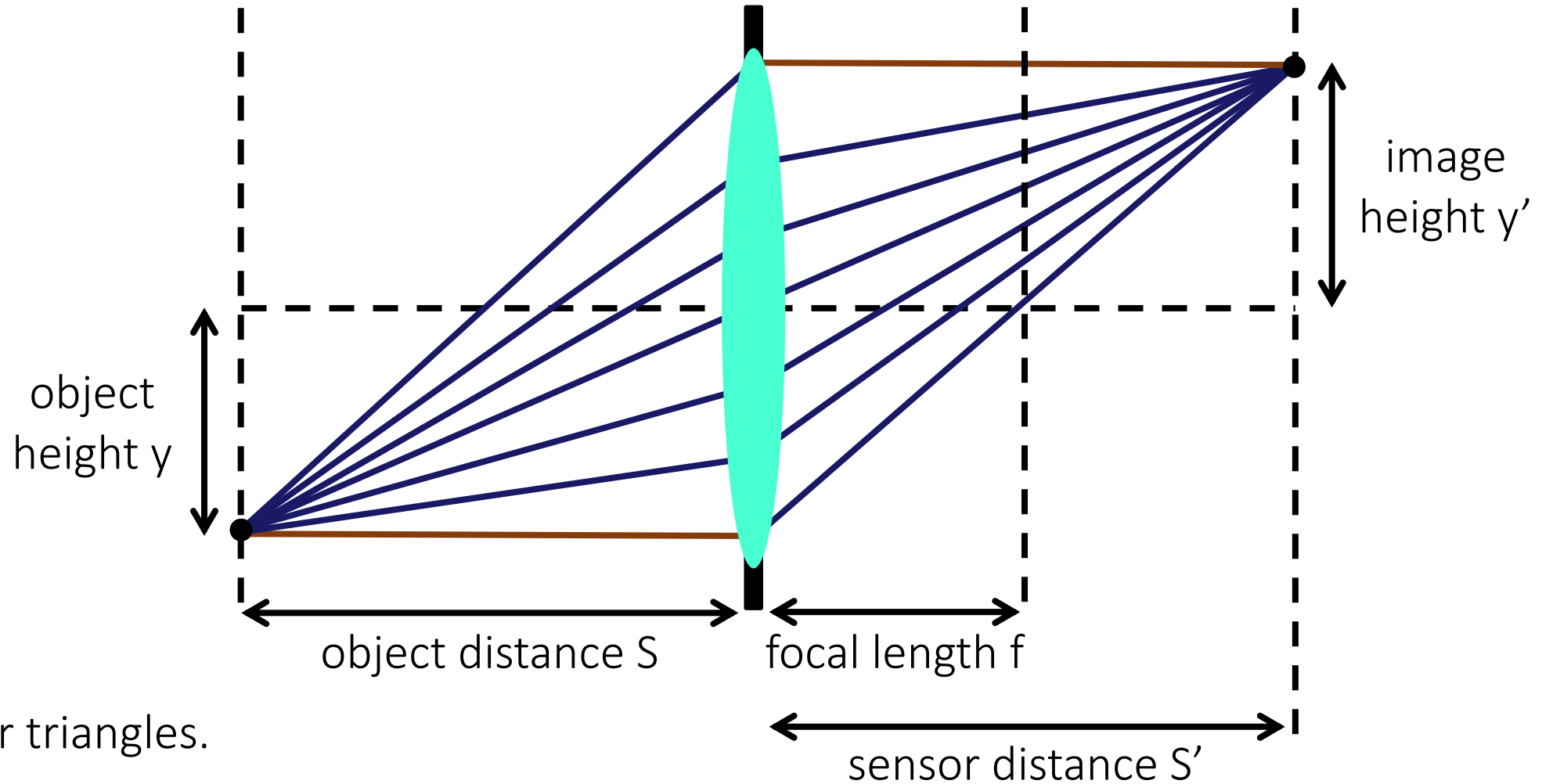
Gaussian lens formula

How can we relate scene-space (S, y) and image space (S', y') quantities?



Gaussian lens formula

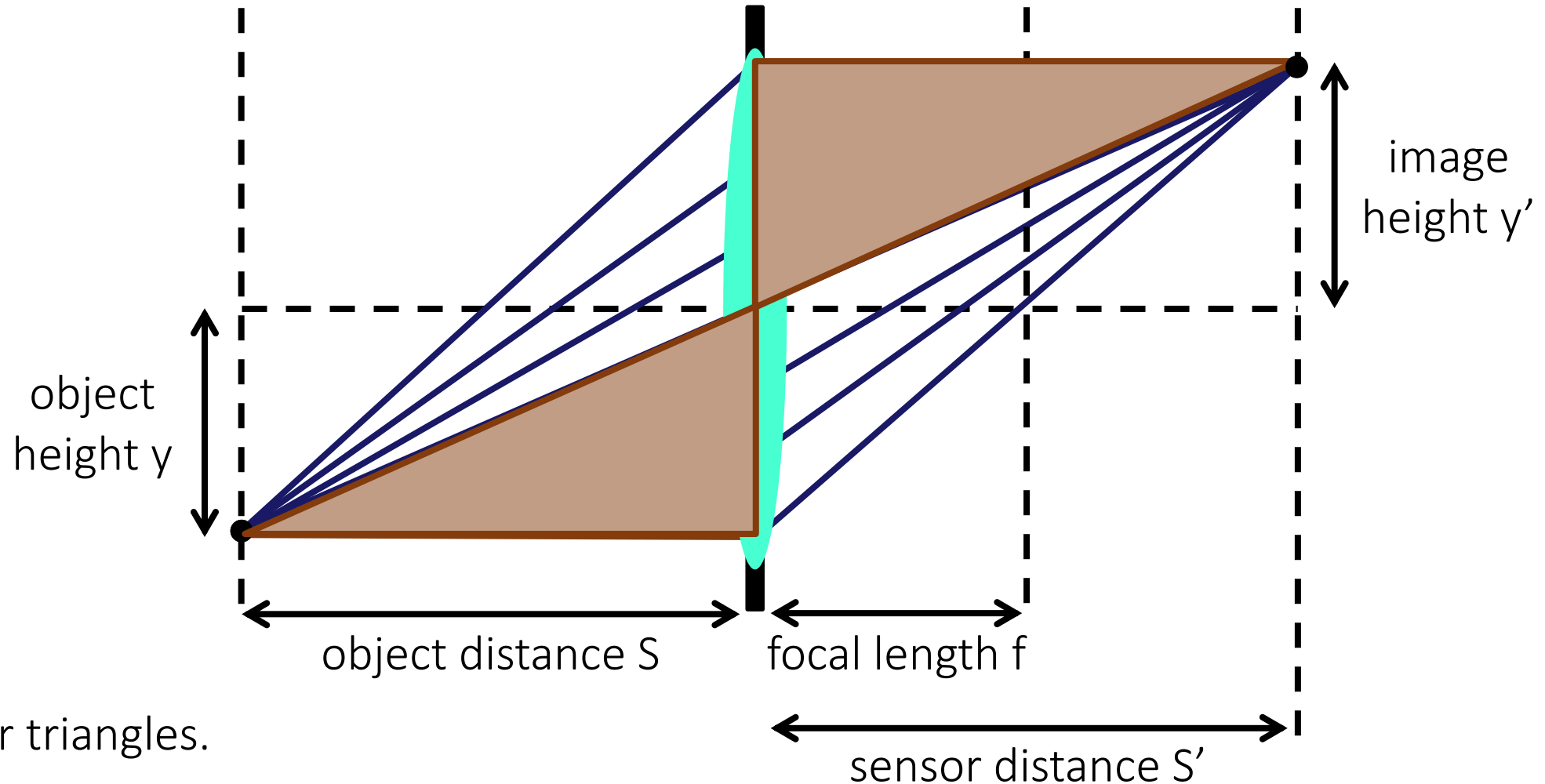
How can we relate scene-space (S, y) and image space (S', y') quantities?



Gaussian lens formula

How can we relate scene-space (S, y) and image space (S', y') quantities?

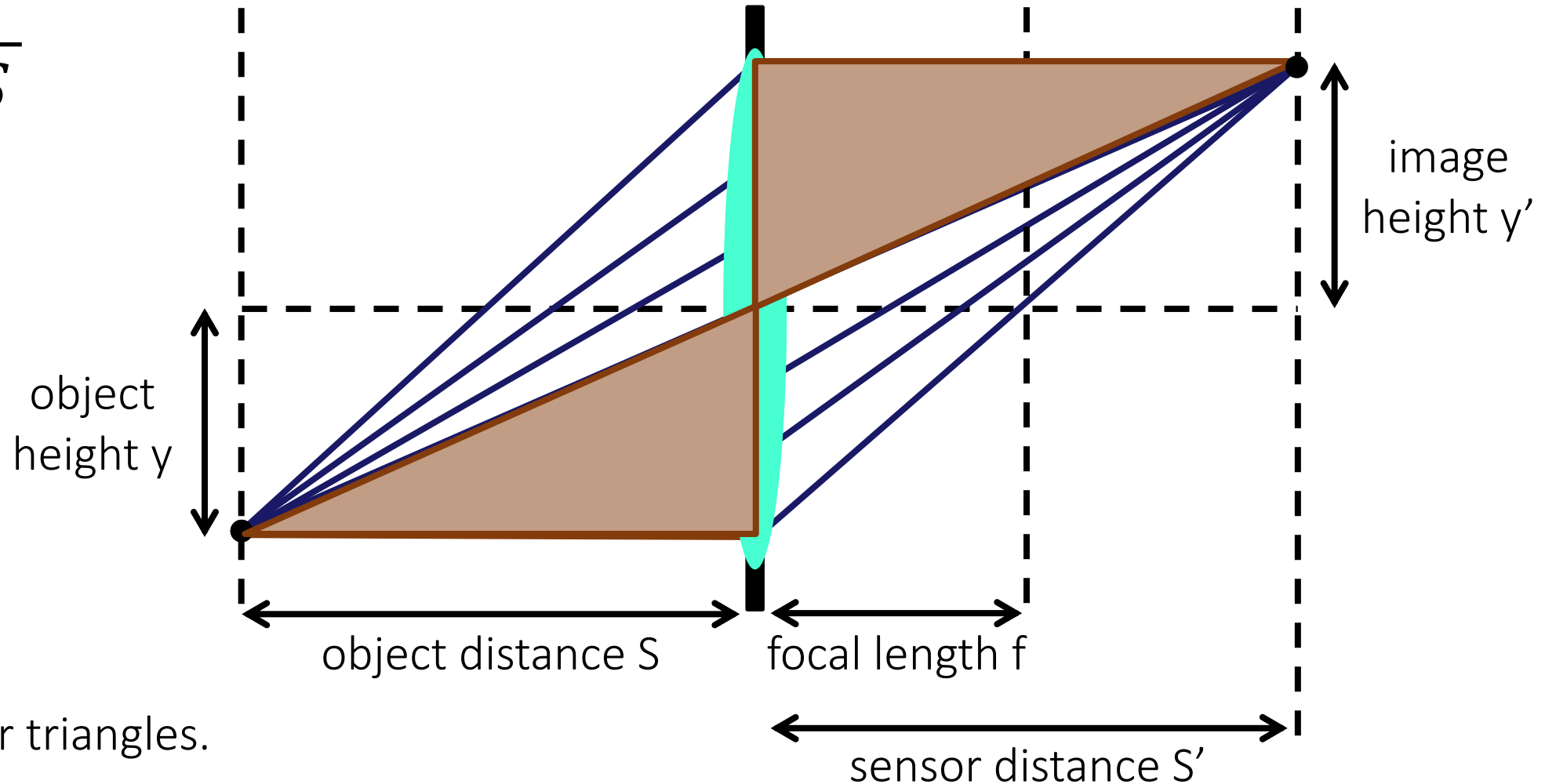
$$\frac{y'}{y} = ?$$



Gaussian lens formula

How can we relate scene-space (S, y) and image space (S', y') quantities?

$$\frac{y'}{y} = \frac{S'}{S}$$

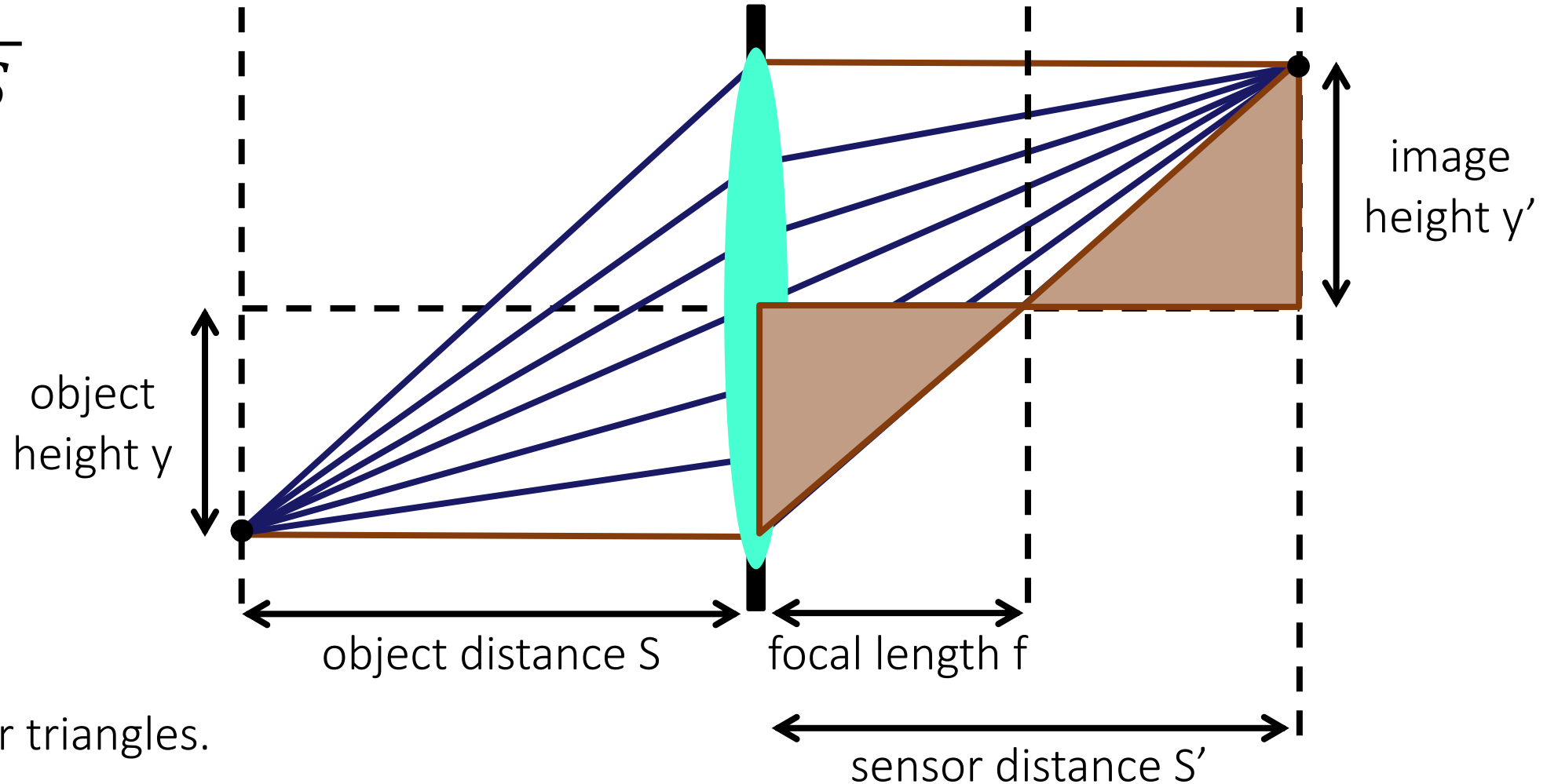


Gaussian lens formula

How can we relate scene-space (S, y) and image space (S', y') quantities?

$$\frac{y'}{y} = \frac{S'}{S}$$

$$\frac{y'}{y} = ?$$



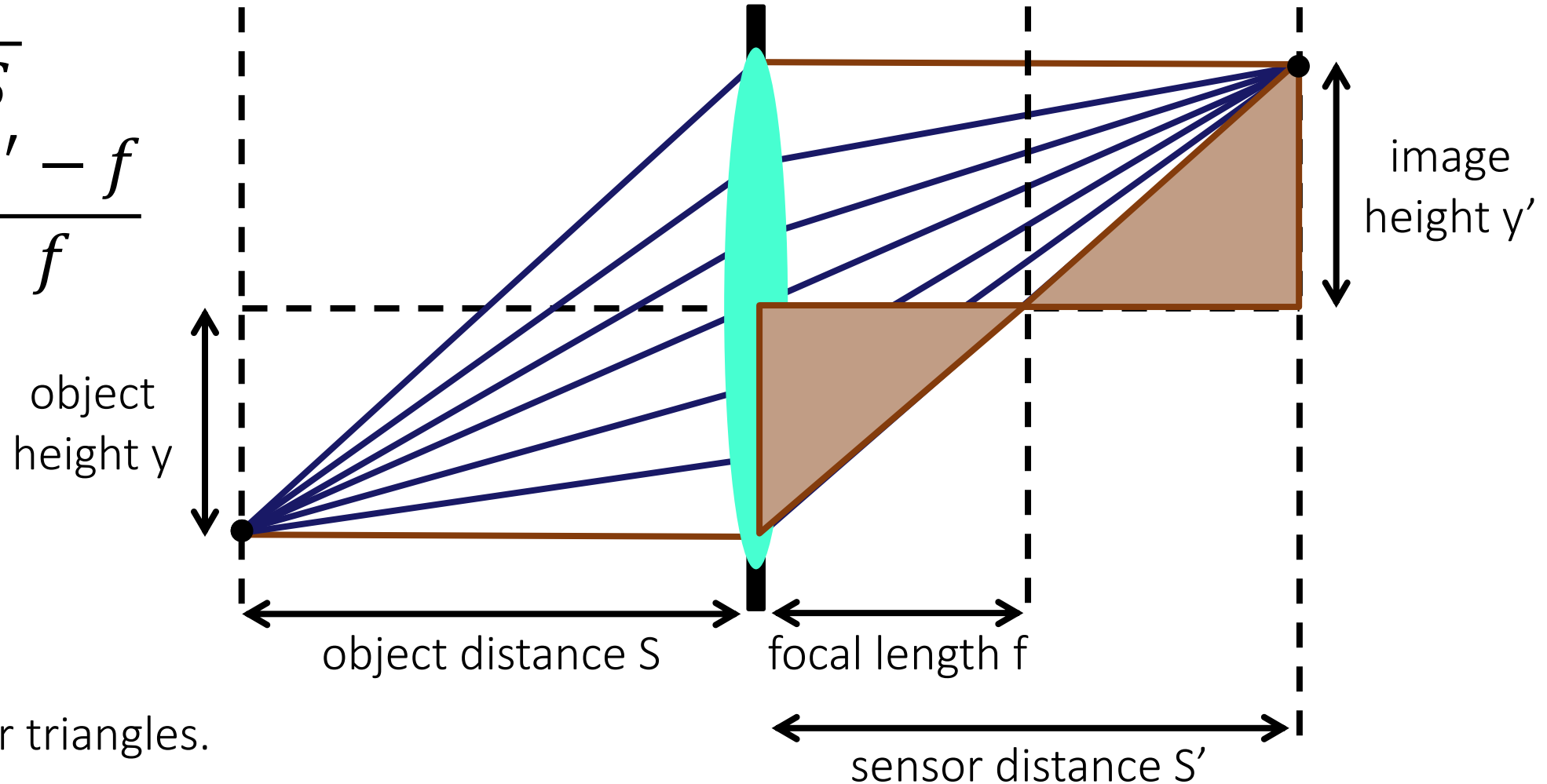
Use similar triangles.

Gaussian lens formula

How can we relate scene-space (S, y) and image space (S', y') quantities?

$$\frac{y'}{y} = \frac{S'}{S}$$

$$\frac{y'}{y} = \frac{S' - f}{f}$$



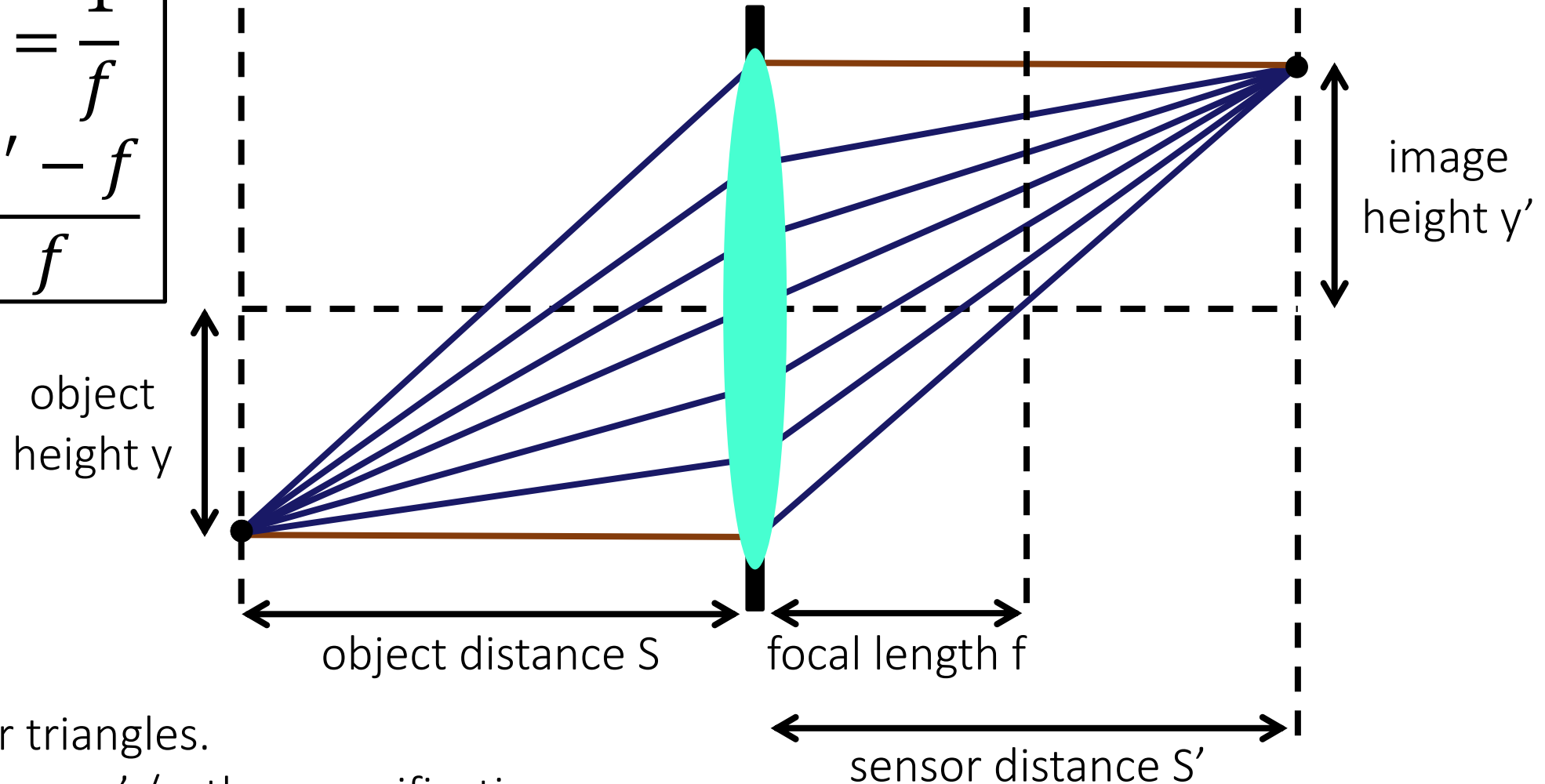
Use similar triangles.

Gaussian lens formula

How can we relate scene-space (S, y) and image space (S', y') quantities?

$$\frac{1}{S'} + \frac{1}{S} = \frac{1}{f}$$

$$m = \frac{S' - f}{f}$$



Use similar triangles.

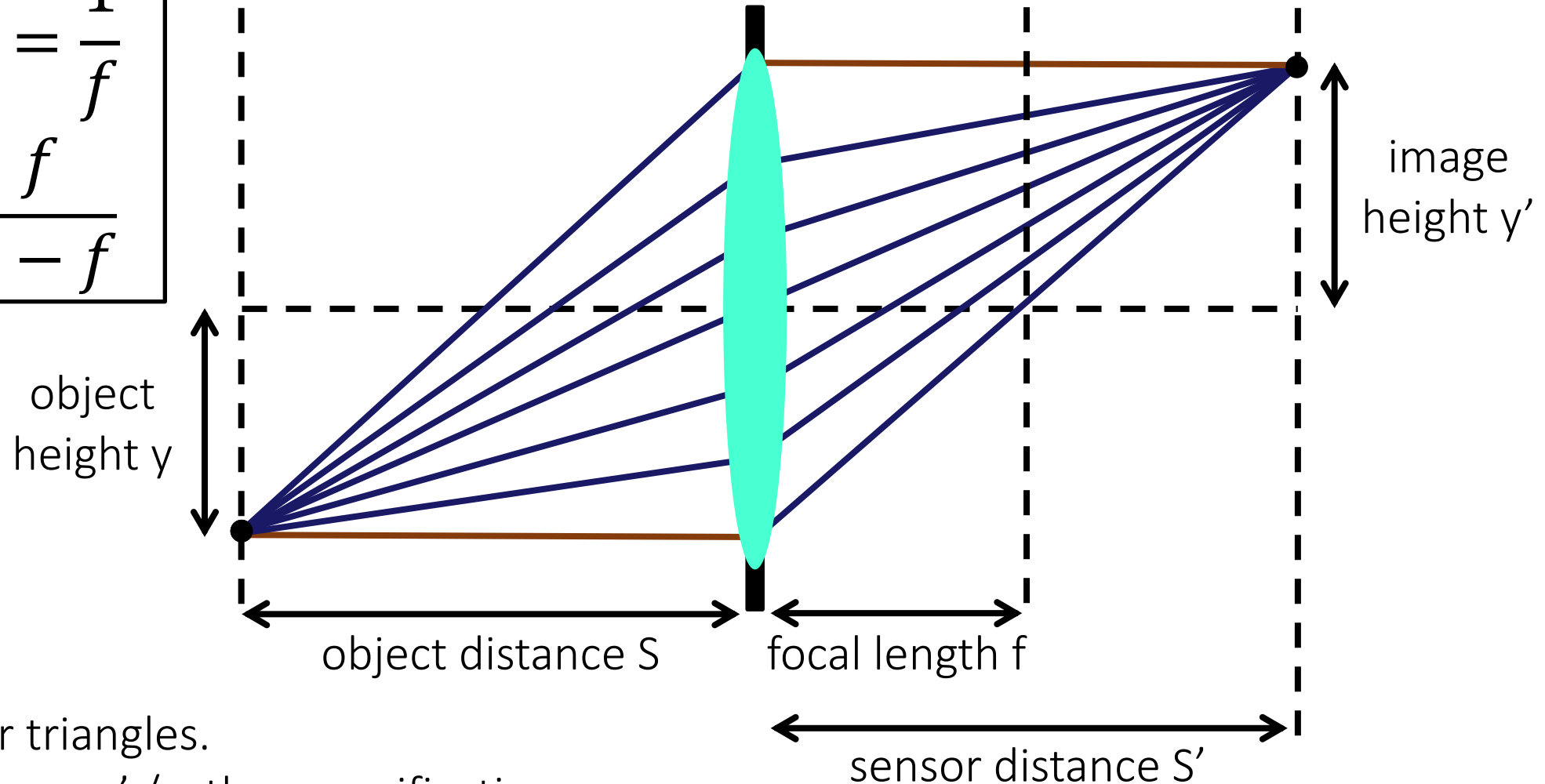
- We call $m = y' / y$ the magnification.

Gaussian lens formula

How can we relate scene-space (S, y) and image space (S', y') quantities?

$$\frac{1}{S'} + \frac{1}{S} = \frac{1}{f}$$

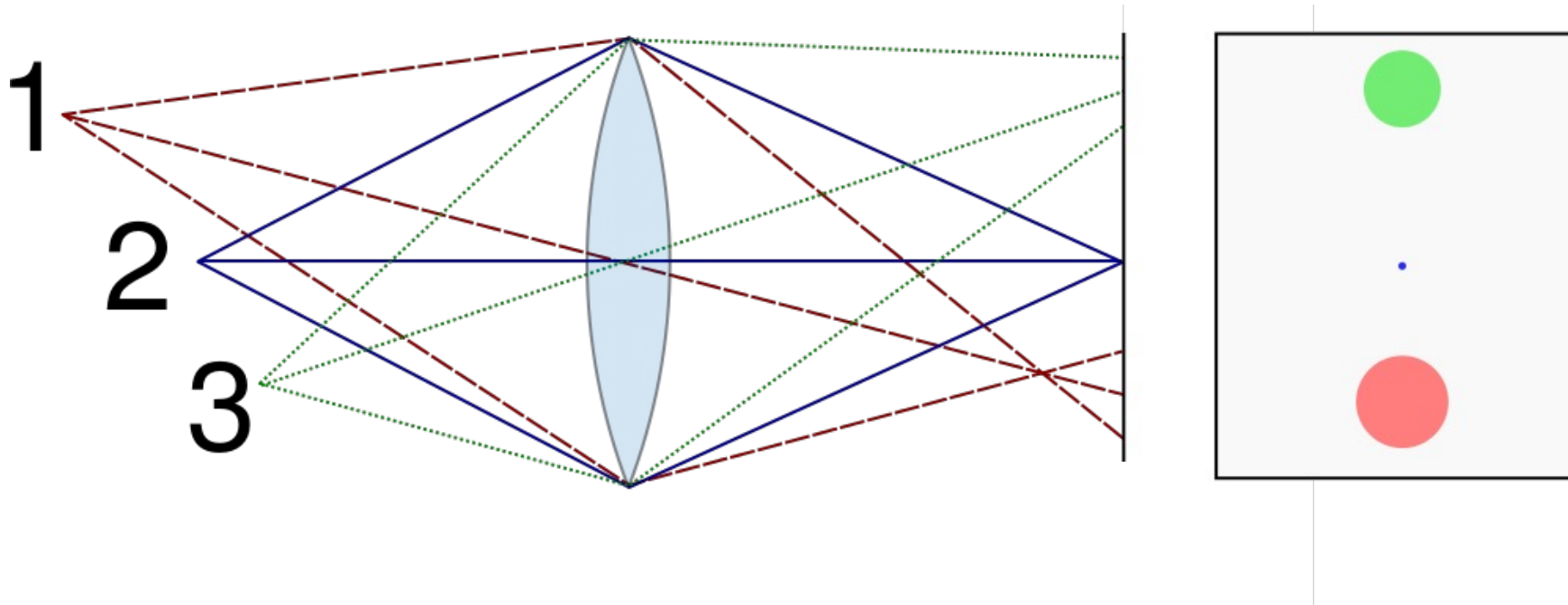
$$m = \frac{f}{S - f}$$



Use similar triangles.

- We call $m = y' / y$ the magnification.

Depth of field



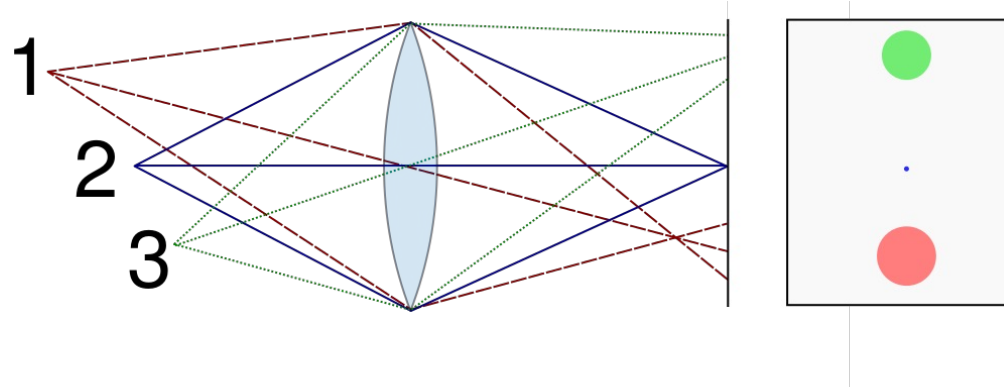
- For a fixed focal length and image plane, there is a specific distance at which objects are "in focus"
 - Other points project to a "circle of confusion" in the image

Depth of field

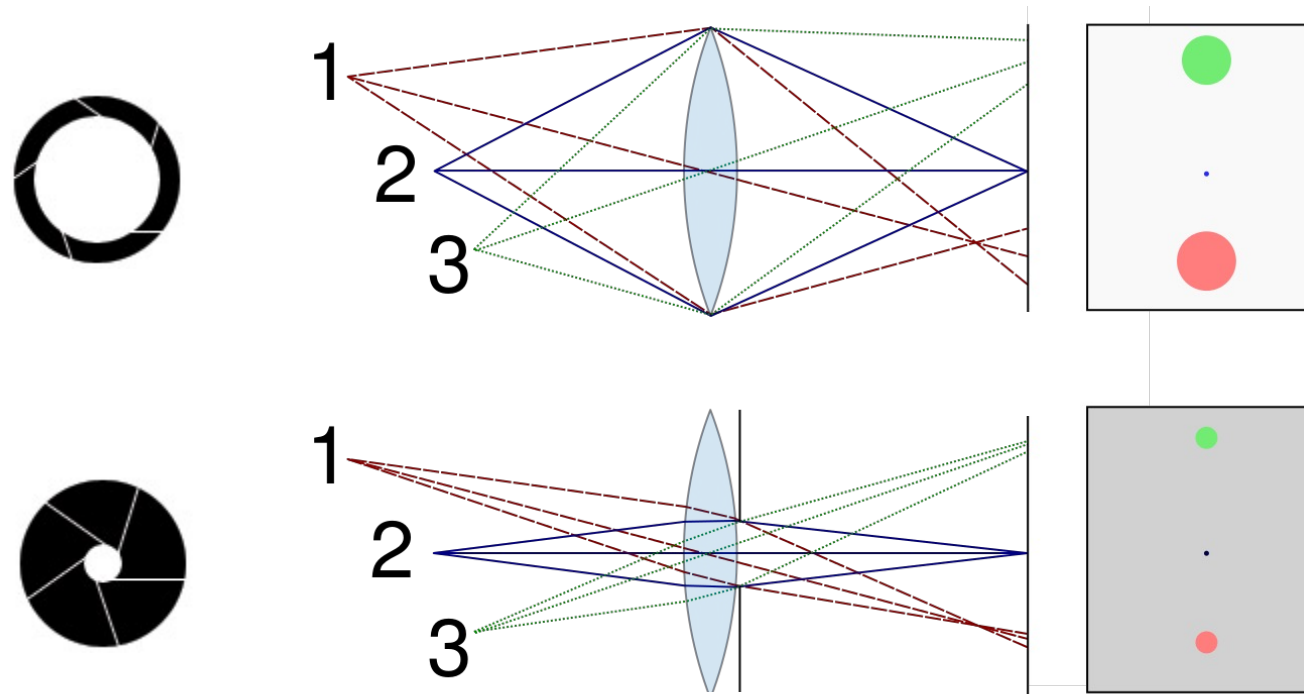
- Depth of field is the distance between the nearest and farthest objects in a scene that appear acceptably sharp in an image ([Wikipedia](#))



Controlling depth of field



Controlling depth of field



Changing the *aperture* size affects depth of field

- A smaller aperture increases the range in which the object is approximately in focus
- But small aperture reduces amount of light – need to increase *exposure*

Varying the aperture



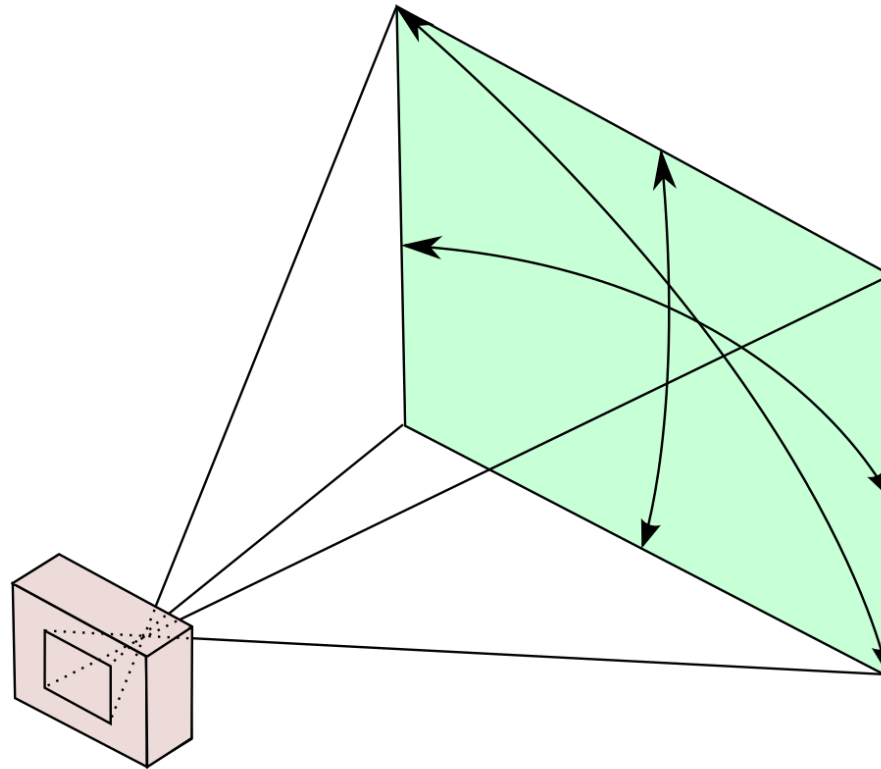
Large aperture = small DOF



Small aperture = large DOF

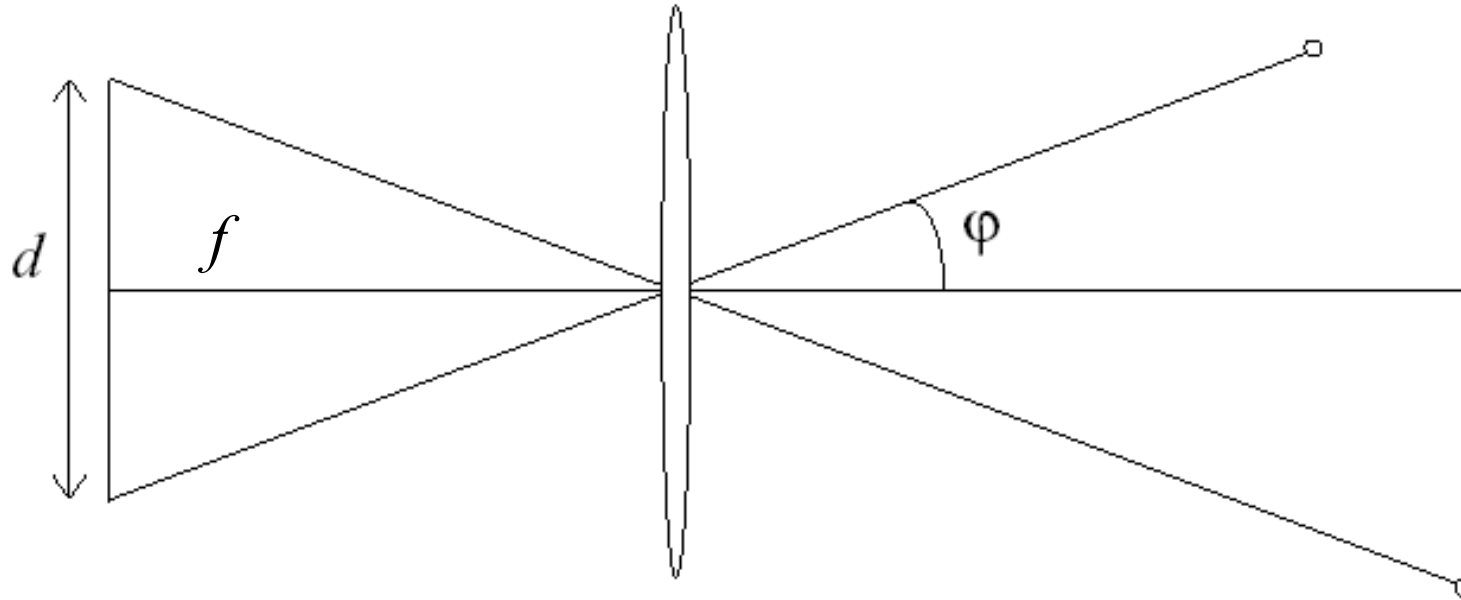
Field of view

- The field of view is the angular extent of the world observed by the camera ([Wikipedia](#))
- What determines the FOV?



Field of view

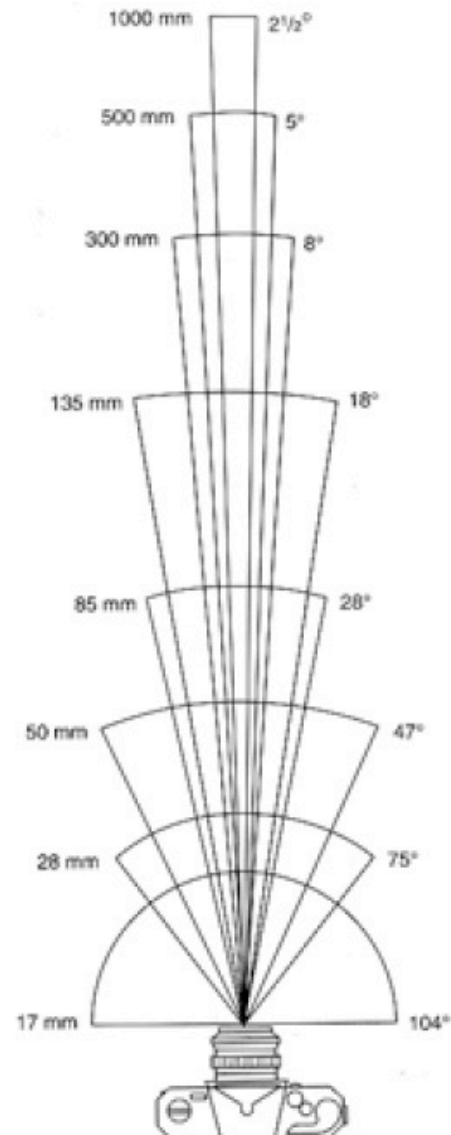
- The field of view is the angular extent of the world observed by the camera ([Wikipedia](#))
- What determines the FOV?
 - Focal length (f), length of the sensor (d):



$$\varphi = \tan^{-1} \frac{d}{2f}$$

- Larger focal length = smaller FOV

Field of view



17mm



28mm

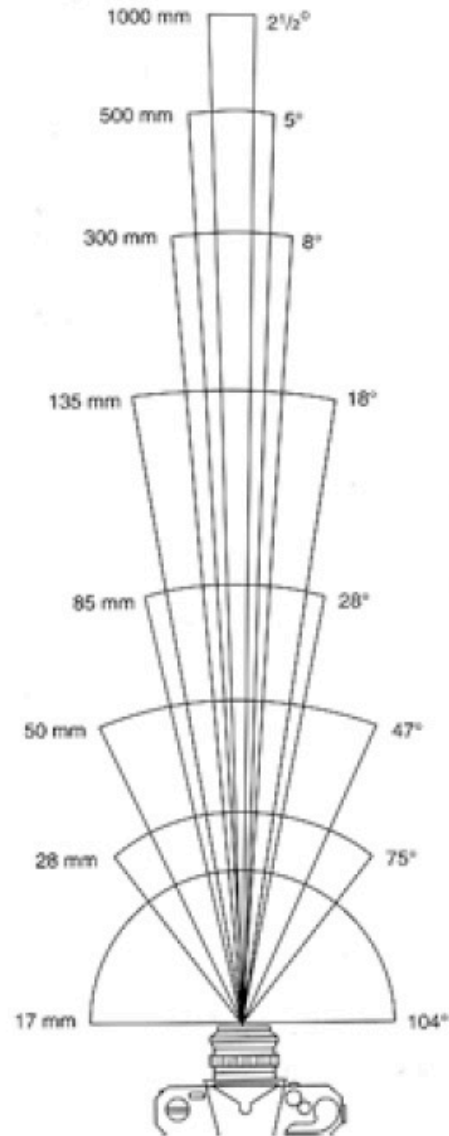


50mm

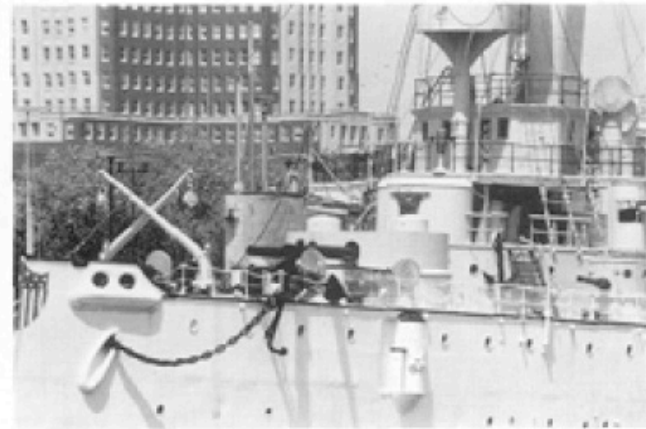


85mm

Field of view



135mm



300mm

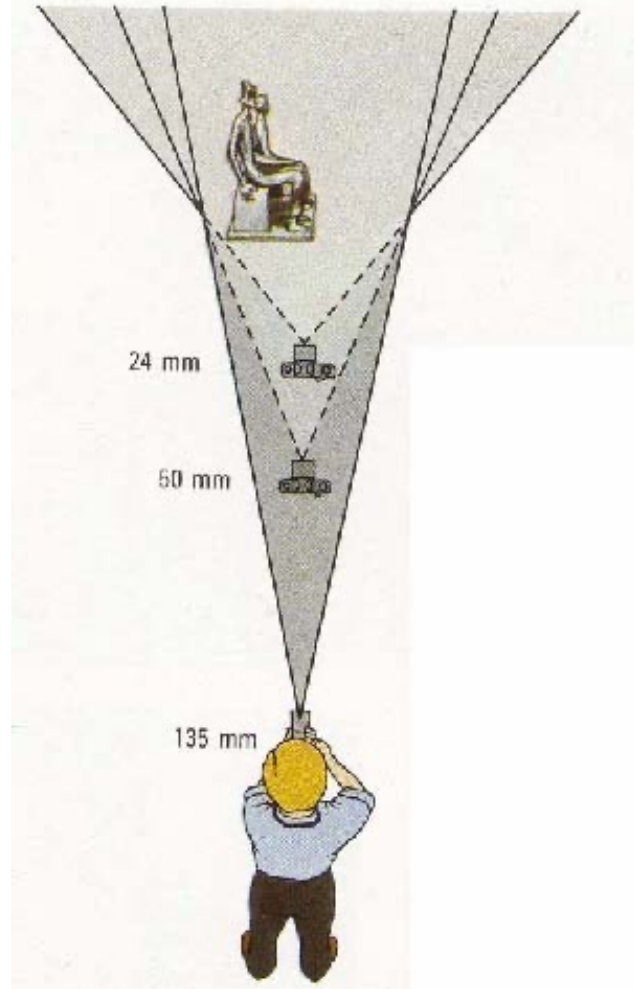


50mm



28mm

Field of view / focal length



Large FOV, small f
Camera close to car



Small FOV, large f
Camera far from the car

Same effect for faces



wide-angle



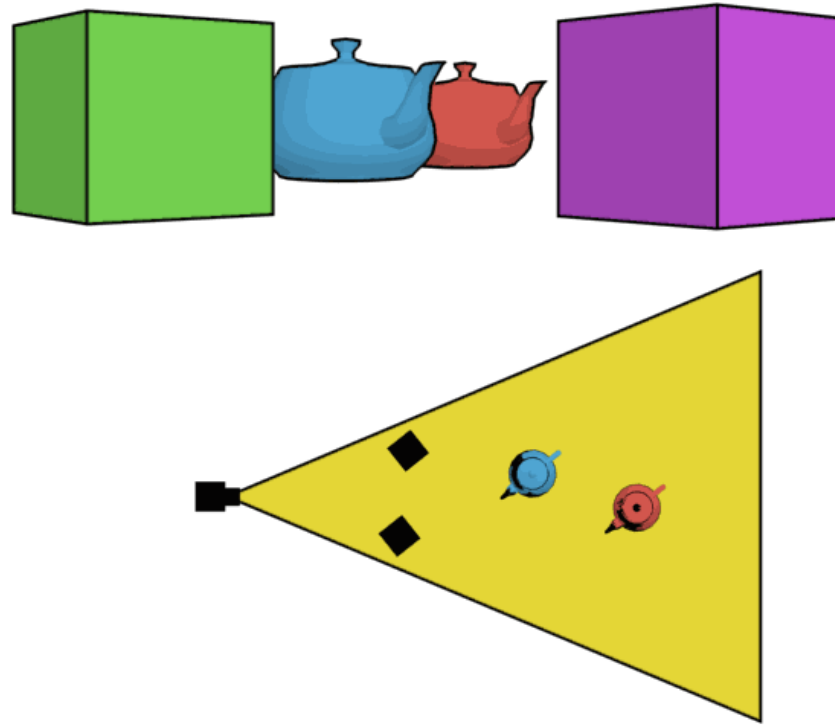
standard



telephoto

The dolly zoom

- Continuously adjusting the focal length while the camera moves away from (or towards) the subject



http://en.wikipedia.org/wiki/Dolly_zoom

The dolly zoom ("The Vertigo shot")

- Continuously adjusting the focal length while the camera moves away from (or towards) the subject

[Example of dolly zoom from *Goodfellas*](#) (YouTube)

[Example of dolly zoom from *La Haine*](#) (YouTube)

marginal distortion

Marginal distortion

- In wide-angle photos, objects on the periphery “look wrong”



Image source:
[A. Hertzmann](#)

Marginal distortion

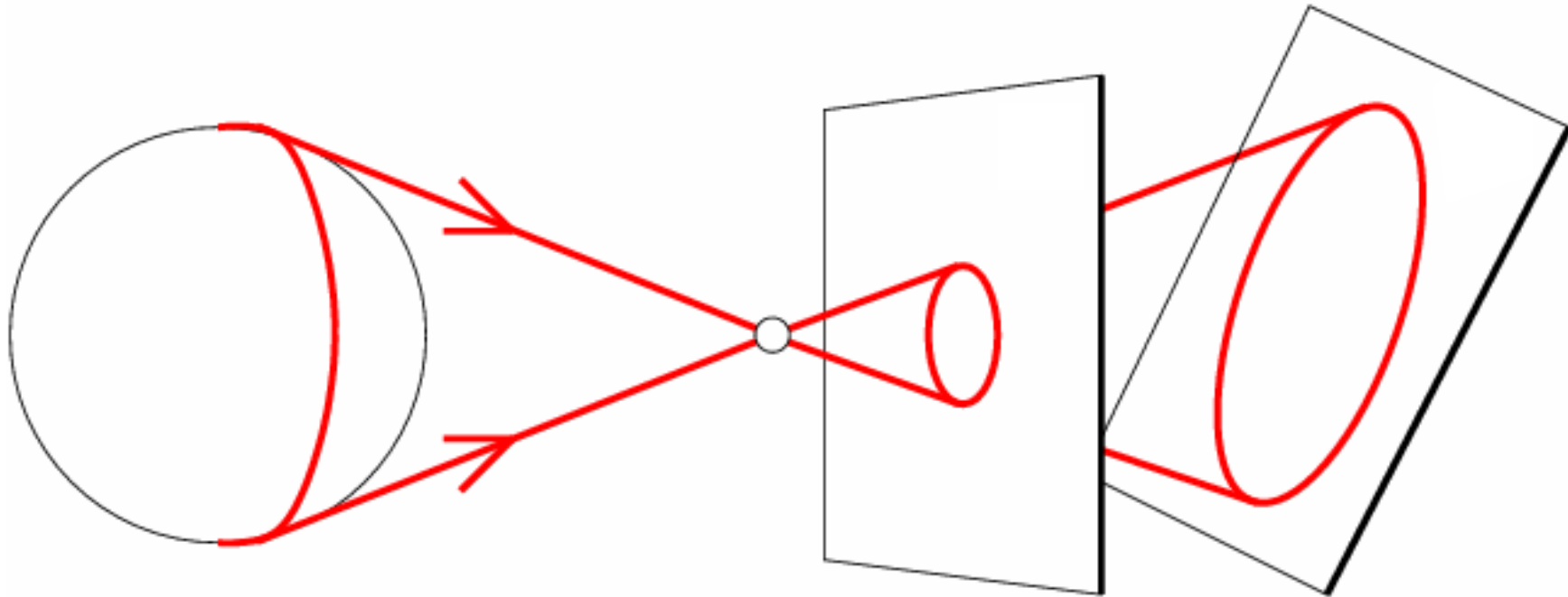
- What is the shape of the projection of a sphere?



Image source: F. Durand

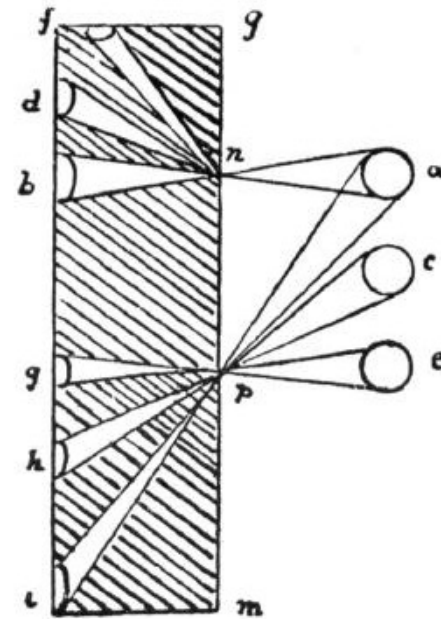
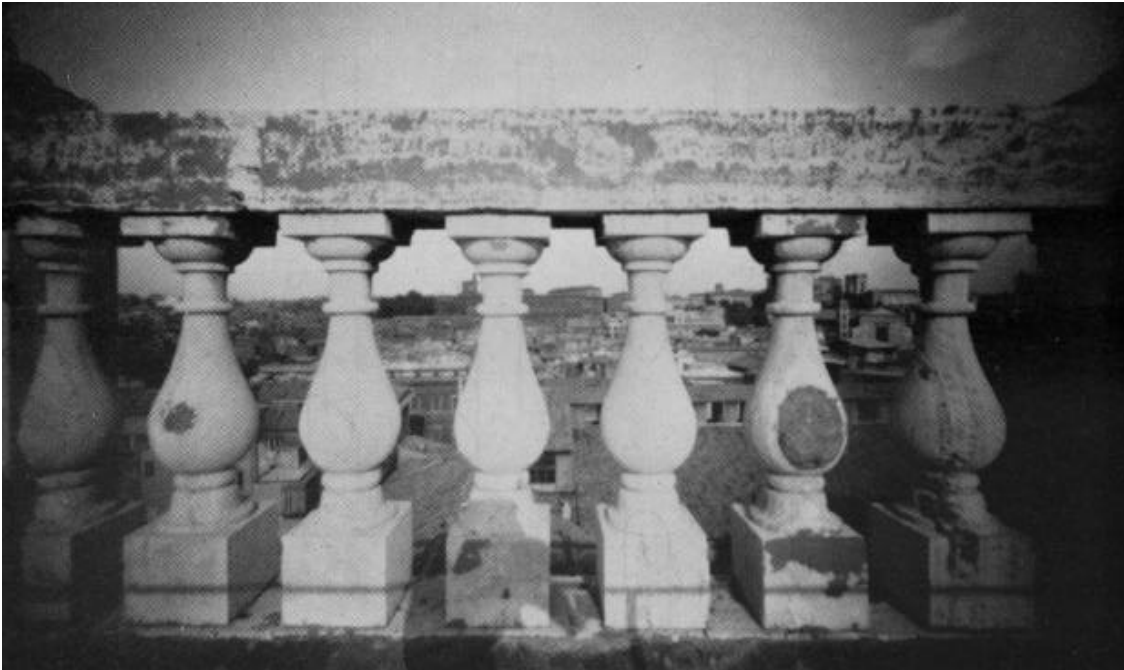
Marginal distortion

- What is the shape of the projection of a sphere?



Marginal distortion

- Are the widths of the projected columns equal?
 - The exterior columns are wider
 - This is not an optical illusion, and is not due to lens flaws
 - Phenomenon pointed out by Leonardo Da Vinci



Marginal distortion

- Which way are these people facing?

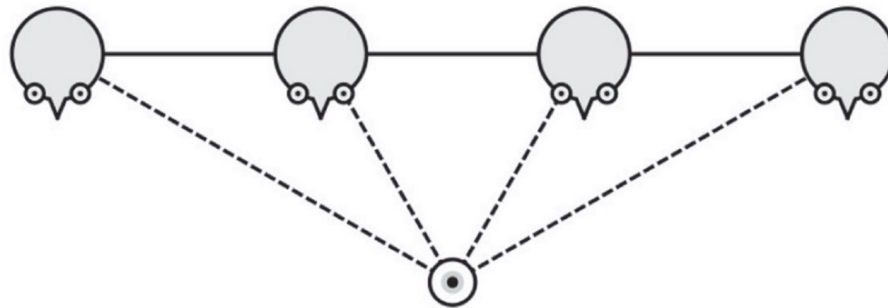


Image source: [Koenderink et al.](#) via [A. Hertzmann](#)

Perspective and art

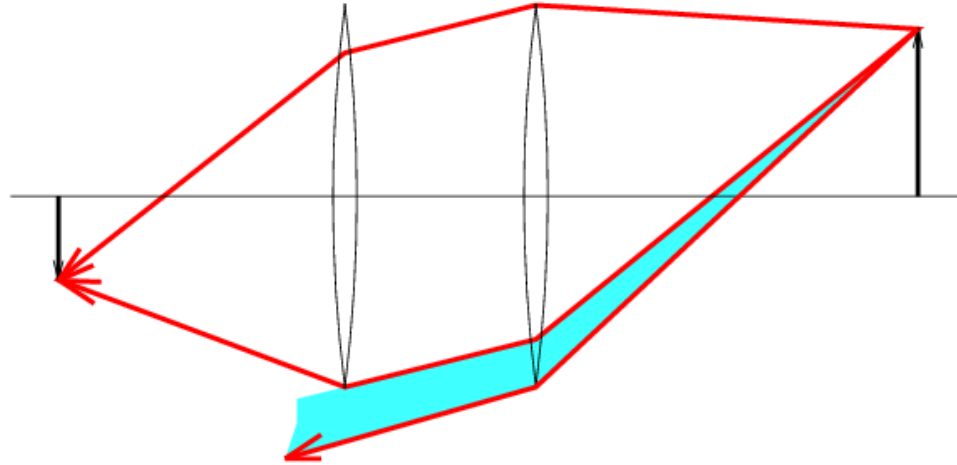


Raphael, [The School of Athens](#), 1509-1511

Perspective and art

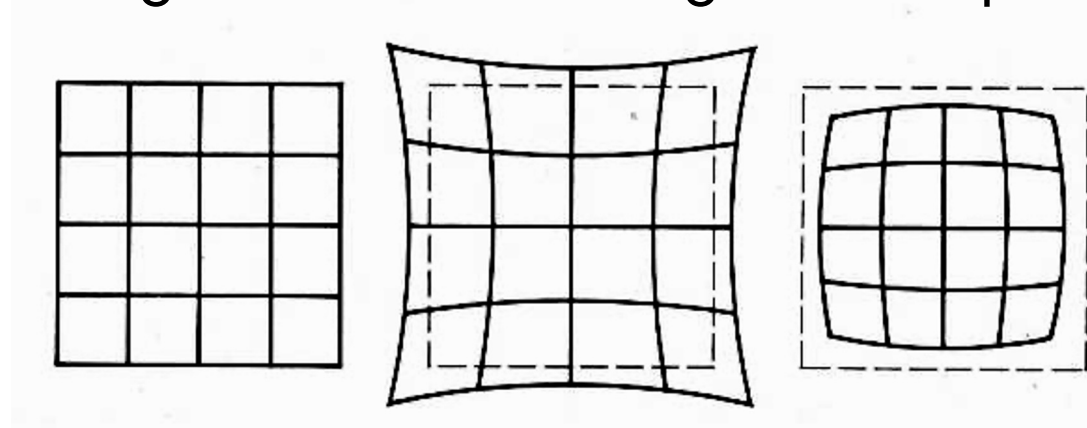
- “Large scenes with many faces are common in art... But, in the entire history of painting, I am unaware of any face depicted with the marginal distortions dictated by linear perspective.”

Vignetting



Radial distortion

- Caused by imperfect lenses
- Distortion is stronger towards the edges of the photo



No distortion

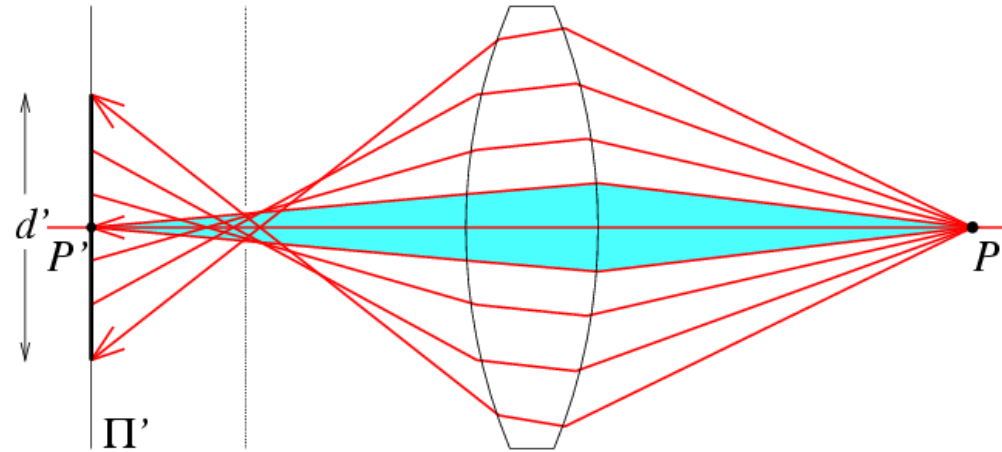
Pin cushion

Barrel



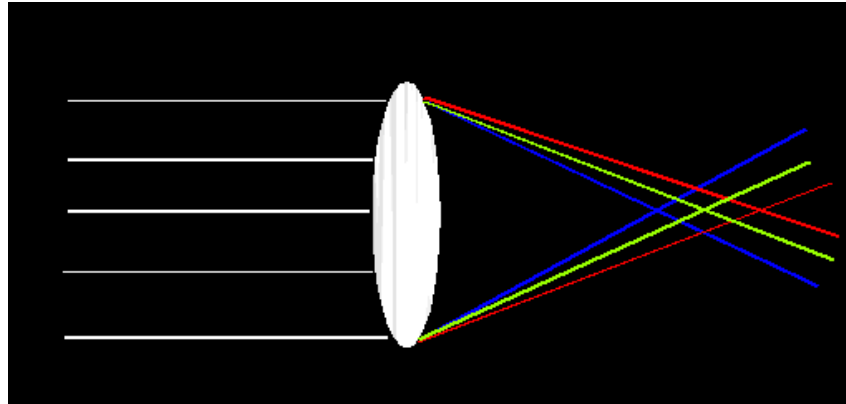
Spherical aberration

- Spherical lenses don't focus light perfectly
- Rays farther from the optical axis focus closer



Lens flaws: Chromatic aberration

- Lens has different refractive indices for different wavelengths: causes color fringing



Near Lens Center

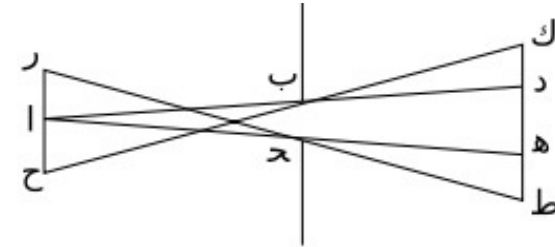


Near Lens Outer Edge

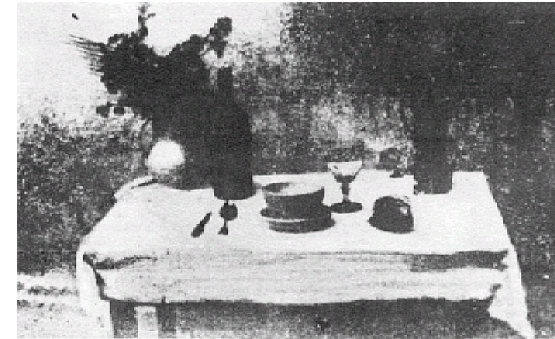


Historic milestones

- **Pinhole model:** Mozi (470-390 BCE), Aristotle (384-322 BCE)
- **Principles of optics (including lenses):** Alhacen (965-1039 CE)
- **Camera obscura:** Leonardo da Vinci (1452-1519), Johann Zahn (1631-1707)
- **First photo:** Joseph Nicéphore Niépce (1822)
- **Daguerreotypes** (1839)
- **Photographic film** (Eastman, 1889)
- **Cinema** (Lumière Brothers, 1895)
- **Color Photography** (Lumière Brothers, 1908)
- **Television** (Baird, Farnsworth, Zworykin, 1920s)
- **First consumer camera with CCD**
Sony Mavica (1981)
- **First fully digital camera:** Kodak DCS100 (1990)



Alhacen's notes



Niepce, "La Table Servie," 1822



Old television camera

https://en.wikipedia.org/wiki/History_of_photography

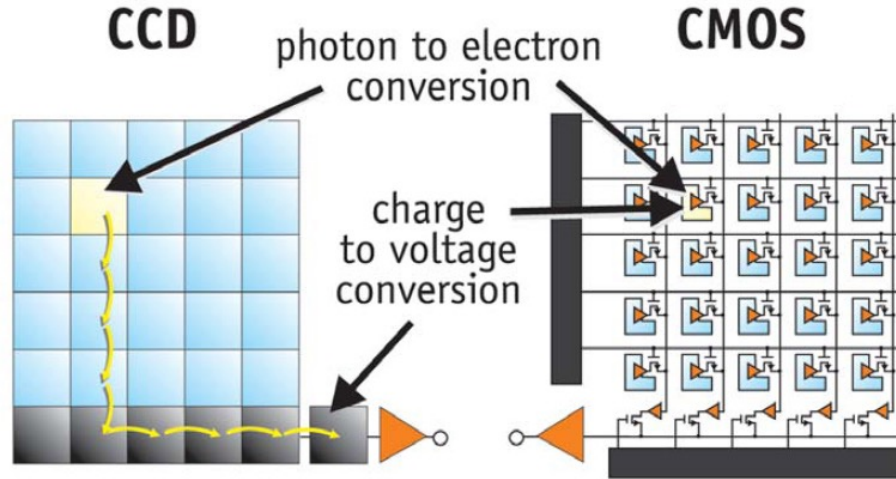
First digitally scanned photograph

- NIST (1957), 176x176 pixels



<http://listverse.com/history/top-10-incredible-early-firsts-in-photography/>

Digital camera sensors



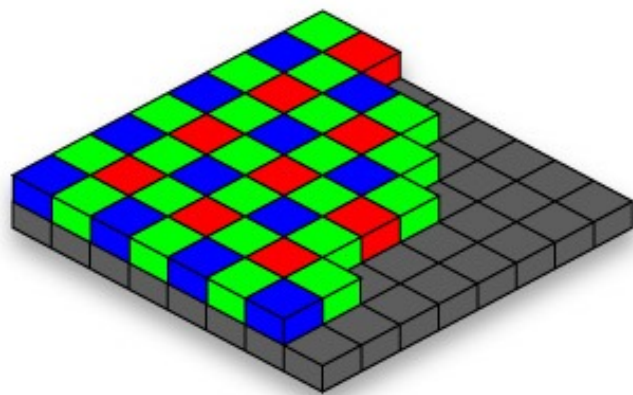
CCDs move photogenerated charge from pixel to pixel and convert it to voltage at an output node. CMOS imagers convert charge to voltage inside each pixel.

- Each cell in a sensor array is a light-sensitive diode that converts photons to electrons
 - Dominant in the past: Charge Coupled Device (CCD)
 - Dominant now: Complementary Metal Oxide Semiconductor (CMOS)

<http://electronics360.globalspec.com/article/9464/ccd-vs-cmos-the-shift-in-image-sensor-technology>

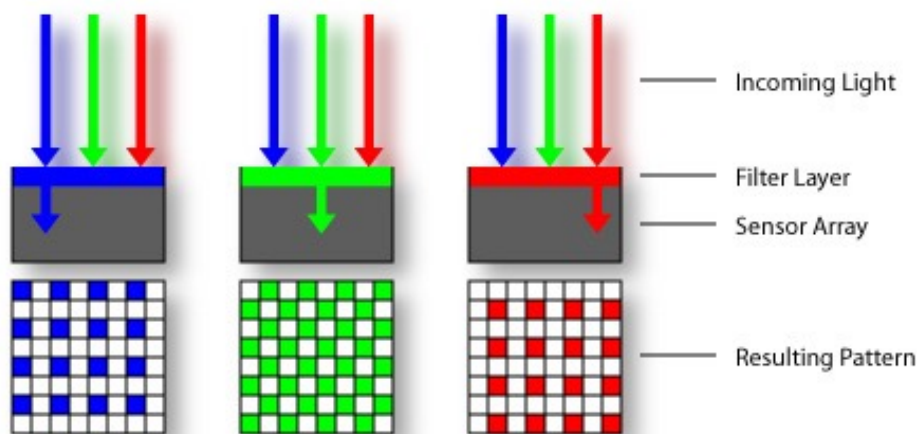
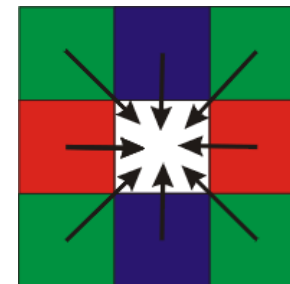
Color filter arrays

Bayer grid (1976)

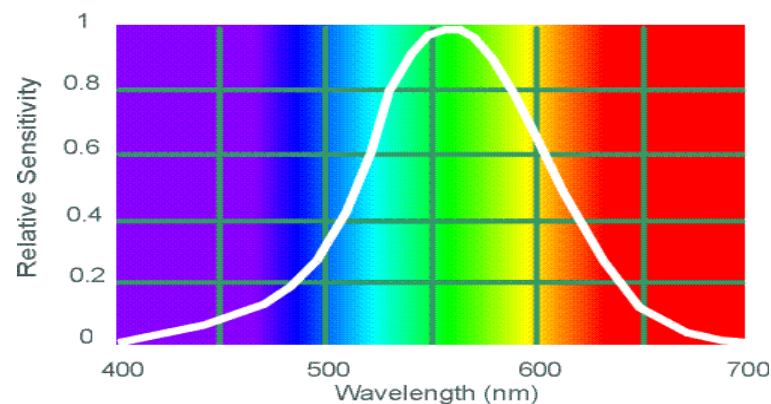


Demosaicing:

Estimation of missing components from neighboring values



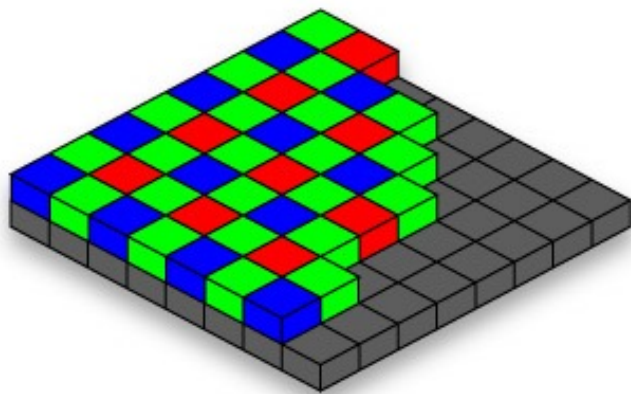
Why more green?



Human Luminance Sensitivity Function

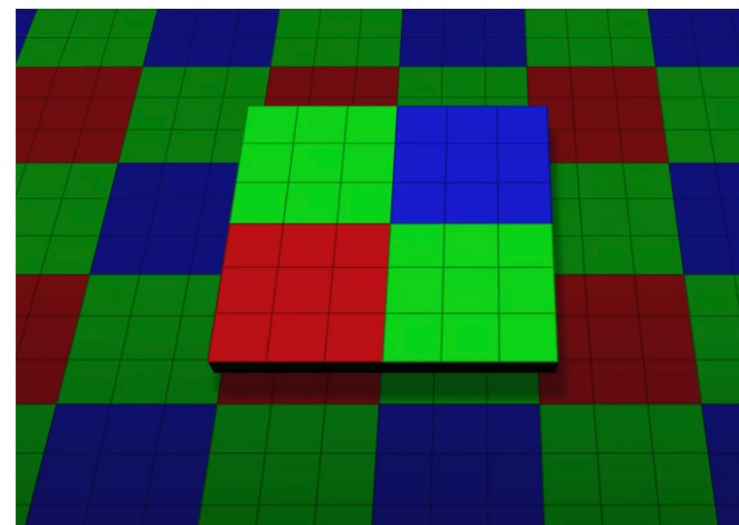
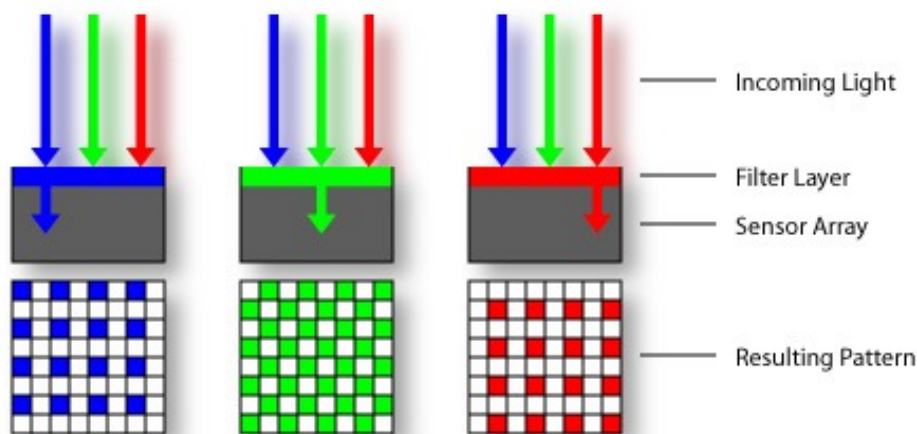
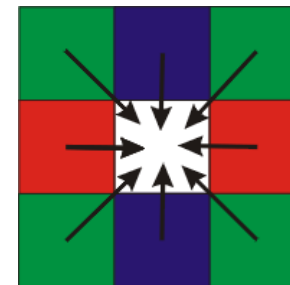
Color filter arrays

Bayer grid (1976)



Demosaicing:

Estimation of missing components from neighboring values



Recent cameraphone technology: [pixel binning](#)

Misc. digital camera artifacts

Noise

- low light is where you most notice noise
- light sensitivity (ISO) / noise tradeoff
- stuck pixels



In-camera processing

- oversharpening can produce halos



Compression

- JPEG artifacts, blocking

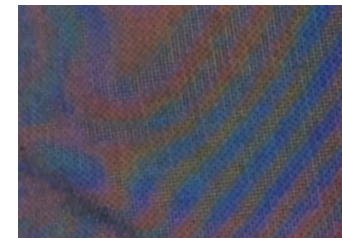


Blooming

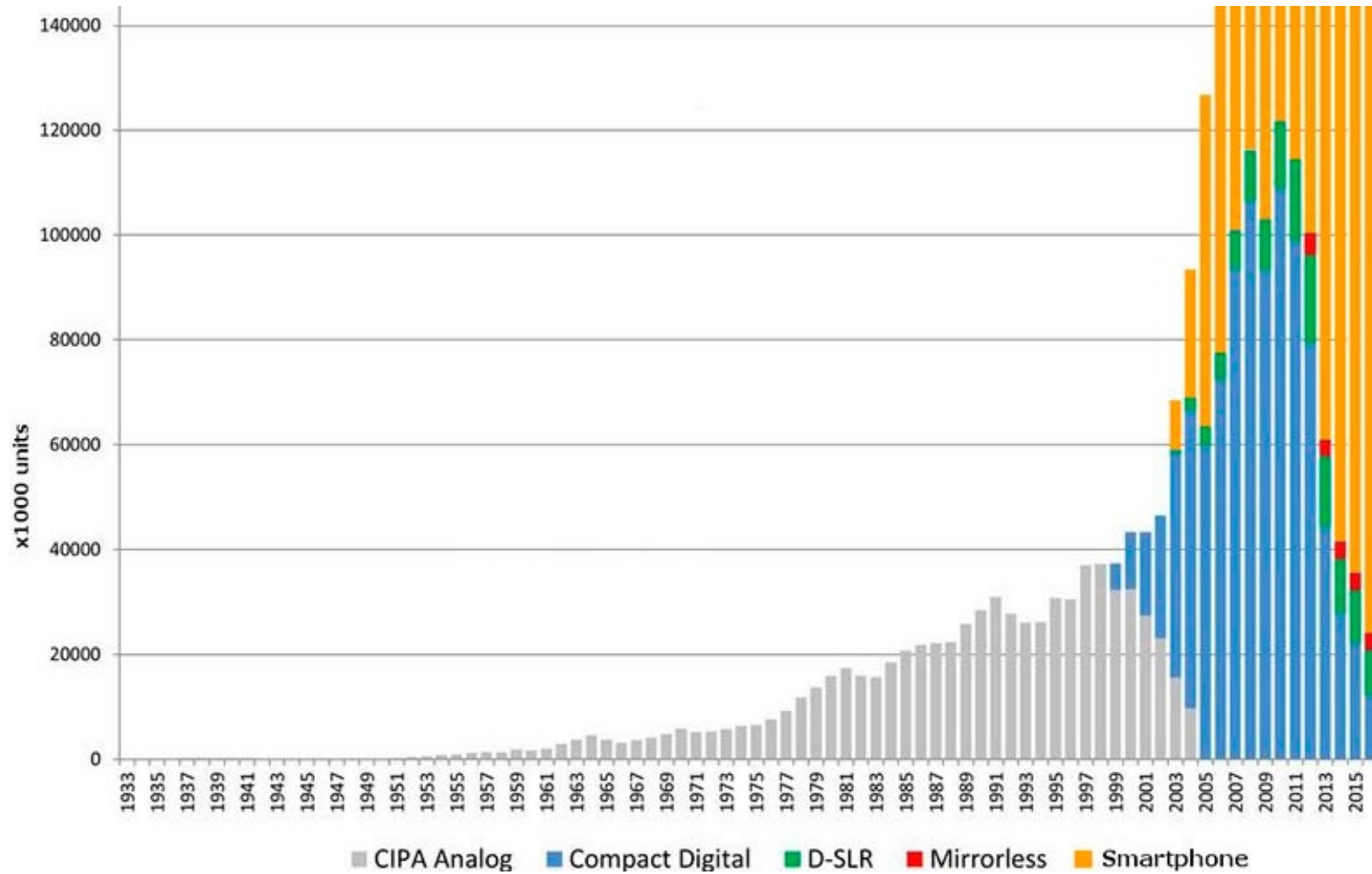
- CCD charge overflowing into neighboring pixels

Color artifacts

- Color moire
- Purple fringing from microlenses



Camera sales over time



[Source](#)

Camera sales over time

The full chart...



[Source](#)

Summary

- Pinhole cameras
- Perspective projection
- Cameras with lenses
- Marginal distortion
- Digital sensors